From Spacetime to Fields: Special Relativity and Classical Field Theory

From two postulates to the Maxwell Lagrangian

Lecture notes

July 23, 2026

Abstract

This document extends the series from mechanics to fields, with every step of the main logical chain displayed. Part I builds special relativity from its two postulates: the invariance of the spacetime interval is *derived*, the Lorentz transformation is obtained from it, and the kinematic consequences — relativity of simultaneity, time dilation, length contraction, velocity composition — follow with their operational definitions kept explicit. Minkowski geometry, four-vectors, and the structure of the Lorentz and Poincaré groups are then developed, and the relativistic particle is treated variationally: the action $S = -mc\int d\tau$ is motivated, its Euler–Lagrange equations solved, and the four-momentum obtained — not defined — as the Noether charge of spacetime translations, the same theorem of the companion mechanics notes run on a new symmetry group. Part II passes from particles to fields: the Lagrangian of the N-mass chain is taken to its continuum limit *at the level of the Lagrangian*, the variational framework of the mechanics notes is reused verbatim for fields, and the Klein–Gordon field appears as the simplest Lagrangian consistent with Lorentz invariance. Noether's theorem for fields localizes conservation laws into continuity equations and produces the energy-momentum tensor; the Hamiltonian formulation introduces the field momentum and the canonical bracket. Part III applies the entire apparatus to electromagnetism: the field tensor, the gauge-invariant Maxwell Lagrangian, all four Maxwell equations from variation and from the Bianchi identity, gauge freedom, and electromagnetic waves. The outlook closes the series: expanded in modes, every field is a family of harmonic oscillators — the opening object of the first document is the atom of field theory. Proofs whose details are purely technical are placed in appendices; every statement used in the main text is either proved there or explicitly pointed to its appendix.

Contents

1 Conventions and Notation

Dots denote time derivatives. Three-dimensional spatial vectors carry arrows ($\vec{x}$, $\vec{v}$); their components carry Latin indices i, j, k running over $1, 2, 3$. Spacetime indices are Greek, $\mu, \nu, \rho, \ldots$, running over $0, 1, 2, 3$, with $x^0 = ct$.

Convention 1.1 (Metric signature)**.** The Minkowski metric is

$$\eta_{\mu\nu} = \mathrm{diag}(+1,\ -1,\ -1,\ -1), \tag{1}$$

the "mostly-minus" convention of Schwartz and of Peskin–Schroeder. Squared timelike intervals are positive.

Convention 1.2 (Summation convention)**.** *This document breaks with the companion mechanics notes, where all sums were written out.* From here on, a repeated index — one upper, one lower — is summed over its range:

$$a^\mu b_\mu \ \equiv\ \sum_{\mu=0}^{3} a^\mu b_\mu = a^0 b_0 + a^1 b_1 + a^2 b_2 + a^3 b_3. \tag{2}$$

An index appearing twice is a *dummy* index and may be renamed freely; an index appearing once must appear, with the same placement, on both sides of every equation. Field-theory formulas become impractically long without this convention; the cost is vigilance about index placement, and the reader should expand a few contractions by hand early on until the shorthand is trusted.

Convention 1.3 (Units)**.** Sections 2–3 keep the speed of light c explicit: while the four-dimensional formalism is being assembled, c is the conversion factor between meters and seconds, and tracking it is a dimension check on every line. At the end of §3 we set $c = 1$ once and for all; the lost factors are recoverable uniquely by dimensional analysis.

The recurring symbols are collected here for reference; each is re-introduced where it first appears.

Symbol	Definition	First appears	Meaning
β	$\beta = v/c$	§2	dimensionless velocity
γ	$\gamma = (1-\beta^2)^{-1/2}$	§2	Lorentz factor
ϕ	$\tanh\phi = \beta$	§2	rapidity
s^2	$s^2 = c^2t^2 - \lvert\vec{x}\rvert^2$	§2	spacetime interval
$\eta_{\mu\nu}$	$\mathrm{diag}(+,-,-,-)$	§3	Minkowski metric
$\Lambda^\mu{}_\nu$	$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}$	§3	Lorentz transformation
τ	$c^2\mathrm{d}\tau^2 = \mathrm{d}x^\mu \mathrm{d}x_\mu$	§3	proper time
∂_μ	$\partial/\partial x^\mu$	§3	spacetime gradient
$\Box$	$\Box = \partial_\mu\partial^\mu$	§3	d'Alembertian
$\phi(x)$	—	§6	scalar field (context separates it from rapidity)
$\pi(x)$	$\pi = \partial\mathcal{L}/\partial\dot{\phi}$	§10	field momentum density
$F_{\mu\nu}$	$\partial_\mu A_\nu - \partial_\nu A_\mu$	§11	electromagnetic field tensor

Cross-references to the companion documents are written as [Mech. §n] for the variational mechanics course and [Osc. §n] for the oscillator reference.

Part I
Spacetime

2 The Two Postulates and the Lorentz Transformation

2.1 A wave equation with a problem

The oscillator reference ended with the wave equation [Osc. §9]: the continuum limit of a chain of masses is a displacement field $u(x,t)$ obeying

$$\frac{\partial^2 u}{\partial t^2} = c_s^2\,\frac{\partial^2 u}{\partial x^2}, \qquad c_s^2 = \frac{T}{\mu}, \tag{3}$$

with T the tension and μ the mass per length. Newtonian mechanics comes with a democracy of inertial frames: the laws take the same form for every observer in uniform motion, frames being related by the *Galilean transformation*

$$\bar{x} = x - vt, \qquad \bar{t} = t. \tag{4}$$

Is the wave equation a law in this sense? Chain rule: for $u(x,t) = \bar{u}(\bar{x},\bar{t})$,

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \bar{x}}, \qquad \frac{\partial}{\partial t} = \frac{\partial}{\partial \bar{t}} - v\,\frac{\partial}{\partial \bar{x}}, \tag{5}$$

so that

$$\frac{\partial^2 u}{\partial t^2} - c_s^2\frac{\partial^2 u}{\partial x^2} \;=\; \frac{\partial^2 \bar{u}}{\partial \bar{t}^2} \;-\; 2v\,\frac{\partial^2 \bar{u}}{\partial \bar{x}\,\partial \bar{t}} \;+\; \left(v^2 - c_s^2\right)\frac{\partial^2 \bar{u}}{\partial \bar{x}^2}. \tag{6}$$

The right-hand side is *not* of the form $\bar{u}_{\bar{t}\bar{t}} - c_s^2\,\bar{u}_{\bar{x}\bar{x}}$: the wave equation changes form under a Galilean boost. For sound this is not a problem — the equation was derived in the rest frame of the *medium*, and the medium's rest frame is genuinely special. A moving observer measures different sound speeds upstream and downstream, and Eq. (6) is precisely the equation that says so.

The problem arrives with light. Maxwell's equations predict electromagnetic waves of speed $c = (\mu_0\varepsilon_0)^{-1/2} \approx 3.00 \times 10^8\,\mathrm{m/s}$ — a wave equation of the form (3) with $c_s \to c$, as Part III will derive. By the logic above, that equation should hold only in the rest frame of light's medium, and measuring the anisotropy of the speed of light should reveal the Earth's motion through it. The Michelson–Morley experiment (1887) and its successors looked and found nothing: the speed of light is the same in every direction, in every season, to (by now) parts in 10^{17}. [experimental input; we take it as given.]

Einstein's move (1905) was to stop treating the null result as a conspiracy to be explained and promote it to a law of nature. Two postulates:

Postulate 2.1 (Relativity)**.** The laws of physics take the same form in every inertial frame. No experiment performed inside a laboratory can detect the laboratory's uniform motion.

Postulate 2.2 (Invariance of the speed of light)**.** There is a finite speed c that is the same in every inertial frame, independent of the motion of the source. Light in vacuum travels at c.

Postulate 2.1 simply retains Galileo's principle. What is abandoned is the Galilean transformation (4) — because, as Eq. (6) shows, it is *incompatible* with Postulate 2.2: under (4) a speed-c signal in one frame has speed $c \pm v$ in another. The task of this section is to find the transformation law that the two postulates force, and the strategy is to first find what the postulates leave *invariant.*

2.2 Setup: events, frames, and the standard configuration

An *event* is a point of spacetime: something with a where and a when, idealized to a point — a firecracker, the coincidence of two particles, one tick of one clock. An inertial frame S assigns coordinates (t, x, y, z) to every event, operationally by a lattice of rulers and synchronized clocks at rest in S; we write the time coordinate as $x^0 = ct$ so that all four coordinates carry the same dimension.

Two frames S and S' are in *standard configuration* if their axes are parallel, S' moves with velocity v along the common x-axis, and the origins coincide: the event $(0, 0, 0, 0)$ has these coordinates in both frames. For the rest of this section we suppress the transverse coordinates y, z and work in the (ct, x) plane; the transverse directions are dealt with in Appendix A (they turn out to be spectators: $y' = y$, $z' = z$).

Two facts about the transformation $(ct, x) \mapsto (ct', x')$ come before the postulates proper:

Lemma 2.3 (Linearity)**.** *The transformation between two inertial frames in standard configuration is linear:*

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \Lambda \begin{pmatrix} ct \\ x \end{pmatrix} \tag{7}$$

for a constant matrix Λ depending only on v.

The physics behind the lemma is the homogeneity of space and time — no event is privileged, so coordinate *differences* must transform the same way wherever they sit — plus continuity. The full argument is technical and lives in Appendix A. [stated; proved in Appendix A.]

2.3 The invariant: derivation of the interval

Define, for any pair of events with separations Δt, Δx in the frame S, the *interval*

$$\Delta s^2 \;=\; c^2\,\Delta t^2 - \Delta x^2. \tag{8}$$

(No claim yet that this is a good idea; the definition is justified by Theorem 2.4 immediately below.) Since the transformation is linear, separations transform like coordinates, and it suffices to study $s^2 = c^2t^2 - x^2$ for events relative to the common origin.

Theorem 2.4 (Invariance of the interval)**.** *For any two inertial frames in standard configuration and any pair of events,*

$$\boxed{c^2\,\Delta t'^2 - \Delta x'^2 \;=\; c^2\,\Delta t^2 - \Delta x^2.} \tag{9}$$

Proof. The proof runs in three steps: the postulates first pin down the *zero set* of s^2, then its scale, and isotropy kills the scale.

Step 1: null separations are preserved. Introduce the null coordinates

$$u = ct - x, \qquad w = ct + x, \qquad \text{so that} \qquad s^2 = u\,w. \tag{10}$$

A pair of events with $u = 0$ (i.e. $x = ct$) can be connected by a light signal moving in the $+x$ direction; a pair with $w = 0$, by one moving in the $-x$ direction. Postulate 2.2 says light moves at c in S' as well: therefore $u = 0 \Rightarrow u' = 0$ and $w = 0 \Rightarrow w' = 0$. (The boost does not swap the two light rays: at $v = 0$ the transformation is the identity, and by continuity in v the rightward ray cannot jump onto the leftward one.)

Step 2: the interval can change at most by an overall factor. In the null coordinates the linear transformation reads $u' = \alpha u + \beta_1 w$, $w' = \beta_2 u + \delta w$. Step 1 forces $\beta_1 = \beta_2 = 0$ (set $u = 0$, vary w, demand $u' = 0$; and conversely). Hence

$$u' = \alpha(v)\, u, \qquad w' = \delta(v)\, w, \qquad \Longrightarrow \qquad s'^2 = u'w' = \alpha(v)\,\delta(v)\; s^2 \;\equiv\; \lambda(v)\, s^2, \tag{11}$$

with $\alpha, \delta > 0$ for a transformation continuously connected to the identity that does not reverse the direction of time. The whole quadratic form is rescaled by a single velocity-dependent number $\lambda(v) > 0$.

Step 3: isotropy forces $\lambda = 1$. Two properties of λ:

- **Reciprocity.** The inverse transformation, from S' back to S, is the boost with velocity $-v$ (Postulate 2.1: S moves at $-v$ as seen from S', and neither frame is privileged). Applying (11) twice, $s^2 = \lambda(-v)\, s'^2 = \lambda(-v)\lambda(v)\, s^2$, so

$$\lambda(v)\,\lambda(-v) = 1. \tag{12}$$

- **Isotropy.** Space has no preferred direction: the physics of a boost to the right at speed $|v|$ is the mirror image of a boost to the left. Formally, let $P : (ct, x) \mapsto (ct, -x)$; then the boost of velocity $-v$ is $P\,\Lambda_v\,P$. Since P leaves s^2 unchanged, the scale factors of Λ_{-v} and Λ_v are equal:

$$\lambda(-v) = \lambda(v). \tag{13}$$

Together, $\lambda(v)^2 = 1$; positivity gives $\lambda(v) = 1$. □

Remark 2.5 (What just happened). Postulate 2.2 by itself only protects the light cone $s^2 = 0$; it is Postulate 2.1 — entering through reciprocity and isotropy — that rigidifies the whole quadratic form. The interval (8) is the replacement for the two quantities Newtonian physics held absolute separately: the time between two events and the distance between them are now frame-dependent, and only the combination $c^2\Delta t^2 - \Delta x^2$ is common property. Figure 1 shows the geometry: the events at fixed interval from the origin form hyperbolae, and every inertial frame's unit ticks lie on the *same* hyperbolae.

2.4 Two routes from the invariant to the transformation

Theorem 2.4 converts the search for the transformation law into a purely algebraic problem: *find all linear maps of the* (ct, x) *plane that preserve* $c^2t^2 - x^2$, then pick out the one that describes a boost of velocity v. In the spirit of the companion notes' double derivation of Hamilton's equations [Mech. §6.2], we solve it twice: once in coordinates adapted to the problem's structure, where the answer explains itself, and once by brute force in the physical coordinates, where every coefficient is worked out by hand.

Route 1: null coordinates. The work is already done. Any linear map that preserves $s^2 = uw$ in particular preserves its zero set, the two null lines, so Steps 1–2 of the proof of Theorem 2.4 apply verbatim: in the null coordinates (10) the map must have the diagonal form $u' = \alpha u$, $w' = \delta w$ with $\alpha, \delta > 0$ — and now, since the interval is preserved exactly,

$$\alpha\,\delta = \lambda = 1, \qquad \text{i.e.} \qquad \delta = \frac{1}{\alpha}. \tag{14}$$

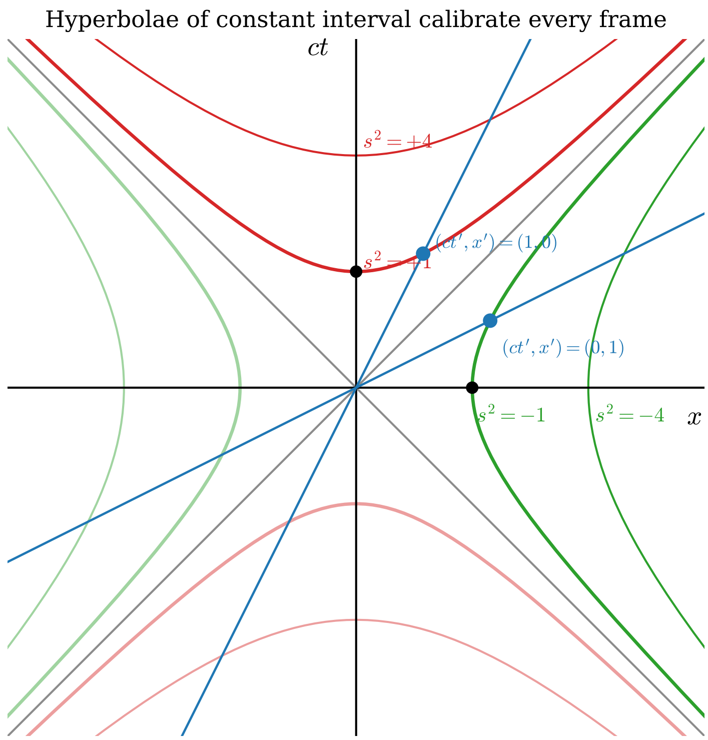

Figure 1: Curves of constant interval $s^2 = c^2t^2 - x^2$ (timelike in red, spacelike in green) and the axes of a boosted frame ($\beta = 0.5$, blue). The boosted frame's unit events — one tick of its clock at its origin, one meter along its ruler at its zero time — lie on the same unit hyperbolae as the unprimed frame's. The hyperbolae are drawn from the exact equations $ct = \pm\sqrt{s^2 + x^2}$ and $x = \pm\sqrt{s^2 + (ct)^2}$.

One free positive number α remains. Since the exponential is a bijection from $\mathbb{R}$ onto the positive reals, we may write $\alpha = \mathrm{e}^{\phi}$ for a unique $\phi \in \mathbb{R}$ — pure repackaging, $\phi = \ln \alpha$, but repackaging with a payoff: composing two such maps *multiplies* the α's, hence *adds* the ϕ's, so the parameter ϕ is the one in which the group law will look simplest (used in §2.5). The one-parameter family is

$$u' = \mathrm{e}^{\phi}\, u, \qquad w' = \mathrm{e}^{-\phi}\, w, \qquad \phi \in \mathbb{R}: \tag{15}$$

each null coordinate is stretched by exactly the factor that shrinks the other, and the product uw survives.

Now translate back. Inverting (10), $ct = \frac{1}{2}(w+u)$ and $x = \frac{1}{2}(w-u)$, and the same relations hold in the primed frame. Substitute (15), then $u = ct - x$, $w = ct + x$:

$$\begin{aligned} ct' = \tfrac{1}{2}\big(w' + u'\big) &= \tfrac{1}{2}\big(\mathrm{e}^{-\phi}(ct+x) + \mathrm{e}^{\phi}(ct-x)\big) \\ &= ct\,\frac{\mathrm{e}^{\phi}+\mathrm{e}^{-\phi}}{2} - x\,\frac{\mathrm{e}^{\phi}-\mathrm{e}^{-\phi}}{2} \;=\; ct\,\cosh\phi - x\,\sinh\phi, \\ x' = \tfrac{1}{2}\big(w' - u'\big) &= \tfrac{1}{2}\big(\mathrm{e}^{-\phi}(ct+x) - \mathrm{e}^{\phi}(ct-x)\big) \\ &= -\,ct\,\frac{\mathrm{e}^{\phi}-\mathrm{e}^{-\phi}}{2} + x\,\frac{\mathrm{e}^{\phi}+\mathrm{e}^{-\phi}}{2} \;=\; -\,ct\,\sinh\phi + x\,\cosh\phi, \end{aligned} \tag{16}$$

where the last step in each line is nothing but the definitions $\cosh\phi = \frac{1}{2}(\mathrm{e}^{\phi} + \mathrm{e}^{-\phi})$, $\sinh\phi = \frac{1}{2}(\mathrm{e}^{\phi} - \mathrm{e}^{-\phi})$: the hyperbolic functions are not imported; they arise directly as the symmetric and antisymmetric combinations of the two exponential stretch factors.

Remark 2.6 (Why hyperbolic functions were inevitable)*.* Compare with the Euclidean plane, feature by feature:

	rotation	boost
preserved form	$x^2 + y^2$ (circles)	$c^2t^2 - x^2$ (hyperbolae)
matrix	$\begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}$	$\begin{pmatrix}\cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi\end{pmatrix}$
parameter	angle θ, additive	rapidity ϕ, additive
level-set parametrization	$(\cos\theta, \sin\theta)$	$(\cosh\phi, \sinh\phi)$
function family	trigonometric	hyperbolic
eigenvalues	$\mathrm{e}^{\pm\mathrm{i}\theta}$ (complex)	$\mathrm{e}^{\pm\phi}$ (real; Remark 2.11)

A rotation slides points along the circles that its invariant form draws; Eq. (16) slides events along the hyperbolae of Fig. 1, which its invariant form draws — a *hyperbolic rotation*, with $\cosh^2\phi - \sinh^2\phi = 1$ playing the role of $\cos^2\theta + \sin^2\theta = 1$. Every row of the table follows from the single sign difference in the preserved form, and the last row — real eigenvalues where rotations have complex ones — is the difference with the most physical content, as Remark 2.11 will show.

It remains to identify the parameter ϕ, called the *rapidity*, in terms of the velocity. The origin of S' is the worldline $x' = 0$; by (16) that is the line $x = ct\tanh\phi$. But the origin of S' moves at velocity v by definition of the standard configuration, i.e. along $x = vt$. Hence

$$\boxed{\tanh\phi = \frac{v}{c} \equiv \beta, \qquad \cosh\phi = \frac{1}{\sqrt{1-\beta^2}} \equiv \gamma, \qquad \sinh\phi = \gamma\beta.} \tag{17}$$

(The last two follow from the first and $\cosh^2 - \sinh^2 = 1$.) Substituting into (16) gives the boxed result of Theorem 2.8 below.

Route 2: undetermined coefficients. Now the same problem with no change of variables: write the general linear map in the physical coordinates and let the invariant grind out the coefficients.

(1) Ansatz. By Lemma 2.3,

$$ct' = A\,ct + B\,x, \qquad x' = D\,ct + E\,x, \tag{18}$$

with four constants depending on β only.

(2) Motion of the moving origin. The worldline $x' = 0$ is, by definition of the standard configuration, the line $x = \beta\,ct$. Substituting: $0 = D\,ct + E\beta\,ct$ for all t, so $D = -\beta E$ and

$$x' = E\,(x - \beta\,ct). \tag{19}$$

One coefficient down, three to go.

(3) Balance the invariant. Demand $(ct')^2 - x'^2 = (ct)^2 - x^2$ *identically* in (ct, x). Expand the left side,

$$(A\,ct + Bx)^2 - E^2(x - \beta\,ct)^2, \tag{20}$$

and match the coefficients of $(ct)^2$, x^2, and the cross term $(ct)\,x$ separately — an identity between quadratic forms is three scalar equations:

$$\begin{aligned} A^2 - \beta^2 E^2 &= 1 \qquad && [(ct)^2 \text{ term}] \\ B^2 - E^2 &= -1 && [x^2 \text{ term}] \\ AB + \beta E^2 &= 0 && [(ct)\,x \text{ term}] \end{aligned} \tag{21}$$

(4) Solve. Square the third equation, $A^2B^2 = \beta^2E^4$, and substitute the first two in the forms $A^2 = 1 + \beta^2E^2$ and $B^2 = E^2 - 1$:

$$(1 + \beta^2E^2)(E^2 - 1) = \beta^2E^4 \quad\Longrightarrow\quad E^2 - 1 - \beta^2E^2 = 0 \quad\Longrightarrow\quad E^2 = \frac{1}{1-\beta^2}. \tag{22}$$

So $E = \gamma$, and back-substituting,

$$A^2 = 1 + \beta^2\gamma^2 = (1-\beta^2)\gamma^2 + \beta^2\gamma^2 = \gamma^2 \;\Rightarrow\; A = \gamma, \qquad B = -\frac{\beta E^2}{A} = -\gamma\beta. \tag{23}$$

The signs: the equations (21) are quadratic and admit sign flips of (A, B) and of E; the positive roots are selected by continuity to the identity at $\beta = 0$, the other choices being spatial reflection and/or time reversal, excluded by the standard configuration. The result is $ct' = \gamma(ct - \beta x)$, $x' = \gamma(x - \beta ct)$ — the two routes meet.

Remark 2.7 (What the change of coordinates achieved). Route 2 is three coupled quadratic equations and an elimination; Route 1 is two decoupled one-line equations, (15). The difference is entirely in the coordinates: the null variables u, w are adapted to the preserved form ($s^2 = uw$ has no diagonal terms to balance), and in them the transformation is diagonal. This is the same simplification that normal coordinates achieve for coupled oscillators [Osc. §6]: pick the variables the structure prefers, and the coupled system falls apart. Remark 2.11 will name the sense in which the null coordinates are exactly the "normal coordinates" of a boost.

Collecting the result of both routes:

Theorem 2.8 (Lorentz transformation)**.** *Two inertial frames in standard configuration with relative velocity $v = \beta c$, $|\beta| < 1$, are related by*

$$\boxed{ct' = \gamma\,(ct - \beta x), \qquad x' = \gamma\,(x - \beta\, ct), \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}},} \tag{24}$$

equivalently by the rapidity form (16) *with* $\tanh\phi = \beta$. *The inverse transformation is the same formula with* $\beta \to -\beta$.

Remark 2.9 (Sanity checks). (i) *Nonrelativistic limit.* For $\beta \ll 1$, $\gamma = 1 + \frac{1}{2}\beta^2 + O(\beta^4)$, and (24) becomes $x' = x - vt + O(\beta^2)$, $t' = t - vx/c^2 + O(\beta^2)$: the Galilean transformation is recovered in the regime where it was ever tested, with the first correction hitting the *time* coordinate — the term vx/c^2 is the relativity of simultaneity, previewed. (ii) *Light stays light.* If $x = ct$ then $x' = \gamma(ct - \beta ct) = ct'$, as built in. (iii) *The speed limit.* γ is real only for $|\beta| < 1$: no inertial frame moves at or above c relative to another. Rapidity makes the limit natural — ϕ ranges over all of $\mathbb{R}$ while $\beta = \tanh\phi$ is confined to $(-1, 1)$.

2.5 Composition: the group structure and velocity addition

What happens when boosts are stacked — a passenger walking in a moving train? Compose two transformations of the form (16) with rapidities ϕ_1, ϕ_2. The matrices are

$$\Lambda(\phi) = \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix}, \tag{25}$$

and the hyperbolic addition formulas $\cosh(\phi_1+\phi_2) = \cosh\phi_1\cosh\phi_2 + \sinh\phi_1\sinh\phi_2$, $\sinh(\phi_1+\phi_2) = \sinh\phi_1\cosh\phi_2 + \cosh\phi_1\sinh\phi_2$ give, by direct multiplication,

$$\boxed{\Lambda(\phi_2)\,\Lambda(\phi_1) = \Lambda(\phi_1 + \phi_2)} \tag{26}$$

— boosts along a common axis form a one-parameter group, and *rapidities simply add*, exactly as rotation angles add in the plane. Velocities do not:

Corollary 2.10 (Velocity composition)**.** *If S' moves at β_1 relative to S, and a body moves at β_2 relative to S' (both along x), its velocity relative to S is*

$$\boxed{\beta_{12} = \tanh(\phi_1 + \phi_2) = \frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}.} \tag{27}$$

Proof. Immediate from (26) and the addition formula for tanh. □

The denominator does the work: for $\beta_1, \beta_2 \in (0, 1)$ it exceeds 1 just enough to keep $\beta_{12} < 1$, and if either input is 1 the output is 1 — light composed with anything is light. Figure 2 composes eight successive boosts of $\beta = 0.5$: the Galilean sum crosses the light speed on the second step; the true velocity approaches it and never arrives, while the rapidity grows linearly.

Remark 2.11 (The light cone is an eigendirection). Equation (15) was secretly an eigendecomposition. The characteristic polynomial of $\Lambda(\phi)$ is

$$\det\bigl(\Lambda(\phi) - \lambda\,1\bigr) = (\cosh\phi - \lambda)^2 - \sinh^2\phi = \lambda^2 - 2\lambda\cosh\phi + 1, \tag{28}$$

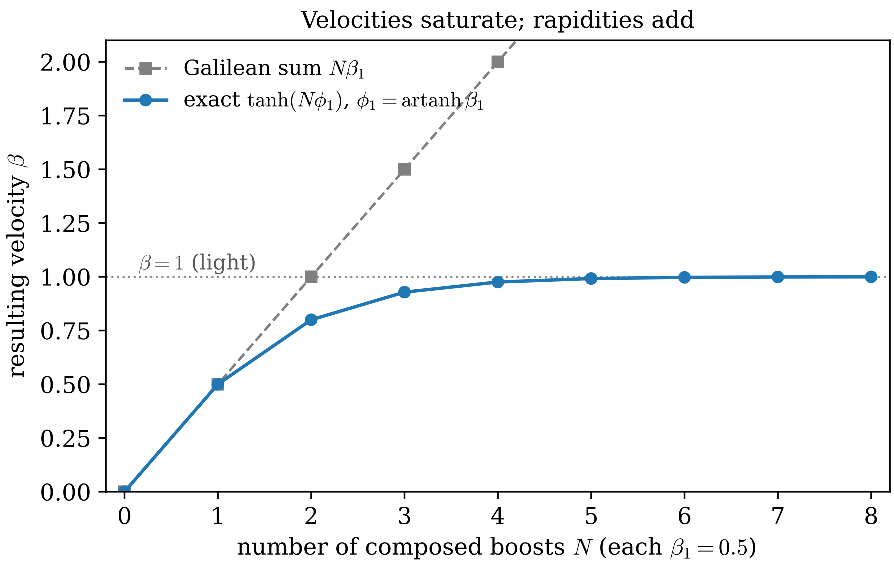

Figure 2: Composing N successive boosts of $\beta_1 = 0.5$. Gray squares: the Galilean prediction $N\beta_1$. Blue circles: the exact result $\tanh(N\phi_1)$ with $\phi_1 = \mathrm{artanh}(0.5)$, from Eq. (26).

with roots

$$\lambda_\pm = \cosh\phi \pm \sinh\phi = \mathrm{e}^{\pm\phi} : \tag{29}$$

real, positive, and reciprocal. The eigenvectors are the null directions: $(ct, x) \propto (1, -1)$, the line $w = 0$, carries $\lambda = \mathrm{e}^{\phi}$, and $(1, 1)$, the line $u = 0$, carries $\lambda = \mathrm{e}^{-\phi}$ — *the eigenvectors of a boost are the light cone*. The null coordinates of Route 1 are the eigencoordinates, Eq. (15) is the diagonalized boost, and the geometric action is now plain: stretch one light-ray direction by e^{ϕ}, compress the other by $\mathrm{e}^{-\phi}$, with $\det = \mathrm{e}^{\phi}\mathrm{e}^{-\phi} = 1$ — an area-preserving *squeeze* of the plane, sliding every event along its invariant hyperbola (Fig. 1) while the asymptotes stay put. Three payoffs:

- **Postulate 2.2, restated in linear algebra.** The light cone is the common eigendirection of every boost — the directions of spacetime on which all inertial observers agree. "The speed of light is invariant" and "light rays are eigenvectors" are the same sentence in two languages.
- **Composition, in one line.** Composing boosts multiplies eigenvalues on shared eigenvectors: $\mathrm{e}^{\phi_1}\mathrm{e}^{\phi_2} = \mathrm{e}^{\phi_1+\phi_2}$, which is Eq. (26) again — rapidity addition is the multiplicativity of eigenvalues, read on the light cone.
- **The eigenvalue is physical.** The stretch factor

$$\mathrm{e}^{\phi} = \cosh\phi + \sinh\phi = \gamma(1+\beta) = \sqrt{\frac{1+\beta}{1-\beta}} \tag{30}$$

  will return: it is the relativistic Doppler factor, the ratio by which a receding observer sees periods stretched, derived as such in §3. [stated; derived in §3.]

2.6 Consequences with operational care

Each of the three classic kinematic effects is a one-line consequence of Theorem 2.8 — *once the measurement being described is stated precisely.* Most of the traditional confusion comes from skipping that step.

Relativity of simultaneity. Two events are simultaneous in S' when $\Delta t' = 0$. By (24), $c\Delta t' = \gamma(c\Delta t - \beta\,\Delta x)$, so

$$\boxed{\Delta t' = 0 \quad \Longleftrightarrow \quad c\,\Delta t = \beta\,\Delta x.} \tag{31}$$

Events simultaneous in S' are *not* simultaneous in S unless they also happen at the same place. In the spacetime diagram (Fig. 3) the surfaces of simultaneity of S' are the tilted lines $ct = \beta x + \text{const}$: simultaneity is a frame-dependent slicing of spacetime, not a fact about spacetime itself. This effect is first order in β — it is the vx/c^2 term of Remark 2.9 — and it is the engine behind the other two effects, which are second order.

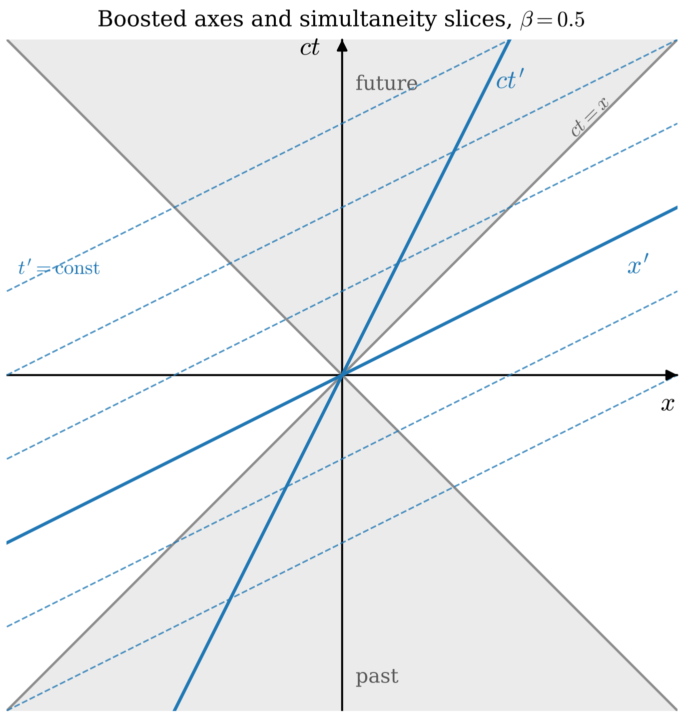

Figure 3: The (x, ct) plane with the light cone of the origin (gray) and the axes of a frame boosted to $\beta = 0.5$ (blue). The ct' axis is the worldline of the moving origin, $x = \beta\,ct$; the x' axis and its dashed translates, $ct = \beta x + \text{const}$, are the moving observer's surfaces of simultaneity, Eq. (31). Both axes close symmetrically onto the light cone, which is shared by all frames.

Time dilation. Take one clock at rest in S', sitting at $x' = 0$, and let it tick twice, the ticks separated by $\Delta t'$ — an interval of *proper time*, time measured by a clock present at both events. What does S record between the same two ticks? Use the inverse of (24) (i.e. $\beta \to -\beta$) with $\Delta x' = 0$:

$$\boxed{\Delta t = \gamma\,\Delta t' \;\geq\; \Delta t'.} \tag{32}$$

A moving clock ticks slowly by the factor γ. The statement is reciprocal — each frame sees the other's clocks run slow — and the reciprocity is consistent precisely because the two frames disagree about which distant events are simultaneous, Eq. (31): the two measurements compare one clock against *different pairs* of synchronized clocks.

Length contraction. Take a rod at rest in S', occupying $x' \in [0, L_0]$; L_0 is its *proper length.* To measure its length in S one must mark the positions of both ends *at the same S-time* — length measurement of a moving object is a simultaneity-laden operation, which is the warning that a frame-dependent answer is coming. Set $\Delta t = 0$ in the second equation of (24): $\Delta x' = \gamma\,\Delta x$, with $\Delta x' = L_0$ and $\Delta x = L$ the measured length. Hence

$$\boxed{L = \frac{L_0}{\gamma} \;\leq\; L_0.} \tag{33}$$

A moving rod is short by the factor γ, along its direction of motion only (the transverse coordinates are untouched, Appendix A).

Example 2.12 (Cosmic-ray muons: the worked number)**.** Muons are produced by cosmic-ray collisions at altitude $h \approx 15\,\text{km}$, with proper lifetime $\tau = 2.197\,\mu\text{s}$, and arrive at the ground copiously — which, pre-relativity, is impossible. Take $\beta = 0.995$, so $\gamma = (1 - 0.995^2)^{-1/2} = 10.0$.

Without time dilation, the characteristic decay length is

$$\beta c\,\tau = 0.995 \times (2.998 \times 10^8\,\text{m/s}) \times (2.197 \times 10^{-6}\,\text{s}) = 655\,\text{m}, \tag{34}$$

and the surviving fraction after 15 km is $\exp(-15\,000/655) = \mathrm{e}^{-22.9} \approx 1.1 \times 10^{-10}$: effectively none.

With time dilation, the lab-frame lifetime is $\gamma\tau$ (Eq. (32): the muon's decay clock is a moving clock), the decay length is $\gamma\beta c\,\tau = 6.56\,\text{km}$, and the surviving fraction is $\exp(-15\,000/6560) = \mathrm{e}^{-2.29} \approx 0.10$: one in ten arrives, consistent with observation. The two exponential laws are plotted in Fig. 4; at ground level they disagree by *nine orders of magnitude.*

The same arithmetic in the muon's own frame is length contraction: the muon's clock runs normally, but the atmosphere is a moving object of proper thickness 15 km, contracted to $15\,\text{km}/\gamma = 1.5\,\text{km}$ (Eq. (33)) — about 2.3 decay lengths, and $\mathrm{e}^{-2.3} \approx 0.10$ again. One phenomenon, two accounts, one answer: the consistency is Theorem 2.4 at work.

2.7 Where this leaves the wave equation

Return to the opening problem. The wave equation (3) is not Galilean invariant — and now we know it was never supposed to be. The transformation that matters is (24), and under *it* the operator $\frac{1}{c^2}\partial_t^2 - \partial_x^2$ is form-invariant, as §3 will show in one line once the four-dimensional gradient is available. Two waves, two conclusions:

- **Light** obeys a wave equation with the invariant speed c. Lorentz invariance is exact, there is no medium, and the equation holds in the same form for every inertial observer.
- **Sound** obeys a wave equation with $c_s = \sqrt{T/\mu}$ — formally the same equation, and formally invariant under "Lorentz" transformations built with c_s in place of c. But this invariance is an accident of the long-wavelength description and is *physically broken*: the medium has a rest frame, detectable by any observer (measure the sound speed in two directions), and the microscopic chain [Osc. §8] is not invariant at all — its dispersion relation deviates from $\omega = c_s k$ at wavelengths approaching the lattice spacing.

This distinction — exact versus emergent, broken-by-the-medium symmetry — returns in §7 when the string is re-derived from a Lagrangian, and it sharpens the question Part II answers: *what does a field theory look like when Lorentz invariance is exact and fundamental?* The simplest answer is the Klein–Gordon field of §8.

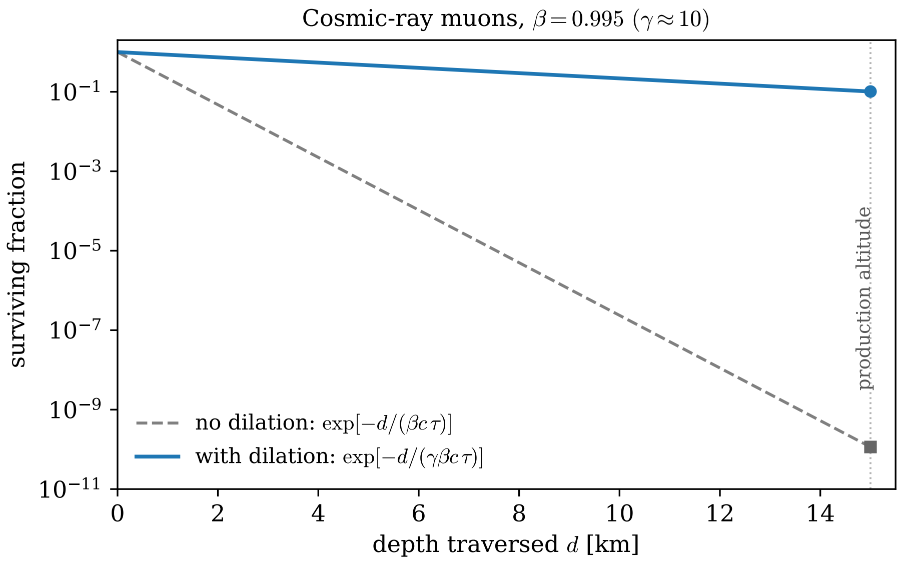

Figure 4: Surviving fraction of cosmic-ray muons ($\beta = 0.995$) versus traversed depth, from the exact exponential laws of Example 2.12. Note the logarithmic scale: at 15 km the prediction without time dilation (gray, dashed) is wrong by nine orders of magnitude.

3 Minkowski Geometry and Four-Vectors

Section 2 did everything in the (ct, x) plane with bare hands. This section builds the apparatus that makes the same physics effortless in four dimensions — the metric, index gymnastics, four-vectors, tensors — and uses it to deliver two results promised earlier: the invariance of the wave operator (§2.7), and the Doppler factor (Remark 2.11). It closes with the structure of the Lorentz group and the switch to natural units.

3.1 The metric and index gymnastics

An event is now a *four-vector* of coordinates,

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct,\ x,\ y,\ z), \tag{35}$$

the index written *upstairs*; such components are called *contravariant*. The interval of Theorem 2.4, extended to all four dimensions (the transverse directions enter with minus signs, being ordinary Euclidean lengths — Appendix A showed they are untouched by the boost), is encoded by the *Minkowski metric* of Convention 1.1:

$$\boxed{s^2 \ =\ \eta_{\mu\nu}\, x^\mu x^\nu \ =\ (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2, \qquad \eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1).} \tag{36}$$

Following Convention 1.2, both μ and ν are summed; the reader meeting the convention for the first time should expand (36) once by hand — sixteen terms, twelve of which vanish because η is diagonal — and never again.

The metric converts upstairs indices to downstairs ones and back. Define the *covariant* components

$$x_\mu = \eta_{\mu\nu}\, x^\nu = (ct,\ -x,\ -y,\ -z), \tag{37}$$

— lowering an index costs a sign on the spatial components — and let $\eta^{\mu\nu}$ denote the matrix inverse of $\eta_{\mu\nu}$,

$$\eta^{\mu\rho}\,\eta_{\rho\nu} = \delta^{\mu}{}_{\nu}, \qquad \eta^{\mu\nu} = \mathrm{diag}(+1,-1,-1,-1) \tag{38}$$

(numerically the same matrix in these coordinates — a coincidence of the diagonal form, not a general fact). *Inverse* is not a choice: demanding that raising undo lowering — $\eta^{\mu\nu}(\eta_{\nu\rho}x^{\rho}) = x^{\mu}$ for every x — is verbatim the statement $\eta^{\mu\nu}\eta_{\nu\rho} = \delta^{\mu}{}_{\rho}$, and the inverse exists, uniquely, because the metric is *nondegenerate*, $\det\eta = -1 \neq 0$ — a genuine requirement, since a degenerate metric would lower indices irreversibly. Then $\eta^{\mu\nu}$ raises indices, $x^{\mu} = \eta^{\mu\nu}x_{\nu}$, and the two operations are inverse to each other.

Remark 3.1 (What the names mean). "Contravariant" and "covariant" are not decorations; they name two opposite transformation behaviors, and the reference point of the "co-" and "contra-" is the *basis*. Expand a vector on basis vectors, $x = x^{\mu}e_{\mu}$; the vector itself is a geometric object that does not care about frames, so if the components change by $x'^{\mu} = \Lambda^{\mu}{}_{\nu}x^{\nu}$, the basis vectors must change by exactly the inverse, $e'_{\mu} = (\Lambda^{-1})^{\nu}{}_{\mu}\,e_{\nu}$, for the expansion to keep describing the same object. Quantities transforming like the basis — with Λ^{-1} — *co*-vary with it and are called covariant; quantities transforming with Λ vary *contra* to it. (§3.5 shows the gradient is the natural citizen of the covariant class, and §3.6 takes the two behaviors as the prototypes for the whole tensor hierarchy.) In special relativity the distinction is bookkeeping rather than content — the metric converts either version into the other at the cost of spatial signs, so x^{μ} and x_{μ} carry the same information — but the bookkeeping is load-bearing: Convention 1.2 sums only upper-lower pairs, and §3.6 shows that pairing is precisely the one whose transformation factors cancel.

The general *inner product* of four-vectors and its expansion in $3+1$ language:

$$\boxed{a\cdot b \;\equiv\; \eta_{\mu\nu}\,a^{\mu}b^{\nu} \;=\; a^{\mu}b_{\mu} \;=\; a_{\mu}b^{\mu} \;=\; a^{0}b^{0} - \vec{a}\cdot\vec{b}.} \tag{39}$$

Which linear maps preserve this structure? Write

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu}\,x^{\nu}. \tag{40}$$

Demanding $s'^2 = s^2$ for *every* event x forces, by the polarization identity $a\cdot b = \frac{1}{4}\big[(a+b)^2-(a-b)^2\big]$, the invariance of every inner product, and hence the condition on the matrix alone:

$$\boxed{\eta_{\mu\nu}\,\Lambda^{\mu}{}_{\rho}\,\Lambda^{\nu}{}_{\sigma} = \eta_{\rho\sigma} \qquad \text{(defining condition of a Lorentz transformation).}} \tag{41}$$

Three immediate consequences of (41) are collected here, before anything is built on top, because the entire calculus of Part I leans on them.

(i) Matrix form. With $\mathsf{\Lambda}$ the matrix $[\Lambda^{\mu}{}_{\nu}]$ (row μ, column ν) and $\boldsymbol{\eta}$ the matrix of the metric,

$$\boxed{\mathsf{\Lambda}^{\mathsf{T}}\,\boldsymbol{\eta}\,\mathsf{\Lambda} = \boldsymbol{\eta}} \tag{42}$$

— the spacetime analogue of $R^{\mathsf{T}}R = 1$ for rotations, with η replacing the identity as the preserved bilinear form. Note carefully: the *transpose*, not the inverse, sits on the left. The superficially similar $\mathsf{\Lambda}^{-1}\boldsymbol{\eta}\,\mathsf{\Lambda} = \boldsymbol{\eta}$ would assert that $\boldsymbol{\eta}$ *commutes* with every Lorentz transformation, which is already false for a single boost (check the 2×2 blocks: $\boldsymbol{\eta}\mathsf{\Lambda}$ flips the sign of $\mathsf{\Lambda}$'s spatial rows, $\mathsf{\Lambda}\boldsymbol{\eta}$ of its spatial columns, and a boost matrix is not invariant under that exchange).

Remark 3.2 (Rotations and Lorentz transformations: one template). Equation (42) is one instance of a template worth seeing whole. For any fixed invertible symmetric matrix G, the solutions of

$$\mathsf{M}^{\mathsf{T}}\,\mathsf{G}\,\mathsf{M} = \mathsf{G} \tag{43}$$

are exactly the linear maps preserving the bilinear form $\langle x, y\rangle = x^{\mathsf{T}}\mathsf{G}\,y$, and they always form a group. The choice $\mathsf{G} = \mathbb{1}$ in n dimensions gives $R^{\mathsf{T}}R = \mathbb{1}$ — the orthogonal group $O(n)$, rotations and reflections, preservers of Euclidean length. The choice $\mathsf{G} = \boldsymbol{\eta}$ gives the Lorentz group, in this classification named $O(1,3)$ — one plus sign, three minus signs — and the general signature $\mathsf{G} = \mathrm{diag}(+\cdots+, -\cdots-)$ gives the family $O(p,q)$. Two things are *shared* across the family, and everything else divides along the signs of G.

Shared: the constraint counting — the template is a symmetric matrix equation, $n(n{+}1)/2$ conditions on n^2 entries, leaving $n(n{-}1)/2$ parameters, one per coordinate plane. For $n = 4$ that is 6 for the rotation group $O(4)$ and 6 for the Lorentz group alike: *same number of planes, same dimension* (the count is signature-blind; §3.8 spends the Lorentz six as three rotations plus three boosts). Shared too: $\det \mathsf{M} = \pm 1$, by taking determinants of the template.

Divided by the sign, with each division already witnessed in this document:

- **Bounded versus unbounded.** A rotation angle lives on a circle; the rapidity lives on $\mathbb{R}$ (Remark 2.9(iii)), $\cosh\phi$ is unbounded where $\cos\theta$ is trapped in $[-1, 1]$, and correspondingly $O(4)$ is a *compact* group (closed and bounded as a set of matrices) while $O(1,3)$ is not — there is no largest boost, and the group is unbounded as a set of matrices. Most of what makes Lorentz representation theory harder than rotation representation theory descends from this single property. [stated; QFT-course territory.]

- **Eigenvalues.** Rotations have unit-modulus eigenvalues $\mathrm{e}^{\pm i\theta}$; boosts have real reciprocal ones $\mathrm{e}^{\pm\phi}$, with null eigenvectors — Remark 2.11, now recognizable as the G-sign showing up in a characteristic polynomial.

- **How many pieces.** $O(n)$ has two connected components ($\det = \pm 1$); $O(1,3)$ has *four* (§3.8), and the doubling has a clean geometric cause: the Euclidean "unit sphere" $\{x : \langle x, x\rangle = 1\}$ is connected, but its Minkowski counterpart is the two-sheeted hyperboloid $t^2 - |\vec{x}|^2 = 1$ of Fig. 1 — and since a continuous family of transformations can never carry a unit timelike vector from one sheet to the other, "which sheet" is a second discrete invariant, alongside det. That is the invariant §3.8 will meet as $\mathrm{sgn}\,\Lambda^0{}_0$.

One more choice of G completes the family: take G *antisymmetric* and the template names the symplectic group — whose preserved pairing is exactly the (q,p) structure underlying the Poisson bracket, so that the canonical transformations of [Mech. §8] are this same template's third appearance in the series. [stated; not developed further here.]

(ii) The inverse, without inverting. Apply the raising and lowering machinery to Λ itself: define $\Lambda_\mu{}^\nu \equiv \eta_{\mu\rho}\,\Lambda^\rho{}_\sigma\,\eta^{\sigma\nu}$, and contract the defining condition (41) with $\eta^{\sigma\nu}$:

$$\eta_{\alpha\beta}\,\Lambda^\alpha{}_\rho\,\Lambda^\beta{}_\sigma\,\eta^{\sigma\nu} = \eta_{\rho\sigma}\,\eta^{\sigma\nu} = \delta_\rho{}^\nu \qquad \Longrightarrow \qquad \Lambda_\alpha{}^\nu\,\Lambda^\alpha{}_\rho = \delta^\nu{}_\rho, \tag{44}$$

i.e.

$$\boxed{\left(\Lambda^{-1}\right)^\nu{}_\mu \;=\; \Lambda_\mu{}^\nu\,:} \tag{45}$$

the inverse transformation is the index-lowered transpose, and no matrix inversion is ever needed. This document adopts (45) as standing notation: from here on the inverse is always written $\Lambda_\mu{}^\nu$, the index positions doing the inverting silently. The horizontal order of Λ's slots is therefore load-bearing — $\Lambda^\mu{}_\nu$ and $\Lambda_\nu{}^\mu$ are inverse-transposes of one another, not the same array, and stacking the indices vertically would destroy the distinction.

(iii) Lowering commutes with transforming. The covariant components $x_\mu = \eta_{\mu\nu} x^\nu$ were defined by the metric; do they inherit a coherent transformation law? Transform the lowered object, and rewrite using (45):

$$a'_\mu = \eta_{\mu\nu}\, a'^\nu = \eta_{\mu\nu}\, \Lambda^\nu{}_\rho\, \eta^{\rho\sigma} a_\sigma = \Lambda_\mu{}^\sigma\, a_\sigma : \tag{46}$$

lowered components transform with the *inverse* — with the basis, in the language of Remark 3.1, whose "covariant" label is hereby proved rather than assigned. Lowering first and transforming second agrees with transforming first and lowering second, and the agreement is (41) once more: one condition, read three ways — inner products preserved, the metric unmoved, index gymnastics frame-proof.

Section 3.8 mines (42) for the structure of the group. Meanwhile, a *four-vector* is, by definition, any quadruple a^μ that transforms like the coordinates, $a'^\mu = \Lambda^\mu{}_\nu a^\nu$ (the definition is systematized, and extended to tensors of arbitrary rank, in §3.6); Eq. (41) then guarantees that $a \cdot b$ is the same number in every frame — a *Lorentz invariant* (or *scalar*). The strategy for the rest of Part I is now set: *build the objects of mechanics out of four-vectors, and equations between four-vectors will hold in every frame at once.*

3.2 Causal structure and proper time

The sign of the interval sorts pairs of events into three invariant classes — invariant because Δs^2 is, Theorem 2.4:

class	condition	invariant meaning
timelike	$\Delta s^2 > 0$	a slower-than-light traveler can attend both events
lightlike (null)	$\Delta s^2 = 0$	exactly a light signal connects them
spacelike	$\Delta s^2 < 0$	no signal connects them; causally independent

Two useful normal forms, each a one-line boost exercise: for a timelike pair there is a frame in which both events happen *at the same place*, separated by pure time $\Delta t = \sqrt{\Delta s^2}/c$; for a spacelike pair, a frame in which they are *simultaneous*, separated by pure distance $\sqrt{-\Delta s^2}$. (For the timelike case, boost with $\beta = \Delta x/(c\Delta t)$, which is < 1 precisely because the pair is timelike; the spacelike case uses $\beta = c\Delta t/\Delta x$.) In the spacelike case the *time order* of the two events is frame-dependent — there are frames in which either precedes the other — which is why any signal between spacelike-separated events would be a causality violation for some observer, and why the light cone of Fig. 3 is the absolute causal boundary: the future and past cones are frame-independent regions, and "elsewhere" is negotiable only in time-ordering, never in reachability.

Now promote the interval from pairs of events to *worldlines.* A material particle traces a curve $x^\mu(t)$ whose every segment is timelike ($|\vec{v}| < c$); the interval along an infinitesimal segment,

$$c^2\, \mathrm{d}\tau^2 \;\equiv\; \mathrm{d}s^2 = c^2 \mathrm{d}t^2 - |\mathrm{d}\vec{x}|^2 = c^2 \mathrm{d}t^2 \left(1 - \beta(t)^2\right), \tag{47}$$

defines the *proper time* $\mathrm{d}\tau$: the time elapsed on an ideal clock riding the segment, since in the clock's momentary rest frame $\mathrm{d}\vec{x} = 0$ and $\mathrm{d}s^2 = c^2 \mathrm{d}t^2_{\mathrm{rest}}$. Integrating,

$$\boxed{\tau \;=\; \int \mathrm{d}\tau \;=\; \int \frac{\mathrm{d}t}{\gamma(t)}, \qquad \gamma(t) = \frac{1}{\sqrt{1-\beta(t)^2}}.} \tag{48}$$

Proper time is a Lorentz scalar — all observers agree on what a given clock reads — and it is the natural parameter along any timelike worldline, a role it takes up in §3.3 and keeps for the rest of the document.

Remark 3.3 (The clock hypothesis). Equation (48) asserts that a clock's rate depends on its instantaneous *speed* and not separately on its acceleration. This is an additional physical assumption — special relativity as derived in §2 relates inertial frames only — and an experimentally supported one: muons circulating in storage rings at $\gamma \approx 29$, under centripetal accelerations of order $10^{18}\,g$, show lifetimes dilated by exactly γ with no acceleration correction. [experimental input.]

Remark 3.4 (The twin "paradox"). Between two fixed timelike-separated events A and B, different worldlines accumulate *different* proper times, and the straight (inertial) worldline accumulates the *most*. Proof: work in the frame where the straight worldline is at rest, so its proper time is the full coordinate time, $\tau_{\text{straight}} = \Delta t$; any other worldline between the same events has $\mathrm{d}\tau = \mathrm{d}t/\gamma \le \mathrm{d}t$ everywhere, with strict inequality wherever it moves. Hence the traveling twin — outbound, turn, return — ages less than the stay-at-home twin (Fig. 5): with $\beta = 0.6$ on both legs, $\tau_{\text{travel}} = \tau_{\text{home}}/\gamma = 0.8\,\tau_{\text{home}}$. This is the *reversed triangle inequality* of Minkowski geometry — the straight path is longest, the single sign in the metric again — and there is no paradox to resolve: the two worldlines are not symmetric, one is straight and one is kinked, and straightness (inertia) is frame-independent. The traveler's turnaround is the geometric fact every observer agrees on.

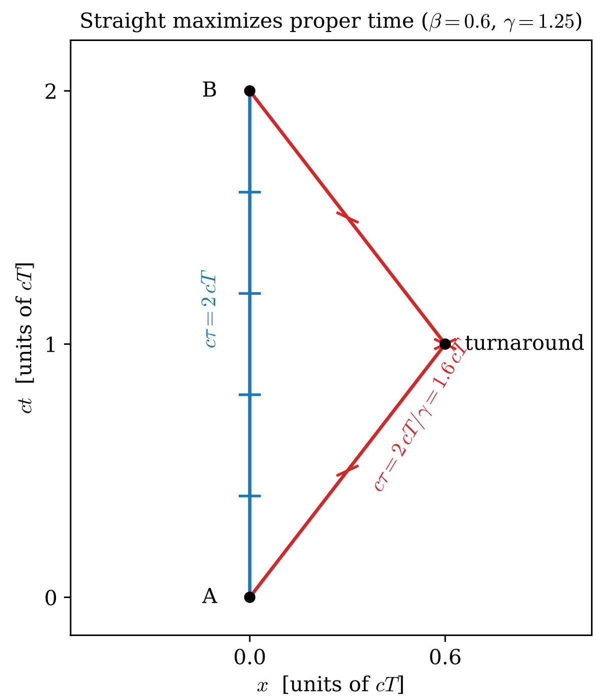

Figure 5: Two worldlines between the same events A and B. Tick marks are placed at equal intervals of *proper* time, $c\,\Delta\tau = 0.4\,cT$, computed from Eq. (48): five intervals fit on the straight worldline, four on the kinked one ($\beta = 0.6$, $\gamma = 1.25$).

Example 3.5 (GPS clocks: the worked number)**.** A GPS satellite orbits at $r = 26\,571\,\text{km}$ with speed $v = \sqrt{GM_\oplus/r} = 3.87\,\text{km/s}$, i.e. $\beta = 1.29 \times 10^{-5}$. By Eq. (48) its clock runs slow relative to ground clocks by the fraction

$$1 - \frac{1}{\gamma} \approx \frac{\beta^2}{2} = 8.3 \times 10^{-11}, \tag{49}$$

which over one day (86 400 s) accumulates to 7.2 μs — and a timing error of 7.2 μs is a ranging error of $c \times 7.2\,\mu\text{s} \approx 2\,\text{km}$. GPS would be useless within hours if this were not corrected. (Gravitational time dilation — general relativity — pushes the other way and harder, $+45.9\,\mu$s per day, for a net satellite clock *gain* of 38.7 μs per day; the system corrects for both. [stated; the gravitational part is outside this document's scope.])

3.3 Four-velocity and four-acceleration

The ordinary velocity $\mathrm{d}x^\mu/\mathrm{d}t$ is *not* a four-vector: t is the zeroth *component* of x^μ, not a scalar, so dividing by $\mathrm{d}t$ ruins the transformation law. The fix is to differentiate with respect to the scalar the worldline carries with it — proper time:

$$\boxed{u^\mu \;\equiv\; \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} = \gamma\,\frac{\mathrm{d}x^\mu}{\mathrm{d}t} = \gamma\,(c,\,\vec{v}),} \tag{50}$$

using $\mathrm{d}t = \gamma\,\mathrm{d}\tau$ from (48). Its norm is fixed once and for all:

$$u \cdot u = \gamma^2\big(c^2 - |\vec{v}|^2\big) = \gamma^2 c^2\big(1-\beta^2\big) = c^2. \tag{51}$$

Every four-velocity is a unit timelike vector (in units of c): the "speed through spacetime" is not a degree of freedom, and the constraint has a derivative worth keeping. Defining the four-acceleration $a^\mu = \mathrm{d}u^\mu/\mathrm{d}\tau$ and differentiating (51) along the worldline:

$$0 = \frac{\mathrm{d}}{\mathrm{d}\tau}(u\cdot u) = 2\,a\cdot u \qquad\Longrightarrow\qquad \boxed{a\cdot u = 0:} \tag{52}$$

four-acceleration is always Minkowski-orthogonal to four-velocity — acceleration can rotate the four-velocity on its hyperboloid $u \cdot u = c^2$ but never change its length. Both facts return in §4, where mu^μ becomes the four-momentum and (51) becomes the mass shell.

3.4 The Doppler effect from the null eigenvalue

Remark 2.11 stated without proof that the boost eigenvalue e^ϕ is the relativistic Doppler factor. The derivation is two lines precisely *because* it runs along the eigendirection.

A source at rest at the origin of S emits light pulses toward $+x$ at proper period T: emission events $(ct, x) = (cnT, 0)$, $n = 0, 1, 2, \ldots$ Each pulse travels the ray $x = x_0 + c(t - t_0)$, along which the null coordinate $u = ct - x$ is *constant*; the n-th pulse is the line $u = cnT$, so successive pulses are separated by

$$\Delta u = cT. \tag{53}$$

An observer receding at velocity β is at rest in the boosted frame S'; by Eq. (15), the pulse spacing in her null coordinate is the eigenvalue-stretched

$$\Delta u' = \mathrm{e}^\phi\,\Delta u = \mathrm{e}^\phi\,cT. \tag{54}$$

She receives the pulses at a fixed position, $\Delta x' = 0$, so the period she measures is $c\,\Delta t'_{\text{rec}} = \Delta(ct' - x') = \Delta u'$. Hence

$$\boxed{\frac{T_{\text{obs}}}{T_{\text{src}}} = \mathrm{e}^\phi = \sqrt{\frac{1+\beta}{1-\beta}}, \qquad \frac{f_{\text{obs}}}{f_{\text{src}}} = \mathrm{e}^{-\phi} = \sqrt{\frac{1-\beta}{1+\beta}}} \tag{55}$$

— redshift for recession ($\phi > 0$), blueshift for approach ($\beta \to -\beta$ flips the exponent), and successive recessions compound by *multiplying* the factors, i.e. adding rapidities: Doppler shifts of collinear relative motions compose exactly as Remark 2.11 promised eigenvalues would.

Remark 3.6 (Against the nonrelativistic Doppler formulas). The classical Doppler effect for sound has *two* formulas — moving source, $f_{\rm obs} = f_{\rm src}/(1+\beta_s)$, versus moving observer, $f_{\rm obs} = f_{\rm src}(1-\beta_o)$ — distinguishable because the medium's rest frame settles who is "really" moving. Equation (55) is a single formula depending only on the *relative* velocity, and expanding it, $\mathrm{e}^{-\phi} = 1-\beta+\frac{1}{2}\beta^2-\ldots$, it interpolates between the two classical answers, splitting the difference at second order. With no medium, no such distinction survives: this is Postulate 2.1 showing up in a lineshape. There is also a purely relativistic *transverse* Doppler shift, a factor $1/\gamma$ at closest approach with no classical analogue at all — it is time dilation (32) observed directly.

3.5 The spacetime gradient and the d'Alembertian

One more object completes the kit: differentiation. Define

$$\partial_\mu \;\equiv\; \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c}\frac{\partial}{\partial t},\; \nabla\right). \tag{56}$$

Why the *downstairs* index? Chain rule: under $x'^\mu = \Lambda^\mu{}_\nu x^\nu$,

$$\partial'_\mu = \frac{\partial}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu}\frac{\partial}{\partial x^\nu} = \Lambda_\mu{}^\nu\,\partial_\nu : \tag{57}$$

the gradient transforms with the *inverse* matrix, in the notation of (45) — precisely the behavior (46) established for lowered components, so the index placement is derived, not decorative. (Check the logic of (57): the old coordinates as functions of the new are $x^\nu = \Lambda_\mu{}^\nu\, x'^\mu$, whence the Jacobian.) Raising the index, $\partial^\mu = \eta^{\mu\nu}\partial_\nu = (\frac{1}{c}\partial_t, -\nabla)$, and contracting gradient with gradient produces a scalar operator, the *d'Alembertian*:

$$\boxed{\Box \;\equiv\; \partial_\mu\partial^\mu = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2.} \tag{58}$$

Because it is a full contraction of Lorentz-covariant objects, $\Box$ is the same operator in every inertial frame — the promise of §2.7 is now a one-line theorem. The wave equation for light, $\Box\,\varphi = 0$, is manifestly form-invariant under Lorentz transformations, in exactly the way Eq. (6) showed it fails to be under Galilean ones; and when field theory needs a relativistic equation of motion in §8, $\Box$ will be the only two-derivative scalar operator available.

3.6 Scalars, vectors, tensors, and the covariance principle

The words "scalar" and "four-vector" have so far been used case-by-case. Before dynamics begins, it is worth assembling the full hierarchy once, systematically — because the central objects ahead carry *two* indices (the field tensor $F_{\mu\nu}$ unifying $\vec{E}$ and $\vec{B}$, the energy-momentum tensor $T^{\mu\nu}$), and the reasoning that will certify them deserves to be set up properly rather than improvised on arrival.

The two prototype behaviors. Everything descends from two objects already in hand. The coordinate displacement transforms with Λ,

$$\mathrm{d}x'^\mu = \Lambda^\mu{}_\nu\,\mathrm{d}x^\nu \qquad \text{(contravariant prototype)}, \tag{59}$$

and the gradient transforms with the inverse, Eq. (57),

$$\partial'_\mu = \Lambda_\mu{}^\nu\,\partial_\nu \qquad \text{(covariant prototype)}. \tag{60}$$

Every transformation behavior in the theory is built from copies of these two — one copy per index, upstairs indices behaving like $\mathrm{d}x^\mu$, downstairs like ∂_μ.

Definition 3.7 (Lorentz scalars, vectors, tensors)**.** Under $x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$:

- a **scalar** (rank 0) is a single number assigned the same value in every frame: $s' = s$. Already met: the interval s^2, proper time τ, the norm $u \cdot u$.
- a **contravariant vector** (type $(1,0)$) is a quadruple transforming like the displacement: $a'^{\mu} = \Lambda^{\mu}{}_{\nu}\, a^{\nu}$. Already met: x^{μ}, u^{μ}.
- a **covariant vector** (type $(0,1)$) transforms like the gradient: $b'_{\mu} = \Lambda_{\mu}{}^{\nu}\, b_{\nu}$ (the inverse, per (45)). Already met: ∂_{μ}, and $x_{\mu} = \eta_{\mu\nu} x^{\nu}$.
- a **tensor of type** (p,q) carries p upstairs and q downstairs indices, each transforming with its prototype's factor:

$$T'^{\mu_1\cdots\mu_p}{}_{\nu_1\cdots\nu_q} = \Lambda^{\mu_1}{}_{\rho_1} \cdots \Lambda^{\mu_p}{}_{\rho_p}\, \Lambda_{\nu_1}{}^{\sigma_1} \cdots \Lambda_{\nu_q}{}^{\sigma_q}\, T^{\rho_1\cdots\rho_p}{}_{\sigma_1\cdots\sigma_q}. \tag{61}$$

  Scalars and vectors are the ranks $(0,0)$, $(1,0)$, $(0,1)$ of the one definition.

Why this law and no other. Equation (61) is not an arbitrary decoration; it is the *unique* linear transformation behavior under which index contraction produces frame-independent physics. The engine is one cancellation:

Proposition 3.8 (Contraction)**.** *Setting one upstairs index of a tensor equal to one downstairs index and summing yields a tensor of type* $(p-1, q-1)$*. In particular, a full contraction — no free indices — is a scalar.*

Proof. It suffices to watch one contracted pair; spectator indices carry their factors along unchanged. For a $(1,1)$ tensor, contract and transform:

$$T'^{\mu}{}_{\mu} = \Lambda^{\mu}{}_{\rho}\, \Lambda_{\mu}{}^{\sigma}\, T^{\rho}{}_{\sigma} = \delta_{\rho}{}^{\sigma}\, T^{\rho}{}_{\sigma} = T^{\rho}{}_{\rho} : \tag{62}$$

the Λ of the upstairs slot annihilates the $\Lambda_{\mu}{}^{\sigma}$ of the downstairs slot, by (45). This is why every invariant met so far — $a \cdot b = a^{\mu} b_{\mu}$, s^2, $u \cdot u$, $\Box$ acting on a scalar — was an upper-lower contraction: the pairing of the two prototype behaviors is precisely the pairing that cancels. □

Read backwards, this *derives* the definition: demand that upstairs and downstairs indices exist, that contracting them yields invariants, and that the behavior be linear — then each upstairs index must carry Λ exactly where each downstairs index carries the inverse $\Lambda_{\nu}{}^{\sigma}$, which is Eq. (61).

Where higher ranks come from. Two mechanisms manufacture tensors of rank ≥ 2, and both will be used. *Outer products*: if a^{μ} and b^{ν} are vectors, the sixteen products $a^{\mu} b^{\nu}$ transform as a $(2,0)$ tensor — immediate from the definition. *Linear physical laws*: whenever physics linearly relates one vector to another, the array of coefficients is forced to be a tensor. That is the content of:

Proposition 3.9 (Quotient rule)**.** *Let* $T_{\mu\nu}$ *be an array defined in every frame, and suppose that for* every *four-vector* a^{ν} *the contraction* $c_{\mu} \equiv T_{\mu\nu} a^{\nu}$ *transforms as a covariant vector. Then* $T_{\mu\nu}$ *is a* $(0,2)$ *tensor. (Analogous statements hold for every index structure.)*

Proof. Write what is assumed, in the primed frame and with the primed ingredients: $T'_{\mu\nu}a'^{\nu} = c'_{\mu}$. Express the right side by the known laws for c and then a:

$$T'_{\mu\nu}\,\Lambda^{\nu}{}_{\sigma}\,a^{\sigma} = \Lambda_{\mu}{}^{\rho}\,c_{\rho} = \Lambda_{\mu}{}^{\rho}\,T_{\rho\sigma}\,a^{\sigma}. \tag{63}$$

This holds for *arbitrary* a^{σ}, so the coefficients of a^{σ} agree: $T'_{\mu\nu}\Lambda^{\nu}{}_{\sigma} = \Lambda_{\mu}{}^{\rho}\,T_{\rho\sigma}$. Multiply by $\Lambda_{\kappa}{}^{\sigma}$ to free the last index (using (45) in the form $\Lambda^{\nu}{}_{\sigma}\Lambda_{\kappa}{}^{\sigma} = \delta^{\nu}{}_{\kappa}$): $T'_{\mu\kappa} = \Lambda_{\mu}{}^{\rho}\,\Lambda_{\kappa}{}^{\sigma}\,T_{\rho\sigma}$ — the $(0,2)$ transformation law. □

The quotient rule is how tensors are *detected* in practice: one never checks sixteen transformation equations directly. Part III runs it verbatim — the Lorentz force on a charge is linear in the charge's four-velocity, so the coefficient array $F_{\mu\nu}$ is a tensor by Proposition 3.9, and the transformation rules of $\vec{E}$ and $\vec{B}$ fall out of Eq. (61) for free. [Forward pointer: §11.]

Sums of same-type tensors and products with scalars stay in type; with outer products and contraction, this closes the algebra of operations, and raising or lowering an index — contraction with $\eta^{\mu\nu}$ or $\eta_{\mu\nu}$ — is a special case rather than a new rule.

The invariant tensors. Two tensors have the same components in every frame. The metric: $\eta'_{\rho\sigma} = \Lambda_{\rho}{}^{\mu}\,\Lambda_{\sigma}{}^{\nu}\,\eta_{\mu\nu} = \eta_{\rho\sigma}$, which is the defining condition (41) rearranged — *a Lorentz transformation is precisely one that leaves the components of the metric unchanged.* And the Kronecker delta $\delta^{\mu}{}_{\nu}$, whose invariance is the statement $\Lambda\Lambda^{-1} = 1$. These are the index-shuffling machines available in every frame for free; a third, the totally antisymmetric $\epsilon^{\mu\nu\rho\sigma}$ (invariant under the proper orthochronous component only), joins them in Part III when first needed. [stated; introduced properly in §11.]

Index gymnastics is frame-proof. One structural fact, already proved in pieces, deserves its general statement. Because $\eta_{\mu\nu}$ and $\eta^{\mu\nu}$ are invariant tensors, *raising or lowering any index of any tensor commutes with the Lorentz transformation law*: gymnastics before the boost and gymnastics after the boost give the same components — for vectors this was Eq. (46), and the general case is the same computation with spectator indices riding along. The index notation's entire frame-independence rests on this commutativity, and the commutativity rests on (41): one condition behind the whole calculus.

Fields: a tensor at every event. Part II needs one more notion, planted here. A *scalar field* assigns a scalar to each event: one function ϕ with

$$\phi'(x') = \phi(x) \qquad \text{whenever } x' = \Lambda x \tag{64}$$

— the moved observer, evaluating her function at her label for the *same event*, reads the same number. A vector or tensor field attaches the corresponding object to each event, components transforming by Eq. (61) on top of the relabeling (64). One consequence is used repeatedly later: if ϕ is a scalar field, then $\partial_{\mu}\phi$ is a covariant vector field and $\Box\phi$ is again a scalar field — differentiation respects the hierarchy, which is what makes it possible to write field equations covariantly at all.

Remark 3.10 (The covariance principle, and a terminological hazard). The payoff of the whole subsection is one sentence. *If an equation sets equal two tensors of the same type, and it holds in one inertial frame, it holds in every inertial frame* — for the difference of its two sides is a tensor whose components vanish in one frame, and Eq. (61), being linear, sends zero to zero. This is the operational content of Postulate 2.1 and the standard strategy of the rest of the document: laws of physics will be written as equations between same-type tensors ("in covariant form"), whereupon their frame independence is automatic rather than checked. Terminology, regrettably, overloads one word: a *covariant index* is a downstairs index, while a *covariant equation* is one written in tensor form and hence form-invariant. Both usages are standard; context separates them.

3.7 The grammar of indices

The hierarchy of Definition 3.7 is complete, and everything ahead computes inside it constantly. The computations obey a small, closed set of rules — scattered so far across conventions and propositions; collected here once, as grammar, with the standing traps named. Everything in this subsection is bookkeeping for structures already established; nothing is new physics.

Rule 1: free and dummy indices. In any single term, an index appears either *once* (a *free* index, labeling which component equation is meant) or *exactly twice, once up and once down* (a *dummy* index, summed by Convention 1.2). Grammar of equations: every term of an equation must carry the same free indices, same letters, *same heights.* Dummies are private to their term and may be renamed at will — to any letter not already in use there.

$$\begin{array}{cc} \underbrace{a^\mu = b^\mu + \Lambda^\mu{}_\nu c^\nu}_{\text{legal}} & \underbrace{a^\mu = b_\mu}_{\text{illegal: heights}} \\ \underbrace{a^\mu = b^\nu}_{\text{illegal: free mismatch}} & \underbrace{a^\mu b_\mu c_\mu}_{\text{illegal: }\mu\text{ thrice}} \end{array} \tag{65}$$

An index appearing three times, or twice at the same height, is not "summed anyway" — it is a syntax error, and in practice almost always the fossil of a forgotten rename.

Rule 2: raising and lowering. $\eta_{\mu\nu}$ lowers, $\eta^{\mu\nu}$ raises, one slot at a time, spectator indices untouched; in this signature the operation costs a sign on each spatial component and nothing on the time component. Lower-then-raise is the identity (§3.1: the inverse property is exactly this demand). Contraction with the metric *is* raising/lowering — one mechanism, not two rules. *Example:* for the four-momentum $p^\mu = (E, \vec{p})$, lowering gives $p_\mu = (E, -\vec{p})$, and the contraction is immediately useful: $p^\mu p_\mu = E^2 - |\vec{p}|^2$, the mass shell (83) in one line.

Rule 3: slot order matters. The horizontal position of a slot is part of a tensor's identity: $T^\mu{}_\nu$ means $\eta^{\mu\rho}T_{\rho\nu}$, while $T_\nu{}^\mu$ means $\eta^{\mu\rho}T_{\nu\rho}$, and the two agree only when $T_{\mu\nu}$ is symmetric. The document's standing examples sit at the two extremes:

$$\begin{aligned} \eta^\mu{}_\nu = \eta_\nu{}^\mu = \delta^\mu{}_\nu \quad &\text{(symmetric: stagger harmless)}, \\ F^\mu{}_\nu = -\,F_\nu{}^\mu \quad &\text{(antisymmetric: stagger load-bearing; } F \text{ arrives in §5.3)}, \end{aligned} \tag{66}$$

and for the non-symmetric Λ the distinction is an inverse (Eq. (45)). Stacking indices vertically is therefore banned for any tensor not known to be symmetric. *Example:* let $T_{\mu\nu}$ have a single nonzero component, $T_{01} = 1$. Then $T^0{}_1 = \eta^{00}\,T_{01} = +1$, while $T_1{}^0 = T_{1\rho}\,\eta^{\rho 0} = T_{10}\,\eta^{00} = 0$: same letters, same heights, different slots, different numbers.

Rule 4: contraction pairs one up with one down. Only upper–lower pairs may be contracted — that pairing is the one whose transformation factors cancel (Proposition 3.8 proved it; here it is grammar). The would-be "contraction" $a^\mu b^\mu$ is not a scalar — it transforms with two uncancelled Λ's — and in this signature it is not even the Euclidean dot product of anything; the invariant pairing of two upper-index vectors is $\eta_{\mu\nu}a^\mu b^\nu$, the metric supplying the missing lower slot. Corollary worth memorizing: the Kronecker delta is an *index substitution machine*, $\delta^\mu{}_\nu\, a^\nu = a^\mu$, and its self-contraction counts dimensions,

$$\delta^\mu{}_\mu = 4 \qquad \text{(not 1)}, \tag{67}$$

— a very common numerical slip in the calculus. *Example:* take the event $(t, x) = (1, 0)$ and boost with $\beta = 0.6$ ($\gamma = 1.25$): the new coordinates are $(t', x') = (1.25, -0.75)$. The illegal same-height sum changes, $t'^2 + x'^2 = 2.1875 \neq 1 = t^2 + x^2$, while the lawful contraction holds, $t'^2 - x'^2 = 1.5625 - 0.5625 = 1 = t^2 - x^2$ — the grammar rule and Theorem 2.4 are the same fact.

Rule 5: symmetric contracted with antisymmetric vanishes.

Proposition 3.11. *If $S^{\mu\nu} = S^{\nu\mu}$ and $A_{\mu\nu} = -A_{\nu\mu}$, then $S^{\mu\nu}A_{\mu\nu} = 0$.*

Proof. Rename the dummies, then use both symmetries: $S^{\mu\nu}A_{\mu\nu} = S^{\nu\mu}A_{\nu\mu} = (+S^{\mu\nu})(-A_{\mu\nu}) = -S^{\mu\nu}A_{\mu\nu}$; a number equal to its own negative vanishes. □

Example: for any vector v^μ, the outer product $v^\mu v^\nu$ is symmetric, so $A_{\mu\nu}\, v^\mu v^\nu = 0$ for every antisymmetric A — in particular $F_{\mu\nu}\, u^\mu u^\nu = 0$, which is why the Lorentz force (100) automatically preserves the normalization of the four-velocity: $u_\mu\, \mathrm{d}u^\mu/\mathrm{d}\tau = (q/m)\, F_{\mu\nu}u^\mu u^\nu = 0$, so $u \cdot u = 1$ is maintained along the motion. This three-line fact recurs at essential points later: second partials $\partial_\mu\partial_\nu$ are symmetric, so their contraction with anything antisymmetric vanishes identically — the mechanism behind gauge invariance of $F_{\mu\nu}$ (§5.3) and behind the automatic charge conservation of §12, both of which are this proposition wearing field-theory clothes.

Rule 6: differentiation. The coordinate derivative carries a genuine lower index, ∂_μ (derived in §3.5), and the basic derivative is

$$\partial_\mu x^\nu = \delta_\mu{}^\nu. \tag{68}$$

The one trap: *before differentiating with respect to an indexed variable, rename every dummy that collides with the free index of the derivative.* Canonical example — differentiate a quadratic form:

$$\frac{\partial}{\partial a_\nu}\big(a_\mu a^\mu\big) = \frac{\partial}{\partial a_\nu}\big(\eta^{\rho\sigma}a_\rho a_\sigma\big) = \eta^{\rho\sigma}\big(\delta_\rho{}^\nu\, a_\sigma + a_\rho\, \delta_\sigma{}^\nu\big) = 2\, a^\nu, \tag{69}$$

both slots contributing — the computation quoted at Eq. (138) for $\partial\mathcal{L}/\partial(\partial_\mu\phi)$ is exactly this, with $a \to \partial\phi$. Attempting it without the rename ($\partial(a_\mu a^\mu)/\partial a_\mu$ — three μ's) is Rule 1's syntax error in its most tempting form.

Fluency test, recommended once with pen and paper: expand $T^\mu{}_\nu\, a^\nu b_\mu$ fully for diagonal η (sixteen terms, four survivors), then verify each rule above was used and no step needed anything beyond them. The grammar is small; the entire covariant calculus of this document is sentences in it.

3.8 The Lorentz and Poincaré groups

Return to the matrix form (42) of the defining condition, $\Lambda^{\mathsf{T}}\boldsymbol{\eta}\Lambda = \boldsymbol{\eta}$. Its solutions form a group — the *Lorentz group*, $O(1,3)$: closure is one line (transpose a product), and the inverse is not only guaranteed but already computed, Eq. (45) — and it satisfies the defining condition itself (sandwich (42) between $(\Lambda^{\mathsf{T}})^{-1}$ and Λ^{-1}). Two discrete invariants follow immediately:

- Taking det of (42): $(\det\Lambda)^2 = 1$, so $\det\Lambda = \pm 1$.
- Taking the $\rho = \sigma = 0$ component of (41): $(\Lambda^0{}_0)^2 = 1 + \sum_i (\Lambda^i{}_0)^2 \geq 1$, so $\Lambda^0{}_0 \geq 1$ or $\Lambda^0{}_0 \leq -1$ — a transformation either preserves the direction of time (*orthochronous*) or reverses it, with no middle ground.

Neither sign can change along a continuous path, so the group falls into **four connected components**, labeled by the pair $(\det\Lambda,\ \operatorname{sgn}\Lambda^0{}_0)$ and represented by four familiar transformations:

component	$\det\Lambda$	$\operatorname{sgn}\Lambda^0{}_0$	representative
proper orthochronous	$+1$	$+$	$\mathbb{1}$ (boosts, rotations)
improper orthochronous	-1	$+$	P: $\vec{x} \to -\vec{x}$
improper non-orthochronous	-1	$-$	T: $t \to -t$
proper non-orthochronous	$+1$	$-$	PT: $x^\mu \to -x^\mu$

Only the first component, $SO^+(1,3)$, contains the identity, and everything derived by continuity arguments in §2 lives there; the other three are the first component times P, T, or PT. Whether P and T are symmetries of nature is a question about *dynamics*, not geometry — the weak interaction famously violates both — and the geometric statement here is only that they are disconnected from the identity: no accumulation of small boosts and rotations ever mirrors space or reverses time.

Dimension: the six planes of spacetime. How many continuous parameters does $SO^+(1,3)$ have? Count constraints: Eq. (42) is a symmetric 4×4 matrix equation — 10 independent conditions on the 16 entries of Λ — leaving $\boxed{6}$ parameters. The six have names: a transformation preserving the metric can rotate any of the six coordinate *planes* of spacetime,

$$\underbrace{(xy),\ (yz),\ (zx)}_{\text{3 rotations, angles } \theta} \qquad \underbrace{(tx),\ (ty),\ (tz)}_{\text{3 boosts, rapidities } \phi} \tag{70}$$

— a boost is a rotation that mixes time with one spatial axis, hyperbolically because of the metric's sign there, which is the whole content of Remark 2.6 said group-theoretically. Each plane generates a one-parameter subgroup with additive parameter (angles add; rapidities add, Eq. (26)).

Remark 3.12 (Boosts alone do not close). The three rotations form a subgroup, $SO(3)$. The three boosts do *not*: composing boosts along different axes yields a boost *times a rotation* (the Wigner rotation, physically observable as Thomas precession of accelerated spins). The clean algebraic statement — rotations and boosts as generators, their commutation relations, and why [boost, boost] = rotation — is Lie algebra material and is deferred. [stated; the Lorentz algebra is standard QFT-course territory, cf. Schwartz Ch. 10.]

The Poincaré group. The full symmetry of Minkowski spacetime adds what homogeneity always permitted: translations,

$$x'^\mu = \Lambda^\mu{}_\nu\, x^\nu + a^\mu, \qquad \boxed{6 + 4 = 10 \text{ parameters.}} \tag{71}$$

The number ten is worth remembering now and using later: by Noether's theorem [Mech. §3.3], every continuous symmetry parameter yields a conservation law, and the ten parameters of the Poincaré group will produce exactly the ten conserved quantities of relativistic physics — energy (1), momentum (3), angular momentum (3), and the three boost charges — for the particle in §4 and again, localized, for fields in §9.

3.9 Natural units from here on

The apparatus is assembled, every factor of c has served its purpose as a dimension check, and Convention 1.3 now takes effect.

Convention 3.13 (Natural units)**.** From this point on, $c = 1$: time and length are measured in the same unit, velocities are dimensionless (β and v are now the same symbol), and mass, energy, and momentum share a dimension. Any equation can be restored to SI form uniquely: insert the power of c that makes the dimensions balance. Example: $E = m$ restores to $E = mc^2$ — energy over mass has dimensions of speed squared, and c is the only speed in the formalism. The four-velocity becomes $u^\mu = \gamma(1, \vec{v})$ with $u \cdot u = 1$; the interval, $s^2 = t^2 - |\vec{x}|^2$; the d'Alembertian, $\Box = \partial_t^2 - \nabla^2$.

With geometry and units in hand, dynamics can start: what does a free particle *do* in Minkowski spacetime, and what action principle says so?

4 The Relativistic Particle

This section is where the two companion threads meet: the variational machinery of the mechanics notes runs, unmodified, on the geometry of §§2–3. The plan: find the action a free particle is forced to have, extract its equations of motion (twice — once in lab coordinates for contact with [Mech. §3], once covariantly), and then let Noether's theorem *derive* energy and momentum, which combine into a four-vector. Natural units throughout (Convention 3.13).

4.1 The action, and why it is forced

What action can a free particle in Minkowski spacetime possibly have? Constraints, in decreasing order of negotiability:

(i) S must be a **Lorentz scalar**. By the covariance principle (Remark 3.10), a scalar action yields equations of motion valid in every frame at once — Postulate 2.1 implemented rather than checked.

(ii) S must be built from the **worldline alone**. The particle is free: no fields, no external structure, no preferred events are available as ingredients.

(iii) S must reproduce **Newtonian mechanics** at $v \ll 1$ — fixing normalization, not form.

Conditions (i) and (ii) are extremely restrictive: from the worldline and its *first* derivative, the only reparametrization-invariant scalar functional is the Minkowski arc length — the proper time of Eq. (48). (Scalars built from higher derivatives do exist — the proper acceleration $\sqrt{-a \cdot a}$ is one — but actions containing them yield equations of motion of higher than second order, outside the mechanical framework of [Mech. §2]; "first derivatives only" is the standard assumption of the whole series, made explicit here.) So, up to the constant that (iii) will fix,

$$\boxed{S[x] \;=\; -m\int \mathrm{d}\tau \;=\; -m\int \sqrt{1-v^2}\,\mathrm{d}t, \qquad L(\vec{v}) = -m\sqrt{1-v^2}.} \tag{72}$$

The second form parametrizes the worldline by lab time and reads off the Lagrangian; it is the form on which the machinery of [Mech. §2–3] operates directly.

Fixing the constant. Expand for small speeds:

$$L = -m\sqrt{1-v^2} = -\,m \;+\; \tfrac{1}{2}mv^2 \;+\; \tfrac{1}{8}mv^4 + \dots \tag{73}$$

The additive constant $-m$ never enters the Euler–Lagrange equations; the next term is the Newtonian kinetic energy, provided the prefactor of $\int \mathrm{d}\tau$ is $-m$ with m the particle's mass — minus sign included, since $\sqrt{1-v^2}$ *decreases* with speed. (Restoring units in (73): $-mc^2 + \frac{1}{2}mv^2 + \dots$ — the rest energy appears as an additive constant before dynamics has even started.)

Remark 4.1 (Stationary action is stationary proper time). Since $S = -m\tau$, extremizing the action extremizes proper time, and by Remark 3.4 the straight worldline is the proper-time *maximum* among all worldlines between fixed events. So the variational principle (72) says: *a free particle moves so that its own clock runs the longest* — inertial motion is the stay-at-home twin's path. The equations of motion below merely spell out what Fig. 5 already shows.

4.2 Two routes to the equations of motion

Route 1: lab coordinates. Apply the Euler–Lagrange equations [Mech. §3.1] to $L = -m\sqrt{1-v^2}$, a function of $\vec{v}$ only. The canonical momentum:

$$\vec{p} \;=\; \frac{\partial L}{\partial \vec{v}} = \frac{m\,\vec{v}}{\sqrt{1-v^2}} = \gamma m\,\vec{v}, \tag{74}$$

— the Newtonian $m\vec{v}$ dressed by γ — and since $\partial L/\partial \vec{x} = 0$,

$$\frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = 0: \tag{75}$$

$\gamma m\vec{v}$ is constant, hence v and $\vec{v}$ are, and the worldline is straight. Free particles move uniformly — reassuring, and required: Lemma 2.3 leaned on it.

Route 2: covariant variation. The same, without ever choosing a frame. Parametrize the worldline arbitrarily, $x^\mu(\lambda)$, and write the action as the arc length it is:

$$S = -m\int \sqrt{\eta_{\mu\nu}\,\dot{x}^\mu\dot{x}^\nu}\;\mathrm{d}\lambda, \qquad \dot{x}^\mu \equiv \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}. \tag{76}$$

Vary the worldline, $x^\mu \to x^\mu + \delta x^\mu$ with endpoints pinned, exactly as in the Gateaux framework of [Mech. §2.2]:

$$\delta S = -m\int \frac{\eta_{\mu\nu}\,\dot{x}^\mu\,\delta\dot{x}^\nu}{\sqrt{\dot{x}\cdot\dot{x}}}\;\mathrm{d}\lambda = +\,m\int \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\dot{x}_\nu}{\sqrt{\dot{x}\cdot\dot{x}}}\right)\delta x^\nu\;\mathrm{d}\lambda, \tag{77}$$

integrating by parts and dropping the vanishing boundary term. The variation δx^ν is arbitrary, so by the fundamental lemma [Mech. App. B] the bracket has zero derivative. But $\sqrt{\dot{x}\cdot\dot{x}}\,\mathrm{d}\lambda = \mathrm{d}\tau$, so $\dot{x}^\mu/\sqrt{\dot{x}\cdot\dot{x}} = \mathrm{d}x^\mu/\mathrm{d}\tau = u^\mu$, and the equation of motion is

$$\boxed{\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = 0 \qquad\Longleftrightarrow\qquad x^\mu(\tau) = x^\mu(0) + u^\mu\,\tau:} \tag{78}$$

straight worldlines, now as a manifestly covariant statement — true in all frames because written in four-vectors, per Remark 3.10.

Remark 4.2 (An equation with a built-in redundancy). Equation (78) looks like four equations but carries only three: contracting with u_μ gives $u \cdot \mathrm{d}u/\mathrm{d}\tau = \frac{1}{2}\,\mathrm{d}(u \cdot u)/\mathrm{d}\tau = 0$ identically, by the normalization (51) — one component is an empty identity, not a law. The missing content is exactly the freedom the description carried from the start: the parameter λ was arbitrary (the action, being arc length, is *reparametrization invariant*), and a redundant coordinate in the description yields an empty equation of motion. File the pattern away: in Part III the potential A^μ carries a redundancy of precisely this flavor — gauge freedom — with precisely this consequence (§13).

4.3 Energy and momentum from Noether's theorem

Here is the section's centerpiece, and the reason this document derives four-momentum rather than defining it. The Lagrangian (72) depends on neither t nor $\vec{x}$: the action is invariant under the four spacetime translations, and Noether's theorem [Mech. §3.3] converts each invariance into a conserved quantity.

Time translation → energy. Invariance under $t \to t+\epsilon$ conserves the Jacobi function $h = \vec{p}\cdot\vec{v} - L$ [Mech. §3.3, §6.3]:

$$E \;\equiv\; \vec{p}\cdot\vec{v} - L = \frac{mv^2}{\sqrt{1-v^2}} + m\sqrt{1-v^2} = m\,\frac{v^2 + (1-v^2)}{\sqrt{1-v^2}} = \frac{m}{\sqrt{1-v^2}} = \gamma m. \tag{79}$$

Space translations → momentum. Invariance under $\vec{x} \to \vec{x} + \vec{\epsilon}$ conserves the canonical momentum (74): $\vec{p} = \gamma m \vec{v}$.

Assembly. Stack the four conserved numbers:

$$\big(E,\ \vec{p}\big) = \big(\gamma m,\ \gamma m \vec{v}\big) = m\,\gamma\,(1, \vec{v}) = m\,u^\mu. \tag{80}$$

The right-hand side is a scalar times a four-vector (Eq. (50)) — so the four Noether charges of the four translations are not merely four conserved numbers; they are the components of a conserved *four-vector*,

$$\boxed{p^\mu \;=\; m\,u^\mu \;=\; \big(E,\ \vec{p}\big) = \big(\gamma m,\ \gamma m\vec{v}\big), \qquad \frac{\mathrm{d}p^\mu}{\mathrm{d}\tau} = 0.} \tag{81}$$

Energy is the *time component* of a four-vector: under a boost, E and $\vec{p}$ mix by Eq. (24) exactly as t and $\vec{x}$ do. That one sentence encodes much of relativistic kinematics — and it was derived, not postulated: it follows from translation symmetry through Noether's theorem, the same derivation as in [Mech. §3.3], run on a new Lagrangian.

Expanding the energy for small speeds,

$$E = \gamma m = m + \tfrac{1}{2}mv^2 + \tfrac{3}{8}mv^4 + \dots \tag{82}$$

— Newtonian kinetic energy riding on a constant that Newtonian physics could never see, because Newtonian physics only ever measured energy *differences*. Restoring units, the constant is $E_{\text{rest}} = mc^2$: mass is the energy a particle cannot shed by slowing down.

The mass shell. The norm of (81) is fixed by (51): $p \cdot p = m^2\, u \cdot u = m^2$, i.e.

$$\boxed{E^2 - |\vec{p}|^2 = m^2 \qquad \Longleftrightarrow \qquad E = \sqrt{|\vec{p}|^2 + m^2}.} \tag{83}$$

In momentum space $(E, \vec{p})$ this is a hyperboloid — the *mass shell* (Fig. 6) — the same invariant-hyperbola geometry as Fig. 1, now with the particle's identity m as the invariant. Useful limits: at rest, $E = m$; and as $m \to 0$ the shell collapses onto the light cone of momentum space, $E = |\vec{p}|$, with $v = \partial E/\partial|\vec{p}| \to 1$ — massless particles move at c and have momentum with no mass to show for it. (The action (72) itself degenerates at $m = 0$, where $\mathrm{d}\tau = 0$ along the worldline; the mass-shell relation survives the limit and is taken as the massless particle's definition. [stated; the proper massless action is a QFT-course topic.])

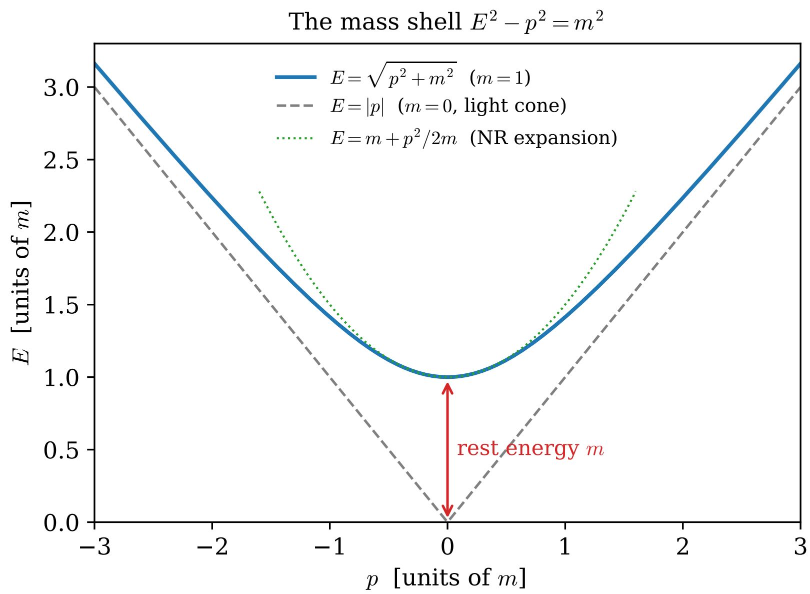

Figure 6: The mass shell (83) in 1+1 momentum space: the exact relativistic dispersion (blue), its massless light-cone asymptote (gray, dashed), and the nonrelativistic parabola $E = m + p^2/2m$ of Eq. (82) (green, dotted), which hugs the shell near $p = 0$ and departs from it at large p. This figure returns, with the axes relabeled (ω, k), as the Klein–Gordon dispersion relation in §8.

Example 4.3 (LHC protons: the worked number)**.** An LHC beam proton has $E = 6.8\,\mathrm{TeV}$ and $m = 0.938\,\mathrm{GeV}$, so

$$\gamma = \frac{E}{m} = 7.25 \times 10^3, \qquad 1 - \beta \approx \frac{1}{2\gamma^2} = 9.5 \times 10^{-9}: \tag{84}$$

the proton trails a light pulse by about $c\,(1 - \beta) \approx 2.9\,\mathrm{m/s}$ — walking pace — while carrying 7250 times its rest energy. On Fig. 6 it sits far up the asymptote, where the shell and the light cone are indistinguishable to one part in 10^8.

Remark 4.4 (The other six charges). Section 3.8 promised ten conservation laws for the ten Poincaré parameters; translations have now supplied four. The three rotations conserve angular momentum $\vec{L} = \vec{x} \times \vec{p}$, exactly as in [Mech. §3.3] — nothing relativistic changes in that derivation. The three boosts conserve the less famous $\vec{K} = E\,\vec{x} - t\,\vec{p}$: indeed $\mathrm{d}\vec{K}/\mathrm{d}t = E\vec{v} - \vec{p} - t\,\dot{\vec{p}} = \gamma m\vec{v} - \gamma m\vec{v} - 0 = 0$. Solving $\vec{K} = \text{const}$ for $\vec{x}$ shows its meaning: the point $\vec{x} = (\vec{K} + t\vec{p})/E$ — the center of energy — moves uniformly. Boost symmetry conserves the *uniformity of the center-of-energy motion*, the relativistic upgrade of the center-of-mass theorem. All six reappear, localized into densities, in §9.

4.4 The relativistic Hamiltonian

The Legendre transform [Mech. §6] completes the particle's description. Invert (74) for the velocity: squaring, $|\vec{p}|^2(1 - v^2) = m^2 v^2$ gives

$$\vec{v} = \frac{\vec{p}}{\sqrt{|\vec{p}|^2 + m^2}}, \tag{85}$$

and the Hamiltonian is

$$H(\vec{p}) = \vec{p}\cdot\vec{v} - L = \frac{|\vec{p}|^2}{\sqrt{|\vec{p}|^2 + m^2}} + \frac{m^2}{\sqrt{|\vec{p}|^2 + m^2}} = \boxed{\sqrt{|\vec{p}|^2 + m^2}} \tag{86}$$

— the mass-shell function itself, as it must be: L has no explicit time dependence, so H equals the conserved energy [Mech. §6.3], and the conserved energy (79) lives on the shell (83). Hamilton's equations close the loop:

$$\dot{\vec{x}} = \frac{\partial H}{\partial \vec{p}} = \frac{\vec{p}}{\sqrt{|\vec{p}|^2 + m^2}} = \vec{v}, \qquad \dot{\vec{p}} = -\frac{\partial H}{\partial \vec{x}} = 0. \tag{87}$$

Remark 4.5 (Is H covariant? No — and instructively so). Route 2 delivered the manifestly covariant Eq. (78); the Hamiltonian route cannot deliver its like, and the loss happens at the first step: the Legendre transform differentiates with respect to *somebody's* time, so writing a Hamiltonian means choosing a slicing of spacetime into moments. Nor is H itself a scalar — it is the *time component* of p^μ, so different inertial observers use different Hamiltonians, related by boosts. Covariance is not broken, but it is demoted from a property visible by inspection to a theorem requiring proof: the ten charges of Remark 4.4 must close, under Poisson brackets, into the algebra of the Poincaré group. In that algebra H and $\vec{K}$ are the generators that *move the time slice*, and the fact that they do not commute is already on display: $\vec{K} = E\vec{x} - t\vec{p}$ carries explicit time dependence, and its conservation is a cancellation,

$$\frac{\mathrm{d}K_i}{\mathrm{d}t} = \frac{\partial K_i}{\partial t} + \{K_i, H\} = -\,p_i + E\,\frac{\partial H}{\partial p_i} = -\,p_i + p_i = 0, \tag{88}$$

with $\{K_i, H\} = p_i \neq 0$ — the classical fingerprint of the commutator $[\hat{K}_i, \hat{H}] \neq 0$ that canonical quantum field theory must live with. The same trade returns at field level in §10: there $H = \int T^{00}\,\mathrm{d}^3x$ is again a 00-component integrated over a chosen slice, canonical quantization inherits the non-manifest covariance, and "the quantum theory is Lorentz invariant" becomes a statement about the algebra of operator charges, verified rather than seen. (A manifestly covariant Hamiltonian formulation of the particle does exist: take the worldline parameter itself as time, whereupon H vanishes identically and the dynamics hides inside the constraint $p\cdot p = m^2$ — the Hamiltonian face of Remark 4.2's reparametrization redundancy. Constrained-dynamics territory; deferred. [stated.])

Remark 4.6 (The square-root problem). $H = \sqrt{|\vec{p}|^2 + m^2}$ is an unpleasant object to promote to a quantum operator: the square root of a differential operator is nonlocal, and awkward to work with. The historical escape was to *square* the relation — demand $E^2 = |\vec{p}|^2 + m^2$ of a wave $\mathrm{e}^{-\mathrm{i}(Et - \vec{p}\cdot\vec{x})}$ — which yields not a Schrödinger-type equation but the *Klein–Gordon equation*, a field equation. That road, from the awkward square root to relativistic fields, is exactly the road this document travels: §8 arrives at Klein–Gordon from the Lagrangian side, and Fig. 6 will be waiting there with its axes relabeled.

5 The Charged Particle

The free particle exhausted what a bare worldline can do. To make anything else happen, the particle needs an environment — and the simplest relativistic environment is a four-vector field $A_\mu(x)$ defined over spacetime, coupled in the simplest way the covariance principle permits. This short section builds that coupling, finds that its redundancy and its equation of motion are both familiar — gauge freedom and the Lorentz force — and establishes two structures that Part III will need: the antisymmetric derivative combination $F_{\mu\nu}$, and the split between canonical and mechanical momentum.

5.1 The coupling

What scalar can be built that is *linear* in a vector field A_μ and local to the worldline? The field must be evaluated on the worldline and contracted with something the worldline provides; the worldline provides $\mathrm{d}x^\mu$. Hence, appended to the free action (72):

$$\boxed{S \;=\; -\,m\int \mathrm{d}\tau \;-\; q\int A_\mu(x)\,\mathrm{d}x^\mu,} \tag{89}$$

with a coupling constant q — the *charge* — measuring how strongly this particular particle couples to this particular field. The interaction term is a Lorentz scalar by construction (one upper index contracted against one lower), and it is reparametrization invariant without help: $\mathrm{d}x^\mu$ needs no $\mathrm{d}\tau$, no metric, no clock. To recognize it, parametrize by lab time. With the traditional names for the components, $A^\mu = (\varphi, \vec{A})$ — so $A_\mu = (\varphi, -\vec{A})$ — the integrand is $A_\mu \mathrm{d}x^\mu = \varphi\,\mathrm{d}t - \vec{A}\cdot\mathrm{d}\vec{x}$, and

$$L = -\,m\sqrt{1-v^2} \;-\; q\,\varphi(t,\vec{x}) \;+\; q\,\vec{A}(t,\vec{x})\cdot\vec{v}. \tag{90}$$

The scalar potential enters like a potential energy; the vector potential enters through a velocity-dependent term — the first Lagrangian of the series in which force will depend on velocity.

5.2 Gauge invariance of the coupling

The field A_μ enters (89) with a redundancy. Shift it by the four-gradient of any scalar function:

$$\boxed{A_\mu \;\longrightarrow\; A_\mu + \partial_\mu\chi.} \tag{91}$$

The interaction changes by

$$-\,q\int \partial_\mu\chi\,\mathrm{d}x^\mu = -\,q\int \mathrm{d}\chi = -\,q\left[\chi(x_B) - \chi(x_A)\right] \tag{92}$$

— a pure endpoint term, which variations with pinned endpoints never see [Mech. §2.2]. Every choice of χ yields the *same* equations of motion: the four functions A_μ overdescribe the physics, and the overdescription is exactly one scalar function's worth. This is the pattern of Remark 4.2 in mirror image — there, redundancy in the *worldline's* description produced an empty equation; here, the redundancy sits in the *background's* description. What the equations of motion can depend on is only the gauge-invariant content of A_μ, and the next subsection watches the variational principle extract precisely that content on its own.

5.3 Equations of motion: the Lorentz force

Route 1: lab coordinates. Run Euler–Lagrange on (90). The canonical momentum picks up a field contribution:

$$\boxed{\vec{P} \;\equiv\; \frac{\partial L}{\partial \vec{v}} \;=\; \gamma m\vec{v} + q\vec{A} \;=\; \vec{p} + q\vec{A},} \tag{93}$$

where $\vec{p} = \gamma m\vec{v}$ is the *mechanical* momentum of (74) — the distinction matters and returns in §5.4. The coordinate derivative of L:

$$\frac{\partial L}{\partial \vec{x}} = -\,q\nabla\varphi + q\,\nabla(\vec{A}\cdot\vec{v}), \tag{94}$$

the gradient acting on the field arguments only ($\vec{v}$ is an independent slot of the Lagrangian, not a field on space). The Euler–Lagrange equation $\mathrm{d}\vec{P}/\mathrm{d}t = \partial L/\partial\vec{x}$ needs the total time derivative of $\vec{A}$ *along the moving particle*, which is the convective derivative:

$$\frac{\mathrm{d}\vec{A}}{\mathrm{d}t} = \frac{\partial\vec{A}}{\partial t} + (\vec{v}\cdot\nabla)\vec{A} \tag{95}$$

— the particle feels the field change both because the field changes and because the particle moves through it. Assembling:

$$\frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = -\,q\left(\nabla\varphi + \frac{\partial\vec{A}}{\partial t}\right) + q\left[\nabla(\vec{A}\cdot\vec{v}) - (\vec{v}\cdot\nabla)\vec{A}\right]. \tag{96}$$

The last bracket is the curl identity $\vec{v}\times(\nabla\times\vec{A}) = \nabla(\vec{A}\cdot\vec{v}) - (\vec{v}\cdot\nabla)\vec{A}$ (valid because $\vec{v}$ carries no position dependence). The equation of motion has organized its right side into two derivative combinations of the potentials, and we *name* them:

$$\vec{E} \;\equiv\; -\nabla\varphi - \frac{\partial\vec{A}}{\partial t}, \qquad \vec{B} \;\equiv\; \nabla\times\vec{A}, \tag{97}$$

whereupon

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}(\gamma m\vec{v}) \;=\; q\,(\vec{E} + \vec{v}\times\vec{B}) :} \tag{98}$$

the Lorentz force law, derived rather than postulated. Two things deserve to be said about the derivation rather than the result. First, $\vec{E}$ and $\vec{B}$ were not inputs: they are the combinations of potential derivatives that the variational principle assembled on its own, and both are inert under the gauge shift (91) — $\nabla\times\nabla\chi = 0$ kills it in $\vec{B}$, and $-\nabla\dot{\chi} - \partial_t\nabla\chi = 0$ in $\vec{E}$ — exactly as §5.2 predicted the equations of motion must arrange. Second, the left side is the *mechanical* momentum: what the force pushes is $\gamma m\vec{v}$, not the canonical $\vec{P}$.

Route 2: covariant form. The same variation without choosing a frame produces the central object of Part III. Vary the interaction in worldline form, $S_{\text{int}} = -q\int A_\mu(x)\,\dot{x}^\mu\,\mathrm{d}\lambda$:

$$\begin{aligned}\delta S_{\text{int}} &= -\,q\int\left[\big(\partial_\nu A_\mu\big)\,\delta x^\nu\,\dot{x}^\mu + A_\nu\,\delta\dot{x}^\nu\right]\mathrm{d}\lambda\\ &= -\,q\int\left[\partial_\nu A_\mu\,\dot{x}^\mu - \frac{\mathrm{d}}{\mathrm{d}\lambda}A_\nu\right]\delta x^\nu\,\mathrm{d}\lambda = -\,q\int\left[\partial_\nu A_\mu - \partial_\mu A_\nu\right]\dot{x}^\mu\,\delta x^\nu\,\mathrm{d}\lambda,\end{aligned} \tag{99}$$

integrating by parts in the second step and using $\mathrm{d}A_\nu/\mathrm{d}\lambda = \dot{x}^\mu \partial_\mu A_\nu$ (the covariant convective derivative) in the third. Adding the free variation from §4.2 and demanding stationarity:

$$\boxed{m\,\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} \;=\; q\,F^\mu{}_\nu\,u^\nu, \qquad F_{\mu\nu} \;\equiv\; \partial_\mu A_\nu - \partial_\nu A_\mu.} \tag{100}$$

The variation *manufactured* the antisymmetric combination $F_{\mu\nu}$ — the only part of $\partial_\mu A_\nu$ that survives, and manifestly gauge invariant, since $\partial_\mu\partial_\nu\chi - \partial_\nu\partial_\mu\chi = 0$. And its tensor character comes with no extra work: the force (100) is linear in the four-vector u^ν with a four-vector result, so by the quotient rule (Proposition 3.9) $F^\mu{}_\nu$ *is* a $(1,1)$ tensor — as promised in §3.6. Its six independent components (antisymmetry kills the diagonal and pairs the rest) are exactly the six numbers (97): $\vec{E}$ and $\vec{B}$ are one tensor seen in $3+1$ pieces, which Part III displays in full (§11).

Example 5.1 (The LHC proton, bent: the worked number)**.** For circular motion at constant speed, γ is constant, and (98) in a pure magnetic field reads $\gamma m\,v^2/r = qvB$, so

$$r = \frac{\gamma m v}{qB} = \frac{p}{qB}\;: \tag{101}$$

the bending radius is set by the *mechanical momentum*, γ included. The proton of Example 4.3 ($p \approx E = 6.8\,\mathrm{TeV}$) rides a dipole field $B = 8.1\,\mathrm{T}$ (the magnets' $8.33\,\mathrm{T}$ design value scaled to the running energy), so

$$r = \frac{6800\,\mathrm{GeV}}{0.300 \times 8.1\,\mathrm{T}}\,\frac{\mathrm{m}}{\mathrm{GeV/T}} \approx 2.80\,\mathrm{km}, \tag{102}$$

the machine's actual bending radius. The cyclotron frequency $\omega = qB/(\gamma m)$ falls with energy — which is why a fixed-frequency cyclotron fails relativistically and the LHC is a *synchro*tron, its fields ramped in step with γ.

Example 5.2 (Constant field: hyperbolic motion)**.** The simplest exact solution of (100): a uniform, static $\vec{E} = E\,\hat{x}$, no $\vec{B}$, charge starting from rest. The potentials $\varphi = -Ex$, $\vec{A} = 0$ give (check against (97): $-\nabla\varphi = E\hat{x}$) the single nonzero pair $F_{01} = \partial_0 A_1 - \partial_1 A_0 = E$, whence $F^0{}_1 = \eta^{00}F_{01} = E$ and $F^1{}_0 = \eta^{11}F_{10} = E$. The covariant force law reduces to two coupled components, with $\alpha \equiv qE/m$:

$$\frac{\mathrm{d}u^0}{\mathrm{d}\tau} = \alpha\,u^1, \qquad \frac{\mathrm{d}u^1}{\mathrm{d}\tau} = \alpha\,u^0 \qquad \Longrightarrow \qquad \frac{\mathrm{d}}{\mathrm{d}\tau}\big(u^0 \pm u^1\big) = \pm\,\alpha\,\big(u^0 \pm u^1\big)\;: \tag{103}$$

the *null combinations* of §2.4 diagonalize the dynamics exactly as they diagonalized the boost. Starting from rest, $u^\mu(0) = (1,0)$, so $u^0 \pm u^1 = \mathrm{e}^{\pm\alpha\tau}$ and

$$u^0 = \cosh\alpha\tau, \qquad u^1 = \sinh\alpha\tau\;: \tag{104}$$

the *rapidity grows linearly in proper time*, $\phi = \alpha\tau$ — each proper second adds the same rapidity increment, and $\beta = \tanh\phi$ approaches but never reaches 1, which is Remark 2.9(iii) in dynamical form. Integrating once more,

$$t = \alpha^{-1}\sinh\alpha\tau, \qquad x = \alpha^{-1}\big(\cosh\alpha\tau - 1\big) \qquad \Longrightarrow \qquad \big(x + \alpha^{-1}\big)^2 - t^2 = \alpha^{-2}\;: \tag{105}$$

the worldline is one of the calibration hyperbolae of Fig. 1 — hence *hyperbolic motion*. The four-acceleration $a^\mu = \mathrm{d}u^\mu/\mathrm{d}\tau = \alpha\,(\sinh\alpha\tau,\,\cosh\alpha\tau)$ satisfies $a\cdot u = 0$ by inspection and has constant invariant norm $\sqrt{-a\cdot a} = \alpha = qE/m$: constant *proper* acceleration — the higher-derivative scalar mentioned in §4.1, realized here by the simplest field. One number, units restored: at proper acceleration g, one year of proper time gives $\phi = g\tau/c \approx 1.03$, hence $\beta \approx 0.77$, with about 1.19 years of coordinate time elapsed and 0.56 light-years covered.

5.4 Canonical versus mechanical momentum

Equation (93) split the momentum in two, and the two halves divide the labor:

	mechanical $\vec{p} = \gamma m\vec{v}$	canonical $\vec{P} = \vec{p} + q\vec{A}$
gauge behavior	invariant	shifts by $q\nabla\chi$
what force acts on	Eq. (98)	—
Noether charge of $\vec{x} \to \vec{x} + \vec{\epsilon}$	—	conserved (when A_μ is $\vec{x}$-independent)
enters Hamilton's equations	—	$\vec{P}$ is the phase-space variable

Neither column can do the other's job: the measurable, gauge-invariant $\vec{p}$ is not the conserved quantity; the conserved $\vec{P}$ is not gauge invariant, and by itself not measurable. The Hamiltonian makes the division explicit. Legendre-transforming (90) with respect to $\vec{P}$ (the inversion runs exactly as in §4.4 with $\vec{p} = \vec{P} - q\vec{A}$):

$$\boxed{H(\vec{x}, \vec{P}) = \sqrt{\big(\vec{P} - q\vec{A}\big)^2 + m^2} \; + \; q\,\varphi.} \tag{106}$$

Comparing with the free $H = \sqrt{|\vec{P}|^2 + m^2}$: the entire interaction is the substitution

$$\vec{P} \to \vec{P} - q\vec{A}, \qquad H \to H - q\varphi \qquad \text{i.e.} \qquad p^\mu \;\to\; p^\mu - qA^\mu \tag{107}$$

— *minimal coupling*, the rule quantum mechanics will inherit verbatim ($\hat{\vec{p}} \to \hat{\vec{p}} - q\vec{A}$ in the Pauli and Dirac equations) and quantum field theory will re-derive from gauge symmetry. That the whole of electromagnetism's action on matter compresses into one substitution is not an accident of this Lagrangian; it is the classical shadow of the gauge principle, and the outlook (§14) says one more word about it.

Part I is complete: kinematics from two postulates, dynamics from one scalar action, energy and momentum from Noether, and a first field — entering as a fixed background to which the particle couples. The obvious next question is the reverse one: what dynamics does A_μ itself obey, and what action prescribes it? Answering requires learning to vary *fields* rather than worldlines, and that is Part II's business — built, like everything in this series, from a chain of masses.

Part II
Classical Field Theory

6 From Chain to Field

Part I closed by asking for the dynamics of a field. This section builds the concept from the series' oldest object: the bead chain of [Osc. §8–9]. There, the continuum limit was taken in the *equations of motion* — a limit in each equation. Here it is taken once, in the *Lagrangian*, and the single limit does more than reproduce the wave equation: it exposes the structural pivot on which all of field theory turns — the demotion of position from dynamical variable to label.

6.1 The chain acquires a Lagrangian

Recall the setting of [Osc. §9.1]: N beads of mass m, spaced a apart on a massless string under tension T between fixed walls, with small transverse displacements $y_j(t)$, $j = 1, \ldots, N$ (and $y_0 = y_{N+1} = 0$ at the walls). There, Newton delivered

$$m\ddot{y}_j = \frac{T}{a}\big(y_{j-1} - 2y_j + y_{j+1}\big). \tag{108}$$

The Lagrangian route needs the one ingredient the free-body diagram never wrote down: the potential energy. It is the work done against tension by the *extra length* of the displaced string. The segment from bead j to bead $j+1$ has length

$$\sqrt{a^2 + (y_{j+1} - y_j)^2} = a\sqrt{1 + \left(\tfrac{y_{j+1}-y_j}{a}\right)^2} = a + \frac{(y_{j+1} - y_j)^2}{2a} + O\big((\Delta y)^4\big), \tag{109}$$

so, to the same small-slope order as (108),

$$U = T \times (\text{excess length}) = \sum_{j=0}^{N} \frac{T}{2a}\big(y_{j+1} - y_j\big)^2, \qquad L = \sum_{j=1}^{N} \tfrac{1}{2}\, m\, \dot{y}_j^2 \; - \; \sum_{j=0}^{N} \frac{T}{2a}\big(y_{j+1} - y_j\big)^2. \tag{110}$$

Check against (108): the coordinate y_j appears in *two* terms of the potential sum — the spring on its left and the spring on its right —

$$\frac{\partial L}{\partial y_j} = -\frac{T}{a}\Big[\big(y_j - y_{j-1}\big) - \big(y_{j+1} - y_j\big)\Big] = \frac{T}{a}\big(y_{j-1} - 2y_j + y_{j+1}\big), \tag{111}$$

and Euler–Lagrange, $\mathrm{d}(\partial L/\partial \dot{y}_j)/\mathrm{d}t = \partial L/\partial y_j$, reproduces (108) exactly. Note for later the mechanism of (111): differencing with *both* neighbors is what turned one difference into a second difference. It is the discrete shadow of an integration by parts, and §7 meets its continuum original.

6.2 Taking the limit once

Refine as in [Osc. §9.2]: $a \to 0$, $N \to \infty$, holding fixed the physical rope — its length $L_{\text{rope}} = (N+1)a$, its mass per length $\mu = m/a$, and its tension T. The displacements merge into a smooth profile, and here is the moment the notation should mark: write

$$y_j(t) \;=\; \phi(x_j, t), \qquad x_j = ja, \tag{112}$$

renaming $y \to \phi$ as the displacement sheds its origin (a transverse direction) and becomes an abstract *field value* — the letter ϕ serves for the rest of the document, and no confusion with the rapidity survives §3's $c = 1$ switch. Then $\dot{y}_j = \dot{\phi}(x_j, t)$ and $(y_{j+1} - y_j)/a \to \partial\phi/\partial x \equiv \phi'$, and the Lagrangian (110), rewritten to expose one factor of a per term,

$$L = \sum_j a \left[\frac{1}{2} \frac{m}{a} \dot{y}_j^2 - \frac{1}{2} T \left(\frac{y_{j+1} - y_j}{a} \right)^2 \right], \tag{113}$$

converges as a Riemann sum, $\sum_j a \to \int \mathrm{d}x$, to

$$\boxed{L = \int_0^{L_{\text{rope}}} \mathrm{d}x\, \mathcal{L}(\dot{\phi}, \phi'), \qquad \mathcal{L} = \tfrac{1}{2}\mu\,\dot{\phi}^2 - \tfrac{1}{2} T\,\phi'^2.} \tag{114}$$

$\mathcal{L}$ is the *Lagrangian density* — Lagrangian per unit length — and its two terms are exactly the kinetic and potential energy densities of the rope. One limit, taken in one object; the equations of motion will be extracted from it once, for all x simultaneously, in §7.

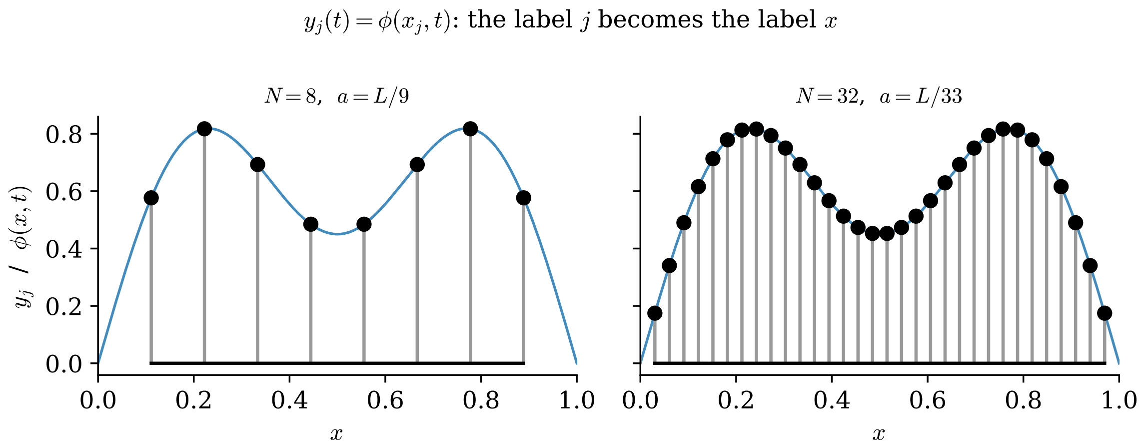

Figure 7: The same smooth profile sampled by two chains ($N = 8$ and $N = 32$ beads). The dynamical variables are the heights y_j; the site index j — equivalently the position $x_j = ja$ — merely *labels* them. Refining the chain makes the label continuous and the collection of heights a function $\phi(x, t)$; nothing about the label's role changes.

6.3 The pivot: x is demoted from variable to label

Now look at what happened to the letter x — the point of the whole construction, easy to use forever without once being said aloud.

In mechanics, position is what the theory is *about*: the $q_i(t)$ are the dynamical variables, and t is the label parametrizing their evolution. In (114) the dynamical variable is ϕ — the field's *value* — and x stands beside t as a second label, saying *which* degree of freedom is meant, exactly as the index j did on the chain (Fig. 7). The chain makes the demotion undramatic: j was never a dynamical variable, and $x_j = ja$ is just j wearing units. A field theory is a mechanical system with a continuous infinity of degrees of freedom, one per point of space, and the whole translation is a dictionary:

	mechanics / chain	field theory
labels degrees of freedom	index i	position x
dynamical variable	$q_i(t)$	$\phi(x,t)$
aggregation	$\sum_i$	$\int \mathrm{d}x$
Lagrangian	$L(q,\dot{q})$	$L = \int \mathrm{d}x\, \mathcal{L}(\phi,\dot{\phi},\phi')$
sensitivity to one d.o.f.	$\dfrac{\partial L}{\partial q_i}$	$\dfrac{\delta L}{\delta \phi(x)}$ (§7)
coupling structure	matrix K_{ij} (near-diagonal)	differential operator $-T\,\partial_x^2$ (local)
normal modes	eigenvectors of K [Osc. §8]	Fourier modes

Two of the rows remain to be justified. The functional derivative $\delta L/\delta\phi(x)$ — "the partial derivative of L with respect to the single degree of freedom sitting at x" — is defined properly in §7. And the mode row is already verified: [Osc. §9.3] showed the chain's normal modes turning into the string's Fourier modes, with the dispersion $\omega = 2\sqrt{T/(ma)}\,\sin(ka/2)$ flattening to $\omega = c_s k$, $c_s^2 = T/\mu$, as $ka \to 0$ — the dictionary's last row, verified before the dictionary existed.

Remark 6.1 (Why relativity insists on fields). Part I gives the pivot a second, independent motivation. Under a boost, t and x mix (Theorem 2.8); but mechanics hardwires an asymmetry between them — t a label, position a dynamical variable — so a mechanics of particles can be made relativistic only against the grain (the covariant formalism of §4 had to demote t itself to a function $x^0(\lambda)$ of an unphysical label to restore the symmetry, and Remark 4.2 was the consequence). Field theory dissolves the asymmetry at the root: t and x sit on the same side of the dictionary, joint labels x^μ on one dynamical variable ϕ, and Lorentz transformations act on labels exactly as §3.6's tensor fields require. The string itself is not relativistic — its $c_s = \sqrt{T/\mu}$ is no invariant speed, and §2.7's conclusion stands — but the *shape* of its description is precisely the shape relativistic dynamics needs, which is why §8 will need to change almost nothing.

The Lagrangian (114) is in hand and the question it raises is sharp: L now takes a whole function $\phi(\cdot,t)$ at each instant — it is a *functional*, in exactly the sense of [Mech. §1] — so what replaces $\partial L/\partial q_i$, and what do its Euler–Lagrange equations look like? That is the next section, and the machinery is already built: the Gateaux framework of [Mech. §2.2] was designed for functionals and does not care that this one's argument is a field.

7 The Field Variational Principle

The Lagrangian (114) takes a function and returns a number; its time integral, the action, is therefore a functional of the field's whole history. Extracting equations of motion from such an object is exactly the problem [Mech. §2] solved — the only novelty is that the "curve" being varied is now a function of *two* labels, and the variation will integrate by parts in both. This section runs the machinery once for a general density, supplies the functional derivative the dictionary promised, gives boundary terms the separate treatment they deserve, and closes two threads: the wave equation (a third time) and the fate of the string's accidental Lorentz invariance.

7.1 Hamilton's principle for fields

Let the density depend on the field's value and first derivatives — general enough for everything in this document:

$$S[\phi] = \int_{t_1}^{t_2} \mathrm{d}t \int_0^{\ell} \mathrm{d}x\, \mathcal{L}\big(\phi,\, \dot{\phi},\, \phi'\big), \tag{115}$$

(writing ℓ for the spatial extent; the infinite line is the limit $\ell \to \infty$ with decay conditions, §7.3). Hamilton's principle,[1] verbatim from [Mech. §3.1]: *the physical field history makes S stationary* among histories with prescribed field configurations at t_1 and t_2 and prescribed spatial boundary behavior.

Vary in the Gateaux sense of [Mech. §2.2]: $\phi \to \phi + \varepsilon\, \eta(x,t)$ with $\eta(x,t_1) = \eta(x,t_2) = 0$, and

$$\delta S[\phi;\eta] = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\right|_0 S[\phi + \varepsilon\eta] = \iint \mathrm{d}t\,\mathrm{d}x \left[\frac{\partial \mathcal{L}}{\partial \phi}\,\eta + \frac{\partial \mathcal{L}}{\partial \dot{\phi}}\,\dot{\eta} + \frac{\partial \mathcal{L}}{\partial \phi'}\,\eta'\right], \tag{116}$$

by the chain rule, with the slot-derivative shorthand of [Mech. §2.2]. Two integrations by parts free η — one in t per term two, one in x per term three:

$$\delta S = \iint \mathrm{d}t\,\mathrm{d}x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) - \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial \phi'}\right)\right]\eta \;+\; \int_{t_1}^{t_2} \mathrm{d}t \left[\frac{\partial \mathcal{L}}{\partial \phi'}\,\eta\right]_{x=0}^{x=\ell}. \tag{117}$$

The temporal boundary term died on the pinned endpoints, as always; the *spatial* boundary term is new, and care keeps it visible — §7.3 is devoted to the ways it vanishes. Granting that it does: η is arbitrary in the interior, the fundamental lemma applies (its proof [Mech. App. B] extends to two integration variables with only notational changes — choose the bump function as a product of two one-variable bumps [stated; the one-line adaptation is left to the reader]), and:

Theorem 7.1 (Field Euler–Lagrange equation)**.** *A field history making* (115) *stationary, with vanishing spatial boundary term, satisfies*

$$\boxed{\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) + \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial \phi'}\right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \qquad \text{at every } (x,t).} \tag{118}$$

Set beside its ancestor $\mathrm{d}(\partial L/\partial \dot{q})/\mathrm{d}t - \partial L/\partial q = 0$: the only change is that the label x, having joined t in the dictionary of §6.3, now also supplies a derivative term — one term per label, symmetrically, a symmetry §8 will make manifest. And the promised reunion: the spatial integration by parts that produced the $\partial_x(\partial\mathcal{L}/\partial\phi')$ term is the continuum original of the two-neighbor mechanism of Eq. (111) — there, y_j sat in two springs and the two first differences assembled a second difference; here, $\phi(x)$ sits under one derivative and the integration by parts moves it, signs and all, to the same effect.

[1] A terminological trap worth disarming once: Hamilton's principle, $\delta\int L\,\mathrm{d}t = 0$, is the variational principle *of Lagrangian mechanics* — the name honors its author (Hamilton, 1834–35), not the function varied. Lagrange himself derived his equations from d'Alembert's principle of virtual work [Mech. §3.2], with no action functional in sight; Hamilton's name thus sits on both sides of the Legendre transform — this principle on the Lagrangian side, Hamilton's equations on the other — by biographical accident. The alternative name "principle of least action" is worse: historically it denotes Maupertuis' principle (fixed energy, free transit time, abbreviated action $\int \vec{p}\cdot\mathrm{d}\vec{q}$), a genuinely different variational statement.

7.2 Defining the functional derivative

Section 6.3 promised a meaning for $\delta L/\delta\phi(x)$ — "the sensitivity of L to the single degree of freedom at x." The Gateaux variation supplies it. For the fixed-time functional $L[\phi] = \int \mathrm{d}x\,\mathcal{L}(\phi,\dot\phi,\phi')$, collect the variation (spatial boundary term disposed of as above) into the form

$$\delta L[\phi;\eta] \;=\; \int \mathrm{d}x\,\frac{\delta L}{\delta\phi(x)}\,\eta(x), \qquad \text{which defines} \qquad \boxed{\frac{\delta L}{\delta\phi(x)} = \frac{\partial\mathcal{L}}{\partial\phi} - \frac{\partial}{\partial x}\left(\frac{\partial\mathcal{L}}{\partial\phi'}\right)} \tag{119}$$

(and $\delta L/\delta\dot\phi(x) = \partial\mathcal{L}/\partial\dot\phi$, no derivative to move). In this notation the field equation (118) is, line for line, mechanics:

$$\frac{\partial}{\partial t}\,\frac{\delta L}{\delta\dot\phi(x)} \;=\; \frac{\delta L}{\delta\phi(x)} \qquad\longleftrightarrow\qquad \frac{\mathrm{d}}{\mathrm{d}t}\,\frac{\partial L}{\partial\dot q_i} = \frac{\partial L}{\partial q_i}. \tag{120}$$

The discrete check. The dictionary row $\partial/\partial q_i \leftrightarrow \delta/\delta\phi(x)$ hides a factor worth exposing. On the chain, Eq. (111) gave

$$\frac{\partial L}{\partial y_j} = \frac{T}{a}\big(y_{j-1} - 2y_j + y_{j+1}\big) = a\cdot T\,\frac{y_{j-1} - 2y_j + y_{j+1}}{a^2} \;\longrightarrow\; a\cdot T\,\phi''(x_j), \tag{121}$$

while (119) applied to $\mathcal{L} = \frac{1}{2}\mu\dot\phi^2 - \frac{1}{2}T\phi'^2$ gives $\delta L/\delta\phi(x) = \partial_x(T\phi') = T\phi''$. So

$$\frac{\partial L}{\partial y_j} \;\longrightarrow\; a\,\frac{\delta L}{\delta\phi(x_j)}\,: \tag{122}$$

the discrete partial derivative is the functional derivative times the *cell size* — sensitivity to one bead is sensitivity-density at its site times the length of chain it represents. The factor of a is the same one that converted $\sum_j$ into $\int \mathrm{d}x$, and it must appear here for $\sum_j(\partial L/\partial y_j)\,\eta_j \to \int \mathrm{d}x\,(\delta L/\delta\phi)\,\eta$ to hold.

7.3 Spatial boundary terms

The term (117) left standing,

$$B = \int_{t_1}^{t_2} \mathrm{d}t\,\left[\frac{\partial\mathcal{L}}{\partial\phi'}\,\eta\right]_{x=0}^{x=\ell}, \tag{123}$$

must vanish for Theorem 7.1 to hold, and each way of killing it is a physical situation. For the string, $\partial\mathcal{L}/\partial\phi' = -T\phi'$ is (minus) the transverse force the string exerts across a cross-section, which gives every option below its physical reading.

- **Fixed ends (Dirichlet).** ϕ prescribed at $x = 0, \ell$; admissible variations have $\eta = 0$ there, and B dies on η. The rope between walls of [Osc. §9]. The boundary *condition* is imposed; the variational principle accepts it.

- **Free ends (Neumann).** Nothing prescribed at the boundary, so η is arbitrary there — and then stationarity itself forces

$$\left.\frac{\partial\mathcal{L}}{\partial\phi'}\right|_{x=0,\,\ell} = 0 \qquad \text{(string: } \phi' = 0 \text{ at the free end)}, \tag{124}$$

  a *natural boundary condition*: the variational principle, given no instruction at the edge, issues one — zero transverse force on the massless end ring, which is Newton's law for an end of zero mass. A principle that generates boundary conditions it was not handed is doing more than repackaging the bulk equation.

- **Periodic.** $\phi(0,t) = \phi(\ell,t)$ — a loop: the two evaluations of B cancel each other. The most convenient choice for mode expansions, used in §14.
- **The infinite line.** $\ell \to \infty$ with $\phi \to 0$ (fast enough) at spatial infinity: $B \to 0$ on decay. The default of field theory proper, and this document's standing assumption from §8 on whenever no boundary is mentioned.

The subsection is short but essential: Noether's theorem for fields (§9) converts symmetries into statements about integrals of densities, and every one of those statements holds *up to the same kinds of boundary terms*, disposed of by the same four mechanisms.

7.4 A third route to the wave equation

For the third route to be genuinely a third route, it must not borrow the density from the chain limit of §6.2 — that would smuggle the beads back in. So build $\mathcal{L}$ directly from the continuous rope: mass per length μ, tension T, transverse displacement $\phi(x,t)$, small slopes.

Kinetic density. The rope element over $[x, x+\mathrm{d}x]$ has mass $\mu\,\mathrm{d}x$ and moves transversely at velocity $\dot\phi(x,t)$, so

$$\mathrm{d}T_{\mathrm{kin}} = \tfrac{1}{2}\,(\mu\,\mathrm{d}x)\,\dot\phi^{\,2} \qquad \Longrightarrow \qquad \text{kinetic density} = \tfrac{1}{2}\,\mu\,\dot\phi^{\,2}. \tag{125}$$

Potential density. The energy cost of a displaced shape is the work done against the tension by the rope's *excess length* — the continuum original of the spring-by-spring tally in Eq. (110). The displaced element has arc length

$$\mathrm{d}s = \sqrt{\mathrm{d}x^2 + \mathrm{d}\phi^2} = \sqrt{1+\phi'^2}\,\mathrm{d}x = \Big(1 + \tfrac{1}{2}\,\phi'^2 + O(\phi'^4)\Big)\mathrm{d}x, \tag{126}$$

an excess of $\frac{1}{2}\phi'^2\,\mathrm{d}x$ per element; pulled out against a tension that is itself unchanged at this order (small slopes, [Osc. §9.1]'s standing approximation),

$$\mathrm{d}U = T\cdot\tfrac{1}{2}\,\phi'^2\,\mathrm{d}x \qquad \Longrightarrow \qquad \text{potential density} = \tfrac{1}{2}\,T\,\phi'^{\,2}. \tag{127}$$

Assembly. Kinetic minus potential, densely:

$$\mathcal{L} = \tfrac{1}{2}\,\mu\,\dot\phi^{\,2} - \tfrac{1}{2}\,T\,\phi'^{\,2}, \tag{128}$$

in agreement with the chain limit (114) — as it must be, and the agreement is a real check: two constructions, one through a lattice and its limit, one never leaving the continuum, land on the same density. Feed it to Theorem 7.1:

$$\frac{\partial\mathcal{L}}{\partial\dot\phi} = \mu\dot\phi, \qquad \frac{\partial\mathcal{L}}{\partial\phi'} = -T\phi', \qquad \frac{\partial\mathcal{L}}{\partial\phi} = 0, \tag{129}$$

so (118) reads $\mu\ddot\phi - T\phi'' = 0$:

$$\boxed{\ddot\phi = c_s^2\,\phi'', \qquad c_s^2 = \frac{T}{\mu}.} \tag{130}$$

The same physics for the third time, by the third route, each shorter than the last — and now each genuinely independent: Newton on beads, then the limit of the equations [Osc. §9.2]; Lagrange on beads (§6.1), then the same limit; and now two energy densities read off the continuum and one variation, all x at once, no bead ever mentioned. The ranking is not aesthetic. The first two routes need beads — a mechanical substructure to apply Newton or Lagrange *to* — and the fields ahead (Klein–Gordon, Maxwell) have none. The third route needs only a density, and choosing densities is exactly what the covariance principle of §3.6 constrains. (The general solution is $\phi = f(x - c_s t) + g(x + c_s t)$ — d'Alembert's form, derived by passing to characteristic variables in [Osc. §9.5]; its physics is not repeated here.)

Remark 7.2 (The status of the string's Lorentz invariance). Rewrite the string density as

$$\mathcal{L} = \tfrac{\mu}{2}\big(\dot\phi^2 - c_s^2\,\phi'^2\big) : \tag{131}$$

up to the factor $\mu/2$, this is the Minkowski norm of the gradient $(\dot\phi, \phi')$ *with c_s in the role of c* — so both the density and its wave equation are form-invariant under "Lorentz" transformations of the labels (t, x) built from c_s. Here, at last, is the precise status of the accident flagged in §2.7: the invariance is a true symmetry *of the long-wavelength equations* and a falsehood about the world they describe. It is broken twice over. Physically, the medium has a rest frame — the frame where (114) was derived — detectable by measuring c_s in both directions, so Postulate 2.1 fails for c_s-boosts. And microscopically, the symmetry does not survive below the lattice scale: the chain's true dispersion $\omega = 2\sqrt{T/(ma)}\,\sin(ka/2)$ [Osc. §9.3] is not the c_s-invariant $\omega = c_s|k|$, and the deviation at $ka \sim 1$ is where the lattice becomes visible. An emergent symmetry, valid to $O(k^2a^2)$, absent from the microscopic theory it descends from. The contrast defines the next section's task: write down a field whose Lorentz invariance is *exact* — symmetry of the Lagrangian under the true Lorentz group of Part I, no medium, no lattice, no preferred frame — and see what the covariance principle leaves standing.

8 Relativistic Scalar Fields: Klein–Gordon

Everything is now assembled: a variational principle for fields (§7), a covariance principle that constrains candidate Lagrangians (§3.6), and a sharp question (Remark 7.2): what does a field theory look like when Lorentz invariance is exact? This section repackages the field Euler–Lagrange equation covariantly, then plays the building-block game — and the game has essentially one answer, the Klein–Gordon field, whose dispersion relation turns out to be an old figure wearing new axis labels.

8.1 The covariant Euler–Lagrange equation

Move to 3+1 dimensions and covariant notation in one step: let the density depend on the field and its four-gradient, $\mathcal{L}(\phi, \partial_\mu\phi)$, and write

$$S[\phi] = \int \mathrm{d}^4x\; \mathcal{L}\big(\phi,\, \partial_\mu\phi\big), \qquad \mathrm{d}^4x = \mathrm{d}t\,\mathrm{d}^3x. \tag{132}$$

The measure takes care of itself: under $x' = \Lambda x$ the Jacobian is $|\det\Lambda| = 1$ (§3.8), so d^4x is Lorentz invariant, and *S is a scalar if and only if $\mathcal{L}$ is a scalar field.* That single sentence is the design constraint for everything ahead.

The variation is §7.1 with more labels and better notation. Vary $\phi \to \phi + \varepsilon\eta$, integrate by parts once per spacetime direction — temporal boundary terms die on pinned endpoints, spatial ones by decay at infinity (§7.3, fourth mechanism, henceforth standing) — and apply the fundamental lemma:

$$\boxed{\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) - \frac{\partial\mathcal{L}}{\partial\phi} \;=\; 0.} \tag{133}$$

Compare with (118): the separate $\partial_t(\partial\mathcal{L}/\partial\dot\phi)$ and $\partial_x(\partial\mathcal{L}/\partial\phi')$ terms — one per label, as §7.1 noted — have fused into a single contracted term, one summed index μ running over all the labels at once. The dictionary of §6.3 put t and x on the same side; Eq. (133) shows what that decision achieves. Note also the index placement: $\partial\mathcal{L}/\partial(\partial_\mu\phi)$ differentiates with respect to a lower-index object and so carries an *upper* μ — it is a four-vector field, contracted by ∂_μ into a scalar equation, every piece certified by §3.6.

8.2 Building scalar Lagrangians

Now the game. $\mathcal{L}$ must be a Lorentz scalar assembled from a scalar field ϕ and its gradient $\partial_\mu\phi$ (§3.6: $\partial_\mu\phi$ is a covariant vector field, and contraction is the only scalar-maker). The full catalogue of blocks:

$$\phi,\ \phi^2,\ \phi^3,\ldots;\qquad \partial_\mu\phi\,\partial^\mu\phi,\ \left(\partial_\mu\phi\,\partial^\mu\phi\right)^2,\ \ldots;\qquad \phi^n\,\partial_\mu\phi\,\partial^\mu\phi;\qquad \text{higher derivatives } (\Box\phi,\ \ldots). \tag{134}$$

Physics thins the catalogue:

- **First derivatives only.** The mechanics of the whole series runs on $L(q,\dot q)$; densities with second derivatives yield higher-order equations of motion whose energies are unbounded below (the Ostrogradsky instability [stated; standard, not proved here]). Discard $\Box\phi$ and friends.
- **At most quadratic.** Quadratic $\mathcal{L}$ means *linear* equations of motion, hence superposition — the defining property of a *free* field, and the natural starting point: every interacting theory in physics is built as a free quadratic core plus higher-power corrections. Discard ϕ^3, $\phi^n(\partial\phi)^2$, … — not as impossible, but as *interactions*, deferred to the outlook.
- **Tidying.** A constant a_0 shifts S without moving its stationary points (though gravity, which weighs $\mathcal{L}$ itself, cares — one sentence of foreshadowing, [stated]). A linear term $a_1\phi$ is removed by the constant shift $\phi\to\phi-a_1/(2a_2)$. The kinetic normalization is fixed to $\frac{1}{2}$ by rescaling ϕ — the field's units are ours to choose.

What survives is a one-parameter family — the most general free scalar field theory:

$$\boxed{\mathcal{L}=\tfrac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi\ -\ \tfrac{1}{2}\,m^2\phi^2,} \tag{135}$$

the parameter written m^2 in anticipation (§8.4 justifies the name). Expanded in 3+1 language,

$$\mathcal{L}=\tfrac{1}{2}\dot\phi^2-\tfrac{1}{2}|\nabla\phi|^2-\tfrac{1}{2}m^2\phi^2, \tag{136}$$

which invites the comparison this Part was built for. The string had $\mathcal{L}=\frac{\mu}{2}(\dot\phi^2-c_s^2\phi'^2)$: same kinetic structure, with $c_s\to c=1$ now *exact* rather than emergent, the overall μ absorbed into the field normalization — and one genuinely new term. Signs are physics, not convention: the relative sign between $\dot\phi^2$ and $|\nabla\phi|^2$ is the metric's signature and is what makes (133) hyperbolic (waves, causal propagation); the sign of the mass term makes $V(\phi)=\frac{1}{2}m^2\phi^2$ a well rather than a hill — the opposite choice, $-m^2\to+\mu^2$, puts every Fourier mode on top of an inverted potential and long-wavelength perturbations grow exponentially (the "tachyonic" instability; when encountered on purpose, it is the engine of spontaneous symmetry breaking — outlook material, [stated]).

Remark 8.1 (What the mass term would mean for a string). The string never had a ϕ^2 term because nothing pushed a displaced bead back toward the axis except its neighbors. Add such a push — mount every bead on its own spring to the equilibrium line, a mattress rather than a rope — and the bead equation gains a term $-\kappa y_j$, whose continuum limit is exactly $-\frac{1}{2}m^2\phi^2$ in the density with $m^2 = \kappa/(\mu a)$ held fixed. The Klein–Gordon field is, mechanically, *a mattress*: a medium whose every element is individually tethered. Each element could oscillate at frequency m even with all its neighbors held still — the physical origin of the mass gap about to appear in the dispersion relation.

8.3 The Klein–Gordon equation

Feed (135) to (133):

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi, \qquad \frac{\partial \mathcal{L}}{\partial \phi} = -m^2\phi, \tag{137}$$

(the first equality is worth one private index-expansion: write $\partial_\mu\phi\,\partial^\mu\phi = \eta^{\rho\sigma}\partial_\rho\phi\,\partial_\sigma\phi$ and differentiate — the two slots contribute equally, canceling the $\frac{1}{2}$). Hence

$$\boxed{\left(\Box + m^2\right)\phi = 0} \qquad \text{— the Klein–Gordon equation.} \tag{138}$$

The d'Alembertian now plays the role promised in §3.5: it is the only two-derivative scalar operator there is, so it *had* to be the one. At $m = 0$ the equation is $\Box\phi = 0$ — the wave equation with the invariant speed, form-invariant in every frame by inspection, closing the arc opened at Eq. (6): this is what the light-like wave equation looks like when it is a law of nature rather than a description of a medium.

8.4 Plane waves and the dispersion relation

Hunt for plane-wave solutions with the covariant ansatz

$$\phi(x) = \mathrm{Re}\ \hat{A}\,\mathrm{e}^{-\mathrm{i}\,k\cdot x}, \qquad k^\mu = (\omega, \vec{k}), \qquad k\cdot x = \omega t - \vec{k}\cdot\vec{x}, \tag{139}$$

(complexification per [Osc. Conv. 1.1]; the exponent is a scalar provided k^μ is a four-vector, so the ansatz survives boosts as a plane wave with boosted k^μ). Each derivative pulls down a factor: $\partial_\mu\,\mathrm{e}^{-\mathrm{i}k\cdot x} = -\mathrm{i}k_\mu\,\mathrm{e}^{-\mathrm{i}k\cdot x}$, so $\Box \to (-\mathrm{i}k_\mu)(-\mathrm{i}k^\mu) = -k\cdot k = -(\omega^2 - |\vec{k}|^2)$, and (138) demands

$$\boxed{\omega^2 = |\vec{k}|^2 + m^2.} \tag{140}$$

This is Fig. 6 with its axes relabeled $(E, p) \to (\omega, \vec{k})$, exactly as Remark 4.6 promised: *the dispersion relation of the field is the energy–momentum relation of a particle of mass m* — which is why the coefficient in (135) was named m^2. Classically this is a suggestive pun: m is, as far as (140) knows, an inverse length, $1/m$ the scale (the *Compton wavelength*, once $\hbar$ is restored as $\hbar/mc$) below which the dispersion straightens onto the light line. The pun becomes physics upon quantization, where $E = \hbar\omega$ and $\vec{p} = \hbar\vec{k}$ turn (140) into (83) literally — the outlook (§14) says how.

Two velocities, one check, one warning:

$$v_{\mathrm{ph}} = \frac{\omega}{|\vec{k}|} = \sqrt{1 + \frac{m^2}{|\vec{k}|^2}} \ > \ 1, \qquad v_{\mathrm{g}} = \frac{\partial\omega}{\partial|\vec{k}|} = \frac{|\vec{k}|}{\omega} \ < \ 1, \qquad v_{\mathrm{ph}}\,v_{\mathrm{g}} = 1. \tag{141}$$

The superluminal phase velocity alarms no one who asks what moves: a single infinite plane wave's crests carry no information (nothing distinguishes one crest from the next), and what transports energy and signals is the group velocity of a wave *packet* [Osc. §4.1's beat logic, one wavelength up] — which is not only subluminal but exactly right: $v_{\mathrm{g}} = |\vec{k}|/\omega = |\vec{p}|/E$ is the velocity of the mass-m

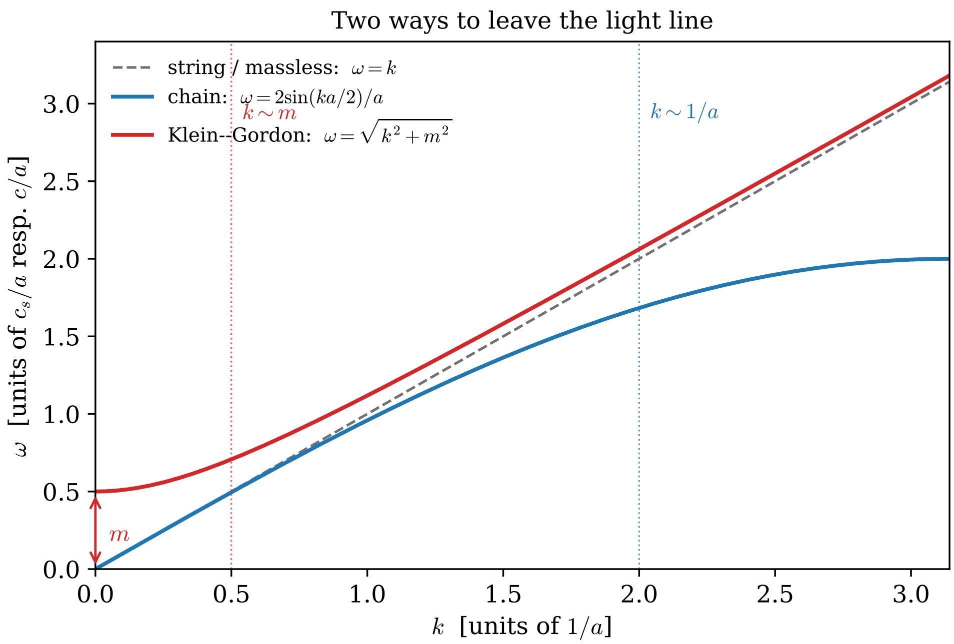

Figure 8: Three dispersion relations, drawn from their exact formulas ($c_s = c = 1$, lattice spacing $a = 1$, $m = 0.5$). The light line $\omega = k$ (gray, dashed) is left in two opposite directions: the chain [Osc. §8.3] bends *below* it at large k, where wavelengths approach the lattice spacing and the medium's discreteness shows; the Klein–Gordon field lifts *above* it at small k, by the mass gap $\omega(0) = m$ — every mattress element oscillates at m even at infinite wavelength (Remark 8.1). Linear, light-like propagation survives only in the window $m \ll k \ll 1/a$: an ultraviolet scale and an infrared scale, bracketing the massless continuum from opposite sides.

particle with that momentum, by Eqs. (81) and (83). The correspondence particle $\leftrightarrow$ packet holds down to the kinematics. [The full story of what can and cannot outrun light in a dispersive medium — front velocity — is genuinely subtle and is flagged, not settled, here.]
Figure 8 assembles the Part's three dispersion relations and reads as a summary of its physics: the lattice scale $1/a$ bounds the wave regime from above (ultraviolet — the chain remembers its beads), the mass m bounds it from below (infrared — the mattress remembers its tethers), and massless light-line propagation is what both cutoffs leave between them. The massive field is *dispersive*: packets built from (140) spread, because their Fourier components travel at different group velocities — in contrast to the massless case, whose general solution $\phi = f(x-t) + g(x+t)$ [Osc. §9.5] carries pulses rigidly forever.

What the field equation cannot yet say is what the field *carries*: energy, momentum, and their currents. That is the subject of Noether's theorem for fields — the next section.

9 Noether's Theorem for Fields

Noether's theorem for mechanics [Mech. §3.3] traded each continuous symmetry for a conserved *number*. For fields the theorem returns with a strictly stronger conclusion: each continuous symmetry yields a conserved *current* — a local accounting of where the conserved stuff sits and how it flows. This section proves the field version, explains why relativity does not merely permit but *demands* the local form, and runs the theorem twice: on an internal symmetry (new in kind — nothing like it exists in Part I) and on spacetime translations, whose four currents assemble into the energy-momentum tensor.

9.1 The theorem

Definition 9.1 (Symmetry of a field theory)**.** A continuous transformation $\phi \to \phi + \varepsilon\,\Delta\phi$ (with $\Delta\phi$ built from ϕ and its derivatives) is a *symmetry* if it changes the Lagrangian density by at most a total divergence,

$$\delta\mathcal{L} = \varepsilon\,\partial_\mu K^\mu \tag{142}$$

for some four-vector K^μ built from the fields — for then δS is a pure boundary term and the equations of motion are unchanged, exactly the logic of §5.2. ($K^\mu = 0$ is the special case of exact invariance.)

Theorem 9.2 (Noether, field version)**.** *If* (142) *holds, then along every solution of the equations of motion* (133) *the current*

$$\boxed{j^\mu \;=\; \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\Delta\phi \;-\; K^\mu} \tag{143}$$

is conserved:

$$\boxed{\partial_\mu j^\mu = 0.} \tag{144}$$

(For several fields, sum the first term over them.)

Proof. Compute the change of $\mathcal{L}$ under the transformation by the chain rule, in the slot-derivative sense of §7.1:

$$\delta\mathcal{L} = \varepsilon\left[\frac{\partial\mathcal{L}}{\partial\phi}\,\Delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\mu(\Delta\phi)\right]. \tag{145}$$

On a solution, the equation of motion (133) converts the first coefficient: $\partial\mathcal{L}/\partial\phi = \partial_\mu\big(\partial\mathcal{L}/\partial(\partial_\mu\phi)\big)$. The two terms then assemble into a total divergence by the product rule:

$$\delta\mathcal{L} = \varepsilon\,\partial_\mu\left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\Delta\phi\right]. \tag{146}$$

Equating with the definition (142) of the symmetry: $\partial_\mu\big[(\partial\mathcal{L}/\partial(\partial_\mu\phi))\Delta\phi - K^\mu\big] = 0$, which is the claim. □

9.2 What "conserved" now means: localization

Unpack $\partial_\mu j^\mu = 0$ in 3+1 language, writing $j^\mu = (\rho, \vec{\jmath})$:

$$\boxed{\frac{\partial\rho}{\partial t} + \nabla\cdot\vec{\jmath} = 0} \qquad \text{— a } \textit{continuity equation:} \tag{147}$$

the density ρ at a point can change only by flux $\vec{\jmath}$ through that point's neighborhood. Integrating over all space and using the divergence theorem,

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\int \rho\,\mathrm{d}^3x = -\oint_\partial \vec{\jmath}\cdot\mathrm{d}\vec{A} \;\longrightarrow\; 0 \tag{148}$$

for fields decaying at infinity (§7.3, fourth mechanism): the global charge $Q = \int j^0\,\mathrm{d}^3x$ is constant, and mechanics' conclusion is recovered as a corollary. But the local statement is the physics: the conserved stuff cannot vanish here and reappear there — it must *flow through the intervening space*, and $\vec{\jmath}$ records the flow.

Remark 9.3 (Why relativity demands the local form). In Newtonian physics, local conservation is a refinement of global conservation; in relativity it is the only coherent version. Suppose a charge could teleport: vanish at event P and appear at a distant event P', simultaneous with P in some frame. P and P' are spacelike separated, so their time order is frame-dependent (§3.2) — boosted observers see the charge appear *before* it vanishes, or find slices of simultaneity (§2.6) on which total charge is doubled or absent. "Total Q at time t" presupposes a simultaneity slicing, and Part I spent a section showing that slicings are negotiable; the only conservation statement all observers can share is the slicing-free, pointwise (147). (Conversely, for a conserved current the global Q comes out the *same* on every slicing — Q is a Lorentz scalar. [stated; the proof integrates $\partial_\mu j^\mu = 0$ over the region between two slices.])

9.3 First run: the complex scalar and its charge

The theorem's first application is a symmetry of a kind Part I never met. Package *two* real Klein–Gordon fields of equal mass into one complex field, $\phi = (\phi_1 + \mathrm{i}\phi_2)/\sqrt{2}$, with density

$$\mathcal{L} = \partial_\mu\phi^*\,\partial^\mu\phi - m^2\,\phi^*\phi \tag{149}$$

(the $\frac{1}{2}$'s of (135) absorbed by the $\sqrt{2}$; and one standing trick, justified once: varying the independent real fields ϕ_1, ϕ_2 is equivalent to varying ϕ and ϕ^* *as if independent* — the two variation bases are related by an invertible linear map, so stationarity in one is stationarity in the other). Varying ϕ^* yields $(\Box + m^2)\phi = 0$; varying ϕ, the conjugate.

The point of the packaging: (149) is exactly invariant under the *global phase rotation*

$$\phi \to \mathrm{e}^{-\mathrm{i}\alpha}\phi, \qquad \phi^* \to \mathrm{e}^{+\mathrm{i}\alpha}\phi^*, \qquad \alpha \text{ constant}, \tag{150}$$

since every term pairs a ϕ with a ϕ^*. This is an *internal* symmetry — it turns the field's *value*, touching the spacetime labels not at all — the first symmetry of the series with no spacetime meaning whatsoever, and the origin of everything the word "charge" means in particle physics. Infinitesimally, $\Delta\phi = -\mathrm{i}\phi$, $\Delta\phi^* = +\mathrm{i}\phi^*$, with $K^\mu = 0$ (exact invariance), and Theorem 9.2 yields:

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\Delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)}\,\Delta\phi^* = (\partial^\mu\phi^*)(-\mathrm{i}\phi) + (\partial^\mu\phi)(+\mathrm{i}\phi^*), \tag{151}$$

$$\boxed{j^\mu = \mathrm{i}\left(\phi^*\,\partial^\mu\phi - \phi\,\partial^\mu\phi^*\right), \qquad \partial_\mu j^\mu = 0.} \tag{152}$$

The conservation deserves one direct check — Noether proved it, but the mechanism is instructive:

$$\partial_\mu j^\mu = \mathrm{i}\big(\phi^*\,\Box\phi - \phi\,\Box\phi^*\big) = \mathrm{i}\big(\phi^*(-m^2\phi) - \phi(-m^2\phi^*)\big) = 0, \tag{153}$$

the cross terms $\partial_\mu\phi^*\partial^\mu\phi$ canceling identically and the mass terms canceling *on shell.*

Remark 9.4 (Charge, not probability)*.* The charge density $j^0 = \mathrm{i}(\phi^*\dot\phi - \phi\dot\phi^*)$ is **not** positive definite — complex conjugating a solution flips its sign. Historically this killed the reading of ϕ as a relativistic wavefunction with j^0 a probability density (the Klein–Gordon equation's first, failed interpretation); rightly read, Q is a *charge*, a quantity that legitimately comes in both signs — field and antifield, particle and antiparticle after quantization. And the current will be needed again: (152) is precisely the kind of object that can sit on the right-hand side of a field equation for A_μ — a conserved J^μ is what Part III's Maxwell Lagrangian will demand as its source, and *this* is where such currents come from.

9.4 Second run: translations and the energy-momentum tensor

Now the central application: the four spacetime translations, $x^\mu \to x^\mu + \varepsilon\, a^\mu$. A scalar field is dragged along, $\phi(x) \to \phi(x - \varepsilon a)$, so

$$\Delta\phi = -\,a^\nu\,\partial_\nu\phi. \tag{154}$$

The density, having no explicit x-dependence, is dragged the same way — and *that is already a total divergence*:

$$\delta\mathcal{L} = -\,\varepsilon\, a^\nu\,\partial_\nu\mathcal{L} = \varepsilon\,\partial_\mu\big(-a^\mu\,\mathcal{L}\big), \qquad K^\mu = -\,a^\mu\mathcal{L}: \tag{155}$$

a quasi-symmetry in the precise sense of (142), with *four* independent parameters a^ν. Theorem 9.2 gives the current

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\left(-a^\nu\partial_\nu\phi\right) \;+\; a^\mu\,\mathcal{L} \;=\; -\,a^\nu\left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\nu\phi - \delta^\mu{}_\nu\,\mathcal{L}\right], \tag{156}$$

and since the four a^ν are independent, the bracket is conserved for each ν separately — four currents, organized as one two-index object:

$$\boxed{T^\mu{}_\nu \;=\; \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\nu\phi \;-\; \delta^\mu{}_\nu\,\mathcal{L}, \qquad \partial_\mu T^\mu{}_\nu = 0 \quad \text{(four equations, one per } \nu\text{)},} \tag{157}$$

the *energy-momentum tensor* (raising the second index, $T^{\mu\nu} = (\partial\mathcal{L}/\partial(\partial_\mu\phi))\partial^\nu\phi - \eta^{\mu\nu}\mathcal{L}$; it is a tensor by the algebra of §3.6, being built from tensors by products and contractions). Its sixteen components record the field's entire mechanics:

component	name	role in (147)
T^{00}	energy density	ρ of the $\nu = 0$ current
T^{i0}	energy flux	$\vec{\jmath}$ of the $\nu = 0$ current
T^{0i}	momentum density	ρ of the $\nu = i$ current
T^{ji}	momentum flux (stress)	$\vec{\jmath}$ of the $\nu = i$ current

and the four global charges are the field's four-momentum,

$$P^\nu = \int T^{0\nu}\, \mathrm{d}^3x = \big(E,\ \vec{P}\big), \tag{158}$$

— §4.3 rerun at field level, with the same symmetry yielding the same conserved quantities, now *localized* into densities and fluxes.

The Klein–Gordon case. For (135), $\partial\mathcal{L}/\partial(\partial_\mu\phi) = \partial^\mu\phi$, so $T^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \eta^{\mu\nu}\mathcal{L}$, and in particular

$$\boxed{T^{00} = \tfrac{1}{2}\,\dot\phi^2 + \tfrac{1}{2}\,|\nabla\phi|^2 + \tfrac{1}{2}\,m^2\phi^2 \ \ \geq\ 0,} \qquad T^{0i} = \dot\phi\,\partial^i\phi. \tag{159}$$

The energy density is a sum of three squares — kinetic, gradient (the tension energy of §7.4, one dimension up), and tether (the mattress springs of Remark 8.1) — and its positivity is not accidental: it is the delayed consequence of the sign choices of §8.2. Flip the kinetic sign and T^{00} is unbounded below; flip the mass sign and the tether term goes negative — §8.2's "signs are physics," made quantitative as a bounded energy. As a check against the series' oldest intuition: the same formula (157) applied to the string density gives energy density $\frac{1}{2}\mu\dot\phi^2 + \frac{1}{2}T\phi'^2$ and energy flux $T^x{}_0 = -T\phi'\dot\phi$ — the transverse force (§7.3) times the transverse velocity, i.e. mechanical power transmitted across the cross-section, exactly as a free-body analysis would say.

Remark 9.5 (The other six, and a symmetry requirement). The remaining Poincaré parameters (three rotations, three boosts; Remark 4.4) yield, by the same theorem, the six currents packaged as $M^{\rho\mu\nu} = x^\mu\,T^{\rho\nu} - x^\nu\,T^{\rho\mu}$. Its conservation is a one-line computation with a condition inside:

$$\partial_\rho M^{\rho\mu\nu} = T^{\mu\nu} - T^{\nu\mu} + x^\mu\,\partial_\rho T^{\rho\nu} - x^\nu\,\partial_\rho T^{\rho\mu} = T^{\mu\nu} - T^{\nu\mu}: \tag{160}$$

angular momentum is conserved *if and only if the energy-momentum tensor is symmetric.* For the scalar field it is, automatically — $\partial^\mu\phi\,\partial^\nu\phi$ and $\eta^{\mu\nu}\mathcal{L}$ are symmetric by inspection. For fields that carry indices of their own the canonical (157) comes out *asymmetric*, the mismatch being the field's intrinsic (spin) angular momentum, and a repair — the Belinfante improvement, adding a total-divergence term that symmetrizes T without touching the charges — exists in general. [stated; Part III meets the concrete case for the electromagnetic field in §13, and the general construction is QFT-course material.]

The Lagrangian story of the free scalar field is now complete: equation of motion, symmetries, currents, energy. What remains of the mechanics dictionary is its second column — momenta, Hamiltonian, brackets — and porting it is the last stop before electromagnetism.

10 Hamiltonian Field Theory

One column of the mechanics dictionary remains untranslated: momenta, Hamiltonian, brackets — the phase-space half of [Mech. Part II]. This section ports it, with one genuinely new ingredient (the Dirac delta, introduced as what it is rather than what it is written as) and one promise kept: Remark 4.5 stated that the particle's trade — manifest covariance given up at the Legendre transform — would recur, unchanged, for fields.

10.1 The field momentum and the Hamiltonian

The dictionary of §6.3 translates $p_i = \partial L/\partial \dot{q}_i$ entry by entry: one momentum per degree of freedom, i.e. one per point,

$$\boxed{\pi(\vec{x}) \;\equiv\; \frac{\delta L}{\delta\dot{\phi}(\vec{x})} \;=\; \frac{\partial\mathcal{L}}{\partial\dot{\phi}}} \tag{161}$$

(the two expressions agree because $\mathcal{L}$ contains no spatial derivatives of $\dot{\phi}$ — nothing for (119)'s integration by parts to move). For Klein–Gordon, $\pi = \dot{\phi}$; for the complex scalar (149), note the crossing, $\pi = \dot{\phi}^*$ and $\pi^* = \dot{\phi}$. The Legendre transform [Mech. §6.1] then runs pointwise — one transform per degree of freedom, aggregated by the dictionary's $\int \mathrm{d}^3x$:

$$\boxed{H = \int \mathrm{d}^3x\,\mathcal{H}, \qquad \mathcal{H} = \pi\,\dot{\phi} - \mathcal{L} \quad (\dot{\phi}\text{ eliminated in favor of }\pi).} \tag{162}$$

For Klein–Gordon, $\dot{\phi} = \pi$ inverts trivially and

$$\mathcal{H} = \tfrac{1}{2}\,\pi^2 + \tfrac{1}{2}\,|\nabla\phi|^2 + \tfrac{1}{2}\,m^2\phi^2 \;=\; T^{00}\big|_{\dot{\phi}\to\pi} : \tag{163}$$

the Hamiltonian density *is* the energy density (159) — the field-theory instance of [Mech. §6.3]'s "$H = T + V$ for a kinetic term purely quadratic in the velocities and a velocity-independent potential," and positive definite for the reasons established in §9.4.

The discrete check, third installment. On the chain, $p_j = m\dot{y}_j = (\mu a)\,\dot{\phi}(x_j) = a\,\pi(x_j)$: the bead's momentum is the momentum *density* times the cell size — the same factor of a as in §7.2, now on the conjugate variable. The two cell factors then combine correctly in the Hamiltonian: $\sum_j p_j^2/(2m) = \sum_j a\,\pi(x_j)^2/(2\mu) \to \int \mathrm{d}x\,\pi^2/(2\mu)$, the kinetic term of (163) (with the string's μ, absorbed to 1 in the Klein–Gordon normalization).

10.2 The Dirac delta as a distribution

Phase-space mechanics needs one more derivative the Lagrangian story never asked for: the sensitivity of a *local* quantity to a local change — $\delta\phi(\vec{y})/\delta\phi(\vec{x})$, the field-theory descendant of $\partial q_i/\partial q_j = \delta_{ij}$. Apply the definition (119) to the evaluation functional $F[\phi] = \phi(\vec{y})$: its variation under $\phi \to \phi + \varepsilon\eta$ is exactly $\varepsilon\,\eta(\vec{y})$, so the sought object is whatever satisfies

$$\int \mathrm{d}^3x\,\frac{\delta\phi(\vec{y})}{\delta\phi(\vec{x})}\,\eta(\vec{x}) = \eta(\vec{y}) \qquad \text{for every test function } \eta. \tag{164}$$

No function does this — a function vanishing away from one point integrates to zero against everything — but (164) itself is a perfectly good definition of a different kind of object: a *distribution*, a linear machine that takes a test function and returns a number, written *as if* it were an integral kernel:

$$\boxed{\frac{\delta\phi(\vec{y})}{\delta\phi(\vec{x})} = \delta^3(\vec{x}-\vec{y}), \qquad \int \mathrm{d}^3x\,\delta^3(\vec{x}-\vec{y})\,f(\vec{x}) = f(\vec{y}).} \tag{165}$$

The integral notation is a fiction with a license: every manipulation performed on δ in this document (integration against smooth functions, integration by parts, change of variables) is the shadow of a legitimate operation on the underlying functional, and Appendix B sets out the small amount of theory that issues the license. The dictionary, meanwhile, closes its last entry with the same cell factor as always:

$$\frac{\partial y_i}{\partial y_j} = \delta_{ij} \qquad \longleftrightarrow \qquad \frac{\delta\phi(x_i)}{\delta\phi(x_j)} = \frac{\delta_{ij}}{a} \;\to\; \delta(x_i - x_j) : \tag{166}$$

the Kronecker delta *per cell volume* is the discrete Dirac delta — check: $\sum_j a\,(\delta_{ij}/a)\,f_j = f_i$ mirrors (165) term for term — and it correctly blows up as $a \to 0$: sensitivity *density* to a single point must be infinite if its integral over a shrinking cell is to stay one.

10.3 Poisson brackets and Hamilton's equations

Port the bracket of [Mech. §8.1] through the dictionary — sum to integral, partials to functional derivatives:

$$\{F, G\} \;=\; \int \mathrm{d}^3x \left[\frac{\delta F}{\delta\phi(\vec{x})}\,\frac{\delta G}{\delta\pi(\vec{x})} - \frac{\delta F}{\delta\pi(\vec{x})}\,\frac{\delta G}{\delta\phi(\vec{x})}\right], \tag{167}$$

inheriting antisymmetry, bilinearity, Leibniz, and Jacobi from the mechanical original (the proofs [Mech. §8.1, App. F] transcribe line by line). The fundamental brackets follow from (165): with $F = \phi(\vec{x})$, $G = \pi(\vec{y})$,

$$\{\phi(\vec{x}),\, \pi(\vec{y})\} = \int \mathrm{d}^3z\; \delta^3(\vec{z}-\vec{x})\,\delta^3(\vec{z}-\vec{y}) = \delta^3(\vec{x}-\vec{y}), \tag{168}$$

$$\boxed{\{\phi(\vec{x}), \pi(\vec{y})\} = \delta^3(\vec{x}-\vec{y}), \qquad \{\phi,\phi\} = \{\pi,\pi\} = 0 \qquad \text{(equal times).}} \tag{169}$$

Dynamics is bracket flow, as in [Mech. §8.2]: $\dot{F} = \{F, H\}$, and in particular

$$\dot{\phi} = \frac{\delta H}{\delta \pi}, \qquad \dot{\pi} = -\frac{\delta H}{\delta\phi}. \tag{170}$$

The Klein–Gordon check. From (163): $\delta H/\delta\pi = \pi$, so $\dot{\phi} = \pi$, recovering the definition; and — here the integration by parts built into (119) acts on the gradient term —

$$\dot{\pi} = -\frac{\delta H}{\delta \phi} = \nabla\cdot(\nabla\phi) - m^2\phi = \nabla^2\phi - m^2\phi. \tag{171}$$

Combining, $\ddot{\phi} = \nabla^2\phi - m^2\phi$ — the Klein–Gordon equation (138), re-derived from the phase-space side. And one sentence that the outlook will inflate: promoting the classical bracket (169) to a commutator, $[\hat{\phi}(\vec{x}), \hat{\pi}(\vec{y})] = \mathrm{i}\,\delta^3(\vec{x}-\vec{y})$, is the *first move* of canonical quantization — the entire apparatus of this section is its starting point.

Remark 10.1 (The slicing trade returns for fields)*.* Remark 4.5 described this trade for the particle; here it recurs for fields, in identical form — which is the point. The momentum (161) differentiates with respect to *somebody's* $\dot{\phi}$; the brackets (169) are *equal-time* brackets, defined on a slice; and the Hamiltonian is $H = \int T^{00}\,\mathrm{d}^3x = P^0$ — the 00-component of a tensor, integrated over the chosen slice, the time component of the field's four-momentum. Different observers slice differently and carry different Hamiltonians. Covariance survives as a theorem with the same statement as before: the ten Poincaré charges — P^μ from §9.4, $M^{\mu\nu}$ from Remark 9.5 — close under the bracket (167) into the Poincaré algebra, boost charges explicitly time-dependent, conservation by the same cancellation as in Remark 4.5. [stated; verifying the algebra is a long computation, not run here.] Canonical quantization inherits every word of this, and "the quantum field theory is Lorentz invariant" is a statement about operator charges, proved, not seen — while the path integral, built on the scalar $S[\phi]$ of §8.1, keeps covariance manifest throughout. The two quantization routes part company exactly here, at a fork Part I's free particle already mapped.

Part III

The Electromagnetic Field

11 The Field Tensor

Part I left electromagnetism half-told: the potentials A^μ entered as a background, the variational principle manufactured the antisymmetric combination $F_{\mu\nu}$ (Eq. (100)), and the quotient rule certified it as a tensor. This section opens Part III by taking the tensor seriously as an object: its components are $\vec{E}$ and $\vec{B}$, its transformation law is the long-sought answer to "how do electric and magnetic fields look from a moving frame," and its two invariants are the only field statements all observers share. The section closes by building the object that will source the dynamics: the four-current.

11.1 The six components

From §5.3, with $A^\mu = (\varphi, \vec{A})$, $A_\mu = (\varphi, -\vec{A})$:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad \vec{E} = -\nabla\varphi - \partial_t \vec{A}, \qquad \vec{B} = \nabla \times \vec{A}. \tag{172}$$

Antisymmetry kills the diagonal and pairs the rest: 6 independent components, and §5.3 claimed they are exactly the six numbers above. Verify by direct evaluation. Time-space:

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = \partial_t(-A^i) - \partial_i \varphi = \left(-\nabla\varphi - \partial_t \vec{A}\right)_i = E_i. \tag{173}$$

Space-space:

$$F_{ij} = \partial_i A_j - \partial_j A_i = -\big(\partial_i A^j - \partial_j A^i\big) = -\,\epsilon_{ijk}\, B_k, \tag{174}$$

(the middle sign from lowering two spatial indices; the last equality is the component form of the curl). Assembled, with rows μ and columns ν:

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{pmatrix}, \qquad F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix} \tag{175}$$

(raising both indices flips only the mixed time-space entries: $F^{0i} = \eta^{00}\eta^{ij}F_{0j} = -F_{0i}$, while $F^{ij} = F_{ij}$). The moral outranks the matrix: *$\vec{E}$ and $\vec{B}$ are not two vector fields.* They are the time-space and space-space slices of one antisymmetric tensor — a split made by a choice of frame, with no invariant meaning — and every "mystery" of their mixing is the ordinary behavior of tensor components under Eq. (61).

11.2 How $\vec{E}$ and $\vec{B}$ transform

The tensor law now delivers what §3.6 promised: the transformation of $\vec{E}$ and $\vec{B}$ for free. Boost along x with velocity β and apply $F'^{\mu\nu} = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma F^{\rho\sigma}$. Two representative components, in full. The (01) entry meets both boost rows:

$$F'^{01} = \Lambda^0{}_0 \Lambda^1{}_1\, F^{01} + \Lambda^0{}_1 \Lambda^1{}_0\, F^{10} = \gamma^2 F^{01} + \gamma^2\beta^2 F^{10} = \gamma^2(1-\beta^2)\, F^{01} = F^{01}, \tag{176}$$

so $E'_x = E_x$: the component along the boost is untouched (the two γ's and the antisymmetry combine to cancel). The (02) entry meets one boost row and one spectator:

$$F'^{02} = \Lambda^0{}_0\, F^{02} + \Lambda^0{}_1\, F^{12} = \gamma\,(-E_y) + (-\gamma\beta)(-B_z) \implies E'_y = \gamma\big(E_y - \beta B_z\big). \tag{177}$$

The remaining components run identically, and the complete result is:

parallel to the boost	transverse to the boost	
$E'_x = E_x$	$E'_y = \gamma\,(E_y - \beta B_z)$	$E'_z = \gamma\,(E_z + \beta B_y)$
$B'_x = B_x$	$B'_y = \gamma\,(B_y + \beta E_z)$	$B'_z = \gamma\,(B_z - \beta E_y)$

Compactly: $\vec{E}_\parallel$, $\vec{B}_\parallel$ inert; $\vec{E}'_\perp = \gamma(\vec{E} + \vec{\beta}\times\vec{B})_\perp$, $\vec{B}'_\perp = \gamma(\vec{B} - \vec{\beta}\times\vec{E})_\perp$. Electric and magnetic fields feed each other under boosts, in exactly the way t and x do.

Remark 11.1 (Magnetism is boosted electricity)*.* The table contains a physical point usually buried in a third course on the subject. Take a single charge at rest: pure Coulomb $\vec{E}$, no $\vec{B}$ anywhere (elementary electrostatics, imported here as input; §12 derives the field equations, and Example 12.2 the field). View it from a moving frame: the table manufactures $\vec{B}'_\perp = -\gamma(\vec{\beta}\times\vec{E})_\perp \neq 0$ — a magnetic field, from nothing but a boost. Magnetism is not a second force; it is what electrostatics looks like from a frame in which the charges move, and its everyday effects are large not because β is large (drift velocities in wires are mm/s, $\beta \sim 10^{-12}$) but because the electrostatic force it corrects is enormous — wires are neutral to extraordinary precision, and the magnetic force between them is the $O(\beta^2)$ relativistic residue of two large, nearly perfectly canceling electric forces. [The quantitative wire analysis — length contraction unbalancing the two charge densities — is a classic worked example; stated here, not run.] Historically the logic ran backwards: Einstein's 1905 paper is titled "On the Electrodynamics of Moving Bodies," and its opening paragraph is precisely the magnet-and-conductor asymmetry this remark dissolves.

11.3 The two invariants

What do all observers agree on? Full contractions (§3.6). With one tensor and the metric, exactly one is available:

$$F_{\mu\nu}F^{\mu\nu} = 2\,F_{0i}F^{0i} + F_{ij}F^{ij} = -2\,\vec{E}^2 + \epsilon_{ijk}\epsilon_{ijl}B_kB_l \implies \boxed{F_{\mu\nu}F^{\mu\nu} = 2\big(\vec{B}^2 - \vec{E}^2\big),} \tag{178}$$

using $\epsilon_{ijk}\epsilon_{ijl} = 2\delta_{kl}$. A second invariant needs the last invariant tensor, now introduced as §3.6 promised: the totally antisymmetric $\epsilon^{\mu\nu\rho\sigma}$, fixed by $\epsilon^{0123} = +1$. It is invariant under the proper orthochronous transformations only — $\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\Lambda^\rho{}_\gamma\Lambda^\sigma{}_\delta\,\epsilon^{\alpha\beta\gamma\delta} = (\det\Lambda)\,\epsilon^{\mu\nu\rho\sigma}$, the definition of the determinant — so objects built with one ϵ flip sign under P or T (*pseudo*-scalars and pseudo-tensors). Define the *dual* tensor and evaluate its slices by the same component patience as (175):

$$\tilde{F}^{\mu\nu} \equiv \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}, \qquad \tilde{F}^{0i} = -B_i, \quad \tilde{F}^{ij} = \epsilon_{ijk}\,E_k \qquad \text{— i.e. } (\vec{E},\vec{B}) \to (\vec{B},\,-\vec{E}): \tag{179}$$

duality trades the two fields. The second invariant:

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = 2F_{0i}\tilde{F}^{0i} + F_{ij}\tilde{F}^{ij} = -2\,\vec{E}\cdot\vec{B} - 2\,\vec{E}\cdot\vec{B} \implies \boxed{F_{\mu\nu}\tilde{F}^{\mu\nu} = -\,4\,\vec{E}\cdot\vec{B}} \tag{180}$$

(a pseudo-scalar: it flips under P, as $\vec{E}\cdot\vec{B}$ visibly does, $\vec{E}$ being a vector and $\vec{B}$ an axial vector). These two numbers are the entire frame-independent content of a field configuration at a point, and they sort configurations into invariant classes: if $\vec{E} \perp \vec{B}$ in one frame, then in all; if $\vec{E} \cdot \vec{B} \neq 0$, both fields are nonzero in *every* frame, and no purely electric or purely magnetic frame exists; if $\vec{E} \cdot \vec{B} = 0$ and $\vec{B}^2 > \vec{E}^2$, some frame sees a pure magnetic field (and symmetrically for $\vec{E}^2 > \vec{B}^2$); and the doubly-null case

$$\vec{E}^2 = \vec{B}^2, \qquad \vec{E} \cdot \vec{B} = 0 \qquad \text{(both invariants zero)} \tag{181}$$

is a class of its own — *null fields*, perpendicular and equal-magnitude in *every* frame, with no rest frame to boost to. Section 13 will find exactly this signature in the electromagnetic wave: light is null, in every sense this document has given the word.

11.4 The four-current

Dynamics needs a source, and §9.3 specified what one must be: a conserved four-vector current. Electromagnetism's version packages charge density and current density,

$$J^\mu = \big(\rho,\ \vec{J}\big), \tag{182}$$

and its four-vector credentials rest on one experimental scalar: *electric charge is Lorentz invariant* (an electron's charge does not depend on its speed — verified to extraordinary precision by the neutrality of atoms, whose electrons move and whose protons' quarks move very differently [experimental input]). For a flow of charges of proper density ρ_0 (density in the local rest frame, a scalar) moving with four-velocity $u^\mu(x)$:

$$J^\mu = \rho_0\, u^\mu = \big(\gamma\rho_0,\ \gamma\rho_0\vec{v}\big) \tag{183}$$

— scalar times four-vector, credentials immediate; the γ in $\rho = \gamma\rho_0$ is length contraction acting on the volume in the denominator of "charge per volume," charge itself inert. A general J^μ is a superposition of such flows. Conservation of charge is then the continuity equation of §9.2,

$$\boxed{\partial_\mu J^\mu = 0 \qquad \Longleftrightarrow \qquad \frac{\partial\rho}{\partial t} + \nabla\cdot\vec{J} = 0,} \tag{184}$$

with Remark 9.3 supplying the reason this *must* be the local statement. For now (184) is an empirical fact about the source; §12 will show the field equations *enforce* it — a source that fails to conserve charge cannot couple to the electromagnetic field at all — and §9.3's Noether current is the standing example of where a microscopic theory gets such a J^μ from.

12 The Maxwell Lagrangian

Now the building-block game of §8.2, replayed for the vector field — with one extra constraint. For the scalar, the covariance principle alone thinned the catalogue; for A_μ, the redundancy met in §5.2 also constrains: *the action may change by at most boundary terms under* $A_\mu \to A_\mu + \partial_\mu\chi$, or the unphysical redundancy would leak into the dynamics. The two constraints turn out to leave essentially one theory standing, and its equations of motion are Maxwell's.

12.1 Building the Lagrangian under two constraints

Candidates quadratic in A_μ (free-field logic, as in §8.2), screened in order:

- **The mass term $A_\mu A^\mu$.** A perfectly good Lorentz scalar — but not gauge invariant: it shifts by $2A^\mu\partial_\mu\chi + \partial_\mu\chi\,\partial^\mu\chi$, no total divergence for generic χ. A photon mass is not merely inelegant; it is *incompatible with the redundancy*, which is why its experimental bound is a test of the framework ($m_\gamma < 10^{-18}\,\text{eV}$ [experimental input]).
- **Generic derivative quadratics** ($\partial_\mu A_\nu\,\partial^\mu A^\nu$, $(\partial_\mu A^\mu)^2$, $\ldots$). Individually gauge-variant. The *only* first-derivative combination inert under the gauge shift is the one the particle's variational principle manufactured in §5.3: $F_{\mu\nu}$, gauge-invariant because $\partial_\mu\partial_\nu\chi$ is symmetric (Proposition 3.11 in action). Scalars built from it: $F_{\mu\nu}F^{\mu\nu}$ and $F_{\mu\nu}\tilde{F}^{\mu\nu}$.
- **The dual candidate $F\tilde{F}$** drops out: it is a total divergence,

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = 2\,\epsilon^{\mu\nu\rho\sigma}\,\partial_\mu A_\nu\,\partial_\rho A_\sigma = 2\,\partial_\mu\big(\epsilon^{\mu\nu\rho\sigma}A_\nu\,\partial_\rho A_\sigma\big), \tag{185}$$

the second equality because pulling ∂_μ out costs a term $\epsilon^{\mu\nu\rho\sigma}A_\nu\partial_\mu\partial_\rho A_\sigma$ that dies on Proposition 3.11 ($\partial_\mu\partial_\rho$ symmetric, ϵ antisymmetric). Boundary terms do not affect the equations of motion, so $F\tilde{F}$ contributes nothing classically. (Its quantum role — topological terms, anomalies — is real and deferred. [stated.])

One field-only candidate survives. Fix its normalization by expanding in 3+1 language via (178):

$$-\tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu} = \tfrac{1}{2}\big(\vec{E}^2 - \vec{B}^2\big), \tag{186}$$

and since $\vec{E}$ carries the time derivatives of the potentials (Eq. (97)), the coefficient $-\frac{1}{4}$ is exactly what gives the kinetic terms the scalar field's $+\frac{1}{2}$ normalization — kinetic minus potential, matching (136).

The source coupling is the linear term $-J^\mu A_\mu$, with J^μ a prescribed external current — and here the gauge constraint becomes decisive. Under the gauge shift,

$$-\,J^\mu\,\partial_\mu\chi = -\,\partial_\mu\big(J^\mu\chi\big) \;+\; \chi\,\partial_\mu J^\mu\,: \tag{187}$$

a total divergence *if and only if* $\partial_\mu J^\mu = 0$. Gauge invariance does not merely tolerate the conservation law of §11.4 — it *demands* it, before a single equation of motion has been derived: a non-conserved current cannot be coupled to the electromagnetic field at all. Assembling:

$$\boxed{\mathcal{L} \;=\; -\tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu} \;-\; J^\mu A_\mu.} \tag{188}$$

Remark 12.1 (Units in Part III). The normalization of (188) fixes the electromagnetic unit system: these are rationalized Heaviside–Lorentz units (with $c = 1$ as always), in which no factors of 4π appear in the field equations. Restoring SI expressions — as Example 13.2 will do — reinstates ε_0 and μ_0 by dimensional analysis, in the same way Convention 1.3 restores c. [stated; the conversion between unit systems is standard.]

12.2 Variation: the inhomogeneous pair

The field Euler–Lagrange equation (133) runs once per component of A_ν — four fields, one shared index. The derivative of the kinetic term is the section's one full index computation, so it is displayed whole. Write $F_{\rho\sigma}F^{\rho\sigma} = 2\big(\partial_\rho A_\sigma\,\partial^\rho A^\sigma - \partial_\rho A_\sigma\,\partial^\sigma A^\rho\big)$; then, with the renaming discipline of Rule 6 (§3.7),

$$\frac{\partial}{\partial(\partial_\mu A_\nu)}\big(\partial_\rho A_\sigma\,\partial^\rho A^\sigma\big) = 2\,\partial^\mu A^\nu, \qquad \frac{\partial}{\partial(\partial_\mu A_\nu)}\big(\partial_\rho A_\sigma\,\partial^\sigma A^\rho\big) = 2\,\partial^\nu A^\mu, \tag{189}$$

(each factor contributes once, the metric raising what the Kronecker deltas deliver), so

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu)} = -\tfrac{1}{4}\cdot 2\left(2\,\partial^\mu A^\nu - 2\,\partial^\nu A^\mu\right) = -\,F^{\mu\nu}, \qquad \frac{\partial\mathcal{L}}{\partial A_\nu} = -\,J^\nu, \tag{190}$$

and (133) reads $\partial_\mu(-F^{\mu\nu}) = -J^\nu$:

$$\boxed{\partial_\mu F^{\mu\nu} = J^\nu.} \tag{191}$$

Unpack with the matrix (175) ($F^{i0} = E_i$, $F^{ij} = -\epsilon_{ijk}B_k$). The $\nu = 0$ component:

$$\partial_i F^{i0} = J^0 \qquad \Longrightarrow \qquad \nabla\cdot\vec{E} = \rho \qquad \text{(Gauss)}, \tag{192}$$

and $\nu = j$:

$$\partial_0 F^{0j} + \partial_i F^{ij} = J^j \qquad \Longrightarrow \qquad \nabla\times\vec{B} - \frac{\partial\vec{E}}{\partial t} = \vec{J} \qquad \text{(Ampère–Maxwell)}, \tag{193}$$

using $\partial_i F^{ij} = -\epsilon_{ijk}\partial_i B_k = (\nabla\times\vec{B})_j$. Two of Maxwell's four equations, from one variation.

The field equations enforce charge conservation. Act with ∂_ν on (191):

$$\partial_\nu J^\nu = \partial_\nu\partial_\mu F^{\mu\nu} = 0 \qquad \text{identically} \tag{194}$$

— Proposition 3.11 once more, symmetric $\partial_\nu\partial_\mu$ against antisymmetric $F^{\mu\nu}$. The requirement of §11.4 is now enforced from both ends: gauge invariance demanded a conserved source before the dynamics existed (187), and the dynamics, once derived, is structurally incapable of sourcing itself from anything else. That one three-line proposition from the grammar section is behind gauge invariance of F, the inertness of $F\tilde{F}$, and charge conservation — three results, one identity.

Example 12.2 (The static point charge)**.** The simplest solution of (191). Put a single charge q at rest at the origin: $\rho = q\,\delta^3(\vec{x})$, $\vec{J} = 0$ — the delta of §10.2 placing the whole charge at one point (§12.4 exhibits this as the static case of a worldline current). Seek a static, spherically symmetric field — the source distinguishes no direction and no time — so try $\vec{E} = E(r)\,\hat{r}$ with $\vec{B} = 0$. Integrate Gauss's law over a ball of radius r and apply the divergence theorem, as in §9.2:

$$\oint_{S_r} \vec{E}\cdot \mathrm{d}\vec{A} = 4\pi r^2\,E(r) = \int_{B_r} q\,\delta^3(\vec{x})\;\mathrm{d}^3x = q, \tag{195}$$

so

$$\boxed{\vec{E} = \frac{q}{4\pi r^2}\,\hat{r}} \qquad \Big(\text{equivalently } \varphi = \frac{q}{4\pi r},\ \ \vec{A} = 0:\ \text{check}\ \ -\nabla\varphi = \vec{E},\ \vec{B} = \nabla\times\vec{A} = 0\Big). \tag{196}$$

The 4π of Heaviside–Lorentz units (Remark 12.1) sits in the field, not in the field equation. This is the pure-electric configuration that Remark 11.1 boosted; that it is the *only* static solution decaying at infinity is a uniqueness statement not proved here. [stated.]

12.3 The homogeneous pair: geometry, not dynamics

Two Maxwell equations remain, and no variation will produce them — because they are not dynamics. They are *identities*, true of any F built from a potential. Substitute $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ into the cyclic sum:

$$\begin{aligned}\partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} &= \big(\partial_\mu\partial_\nu A_\rho - \partial_\mu\partial_\rho A_\nu\big) + \big(\partial_\nu\partial_\rho A_\mu - \partial_\nu\partial_\mu A_\rho\big) + \big(\partial_\rho\partial_\mu A_\nu - \partial_\rho\partial_\nu A_\mu\big)\\ &= 0,\end{aligned} \tag{197}$$

the six terms canceling in pairs (first with fourth, second with fifth, third with sixth — equality of mixed partials each time). This is the *Bianchi identity*; contracting it with $\frac{1}{2}\epsilon^{\lambda\mu\nu\rho}$ packages the same statement as

$$\boxed{\partial_\mu \tilde{F}^{\mu\nu} = 0,} \tag{198}$$

(ϵ picks out exactly the totally antisymmetric part of ∂F, which is what the cyclic sum is). Unpack with $\tilde{F}^{i0} = B_i$, $\tilde{F}^{ij} = \epsilon_{ijk}E_k$ (§11.3): the $\nu = 0$ component is $\nabla \cdot \vec{B} = 0$ (no magnetic charge), and $\nu = j$ is $\nabla \times \vec{E} + \partial_t \vec{B} = 0$ (Faraday). All four of Maxwell's equations are now in hand, and the division is worth displaying:

pair	origin	content
$\partial_\mu F^{\mu\nu} = J^\nu$	variation of (188)	dynamics; couples to matter
$\partial_\mu \tilde{F}^{\mu\nu} = 0$	identity of $F = \partial A - \partial A$	geometry; no source possible

Half of electromagnetism is a law of motion; the other half is the tautological cost — and reward — of describing the field by a potential. (The converse also holds: any F satisfying the Bianchi identity is, locally, $\partial_\mu A_\nu - \partial_\nu A_\mu$ for some A — the Poincaré lemma [stated; a real theorem, with global exceptions on topologically nontrivial regions that become physics in their own right — magnetic monopoles, Aharonov–Bohm].)

12.4 One coupling, two faces

Part I coupled a *particle* to the field through $-q\int A_\mu\,\mathrm{d}x^\mu$; this Part couples the *field* to a current through $-\int \mathrm{d}^4x\, J^\mu A_\mu$. These are the same term. Write the current of a single point charge on a worldline $x^\mu(\lambda)$ — the delta apparatus of §10.2 finally building something:

$$J^\mu(x) = q\int \mathrm{d}\lambda\, \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\, \delta^4\big(x - x(\lambda)\big) \tag{199}$$

(reparametrization invariant, since $\mathrm{d}\lambda$ cancels; and the $\mu = 0$ component integrates over the δ in time to give $J^0 = q\,\delta^3(\vec{x} - \vec{x}(t))$ — a point charge's density, correctly). Insert into the field-side coupling; the δ^4 collapses the spacetime integral:

$$-\int \mathrm{d}^4x\; J^\mu A_\mu = -\,q\int \mathrm{d}\lambda\, \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\, A_\mu\big(x(\lambda)\big) = -\,q\int A_\mu\,\mathrm{d}x^\mu \tag{200}$$

— the particle-side coupling of Eq. (89), recovered exactly. One interaction term, two Euler–Lagrange addressees: vary the *worldline* through it and the Lorentz force emerges (§5.3); vary the *potential* through it and the source of Maxwell's equations emerges (§12.2). A single term in S supplies both action and reaction. And when the source is itself a field rather than a particle, §9.3 has already built the J^μ that supplies the coupling's other side — the conserved $U(1)$ current, whose conservation is exactly what (187) requires of any source.

What the Lagrangian has not yet delivered is the thing electromagnetism is famous for: waves. They are hiding inside (191), behind the gauge redundancy — and extracting them is the next section's business.

13 Gauge Freedom and Electromagnetic Waves

Maxwell's equations are in hand; light is not yet. This section extracts it, and the extraction is an exercise in managing the redundancy: the wave equation for A^μ appears only after the gauge freedom is used well, the physical content of the wave — two transverse polarizations — appears only after the redundancy is counted carefully, and the section closes with the field's energy-momentum tensor, which settles two outstanding items at once (Remark 9.5's concrete Belinfante case, and §11.3's claim that light is null).

13.1 Fixing a gauge: the Lorenz condition

Write the dynamical Maxwell pair (191) in terms of the potential:

$$\partial_\mu F^{\mu\nu} = \partial_\mu\big(\partial^\mu A^\nu - \partial^\nu A^\mu\big) = \Box A^\nu - \partial^\nu\big(\partial_\mu A^\mu\big) = J^\nu. \tag{201}$$

The first term is four Klein–Gordon operators at $m = 0$; the second term couples the components and blocks the wave reading. But $\partial_\mu A^\mu$ is exactly the kind of thing gauge freedom can reach: under $A_\mu \to A_\mu + \partial_\mu \chi$,

$$\partial_\mu A^\mu \;\longrightarrow\; \partial_\mu A^\mu + \Box\chi, \tag{202}$$

so choosing χ to solve the sourced wave equation $\Box\chi = -\,\partial_\mu A^\mu$ (solvable; standard [stated]) lands the potential in the *Lorenz gauge*,[2]

$$\partial_\mu A^\mu = 0, \tag{203}$$

where (201) collapses to

$$\boxed{\Box A^\nu = J^\nu\,:} \tag{204}$$

four decoupled, sourced wave equations, one per component — the massless Klein–Gordon field of §8, four copies, plus a source. The arc that began with Eq. (6) closes here: *this* is the wave equation whose Galilean non-invariance started the document, now derived from a Lorentz-scalar action, invariant by construction, no medium anywhere in sight. (What this section does *not* do is solve the sourced equation: the retarded potentials — the field propagating outward from its source at speed c — are the standard next step and are deferred [stated]; from §13.2 on we set $J^\nu = 0$ and study the waves themselves.)

Two housekeeping notes. First, the gauge freedom is not exhausted: (203) survives any further shift with $\Box\chi = 0$ — *residual* freedom, about to be needed. Second, an alternative choice exists:

Remark 13.1 (The Coulomb gauge). One may instead impose $\nabla \cdot \vec{A} = 0$ (Coulomb gauge), whereupon Gauss's law becomes $\nabla^2\varphi = -\rho$ — the scalar potential is determined *instantaneously* by the charge distribution, with no wave equation and no retardation. Nothing acausal follows (φ is not measurable; the gauge-invariant $\vec{E}$, $\vec{B}$ propagate causally [stated; this follows from the retarded solution just deferred]), but manifest covariance is gone: the condition $\nabla \cdot \vec{A} = 0$ names a frame. The trade is the same as in Remark 4.5 — Coulomb gauge gains explicit physical degrees of freedom at the cost of manifest symmetry, Lorenz gauge the reverse — and quantum theory uses both, for exactly those reasons. [stated; this document stays in Lorenz gauge.]

[2] Lorenz, not Lorentz: the condition is due to Ludvig Lorenz (Danish, 1867), not Hendrik Lorentz (Dutch) — and since (203) is built from a four-divergence, the Lorenz condition is Lorentz-invariant, a coincidence that has corrupted spellings for a century.

13.2 Counting the degrees of freedom: two polarizations

Vacuum ($J^\nu = 0$), plane-wave ansatz as in §8.4:

$$A^\mu(x) = \mathrm{Re}\, \varepsilon^\mu\, \mathrm{e}^{-\mathrm{i}k\cdot x}, \tag{205}$$

with a constant *polarization vector* ε^μ. Three conditions now stack, and the count they leave is the physics:

- **Dynamics.** $\Box A^\mu = 0$ demands $k \cdot k = 0$: the wavevector is *null*, $\omega = |\vec{k}|$ — the $m = 0$ dispersion of Fig. 8, light on the light line, as the name always promised. Four amplitudes ε^μ remain.
- **The gauge condition.** $\partial_\mu A^\mu = 0$ demands $k \cdot \varepsilon = 0$: one linear constraint, $4 \to 3$.
- **The residual freedom.** The leftover shifts, $\chi = \mathrm{Re}\,(\mathrm{i}\lambda\, \mathrm{e}^{-\mathrm{i}k\cdot x})$ (which obey $\Box\chi = 0$ precisely because $k^2 = 0$), move the polarization by

  $$\varepsilon^\mu \;\longrightarrow\; \varepsilon^\mu + \lambda\, k^\mu\, : \tag{206}$$

  the component of ε *along* k^μ is pure gauge — present in the description, absent from the physics — $3 \to 2$.

$$\boxed{4 \text{ (components)} \;-\; 1 \text{ (gauge condition)} \;-\; 1 \text{ (residual freedom)} \;=\; 2 \;\text{ physical polarizations.}} \tag{207}$$

Concretely, for $\vec{k} = \omega\hat{z}$: $k \cdot \varepsilon = 0$ kills one combination of $\varepsilon^0, \varepsilon^z$, the residual shift kills the other, and what remains is $\varepsilon^\mu = (0, \varepsilon_x, \varepsilon_y, 0)$ — *transverse* light, two independent directions of it. This is the pattern of Remark 4.2 and §5.2 brought to its conclusion: a redundant description carries components that the equations of motion decline to govern, and a correct count must subtract them *twice* — once for the condition that can be imposed, once for the freedom that survives it.

The fields of the wave close a promise. For the transverse ε above, Eqs. (97) give $\vec{E} \parallel \vec{\varepsilon}_\perp$, $\vec{B} = \hat{k} \times \vec{E}$: mutually perpendicular, perpendicular to the propagation, equal in magnitude. Hence both invariants vanish,

$$\vec{B}^2 - \vec{E}^2 = 0, \qquad \vec{E} \cdot \vec{B} = 0 \tag{208}$$

— the electromagnetic wave is a *null field* in the precise sense of §11.3, with a null wavevector to match: light is null in every sense this document has given the word, in every frame, with no rest frame to boost to — which is where the document came in (Postulate 2.2), now as output rather than input.

13.3 Improving the energy-momentum tensor

Run the canonical recipe (157), summed over the four fields A_ρ as Theorem 9.2 prescribes:

$$T^{\mu\nu}_{\text{can}} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\rho)}\, \partial^\nu A_\rho - \eta^{\mu\nu}\mathcal{L} = -\,F^{\mu\rho}\,\partial^\nu A_\rho + \tfrac{1}{4}\,\eta^{\mu\nu} F_{\rho\sigma}F^{\rho\sigma}, \tag{209}$$

and meet the defect Remark 9.5 predicted for a field with indices of its own: the first term is neither symmetric in $\mu\nu$ nor gauge invariant — a bare $\partial^\nu A_\rho$, not an F. The repair is one total divergence. Add

$$\Delta^{\mu\nu} = \partial_\rho\big(F^{\mu\rho} A^\nu\big) = F^{\mu\rho}\,\partial_\rho A^\nu + \big(\partial_\rho F^{\mu\rho}\big) A^\nu = F^{\mu\rho}\,\partial_\rho A^\nu \quad \text{(free field)}, \tag{210}$$

which changes no charge ($\Delta^{0\nu}$ is a spatial divergence, integrating to zero) and no conservation law ($\partial_\mu \Delta^{\mu\nu} = \partial_\mu \partial_\rho (F^{\mu\rho} A^\nu) = 0$: symmetric derivatives on a $\mu\rho$-antisymmetric object, Proposition 3.11, fourth appearance). The first term of T_{can} then completes to an F:

$$\boxed{T^{\mu\nu} = -\,F^{\mu\rho}\,F^{\nu}{}_{\rho} + \tfrac{1}{4}\,\eta^{\mu\nu}\,F_{\rho\sigma}F^{\rho\sigma}} \tag{211}$$

— symmetric, gauge invariant, conserved, all by inspection: the Belinfante improvement in its simplest working instance. Its physical slices, evaluated with the matrix (175):

$$\boxed{T^{00} = \tfrac{1}{2}\big(\vec{E}^2 + \vec{B}^2\big), \qquad T^{0i} = \big(\vec{E} \times \vec{B}\big)_i} \tag{212}$$

— the classical energy density of the electromagnetic field (positive definite, confirming the sign choice of §12.1's $-\frac{1}{4}$), and the *Poynting vector*, the energy flux, obtained here without a single line of the traditional $\vec{E} \times \vec{B}$ manipulation: it is a component of a tensor that a symmetry built. One structural remark is worth adding, and foreshadows much: the trace vanishes, $T^{\mu}{}_{\mu} = -F^{\mu\rho}F_{\mu\rho} + \frac{1}{4} \cdot 4 \cdot F_{\rho\sigma}F^{\rho\sigma} = 0$ — the fingerprint of a theory with no mass and hence no intrinsic scale. [stated; tracelessness and scale invariance is a QFT-course thread.]

For the plane wave of §13.2, with $|\vec{E}| = |\vec{B}|$ and $\vec{E} \perp \vec{B} \perp \hat{k}$:

$$T^{00} = \vec{E}^2, \qquad T^{0i} = \vec{E}^2\,\hat{k}_i \qquad \Longrightarrow \qquad |\text{flux}| = (\text{density}) \times c\,: \tag{213}$$

the wave's energy travels at exactly the speed of light — as it must, there being nothing else in a null field's kinematics for it to do.

Example 13.2 (Sunlight: the worked number)**.** Restore SI units and point (212) at the sky. The solar constant — the time-averaged Poynting flux at Earth's orbit — is $\langle S\rangle = 1.36\,\text{kW/m}^2$. For a wave of amplitude E_0, the SI average is $\langle S\rangle = \frac{1}{2}\varepsilon_0 c\,E_0^2$, so

$$E_0 = \sqrt{\frac{2\langle S\rangle}{\varepsilon_0 c}} \approx 1.0 \times 10^3\,\text{V/m}, \qquad B_0 = \frac{E_0}{c} \approx 3.4\,\mu\text{T} \tag{214}$$

— a kilovolt per meter of oscillating field in ordinary daylight, and a magnetic partner a tenth of the Earth's field. The momentum flux of the wave is a pressure, $\langle S\rangle/c \approx 4.5\,\mu\text{Pa}$: minute on skin, decisive on comet tails and solar sails, and a direct classical reading of the momentum density that (211) localizes in empty space.

Remark 13.3 (Conservation with sources). The tensor (211) was conserved for the *free* field; with a source, the field alone no longer conserves energy-momentum, and the failure is exactly the exchange with matter. Differentiate (211):

$$\partial_\mu T^{\mu\nu} = -\big(\partial_\mu F^{\mu\rho}\big)F^{\nu}{}_{\rho} \;-\; F^{\mu\rho}\,\partial_\mu F^{\nu}{}_{\rho} \;+\; \tfrac{1}{2}\,F_{\rho\sigma}\,\partial^\nu F^{\rho\sigma}. \tag{215}$$

The first term is $-F^{\nu\rho}J_\rho$ by the inhomogeneous pair (191). The last two cancel by the Bianchi identity: substituting the cyclic sum in the form $\partial^\nu F^{\rho\sigma} = -\partial^\rho F^{\sigma\nu} - \partial^\sigma F^{\nu\rho}$ into the third term turns it, after renaming dummies (Rule 1), into two copies of $+\frac{1}{2}\,F^{\mu\rho}\partial_\mu F^{\nu}{}_{\rho}$, which cancel the middle term. Hence

$$\boxed{\partial_\mu T^{\mu\nu} = -\,F^{\nu\rho}\,J_\rho\,.} \tag{216}$$

The $\nu = 0$ component, in 3+1 language:

$$\frac{\partial}{\partial t}\,\tfrac{1}{2}\big(\vec{E}^2 + \vec{B}^2\big) + \nabla \cdot \big(\vec{E} \times \vec{B}\big) = -\,\vec{E} \cdot \vec{J}\,: \tag{217}$$

the field's energy budget loses, per unit volume, exactly the power the Lorentz force (98) delivers to the charges — §5's force law, seen from the field's side of the one shared coupling (§12.4). Only the sum of field and matter energy-momentum is conserved; the transfer between them is what (98) describes from the matter side.

The classical story is complete: one scalar action per object — the free scalar (135), the Maxwell field (188), the charged particle (89) stitching them together — equations of motion, symmetries, currents, energy, waves. What the classical theory cannot say is why the light arrives in lumps. The outlook locates that boundary.

14 Outlook

This document has run out of classical physics to derive, and the outlook's first job is not a preview but a closing of the series' longest arc. The second job is to mark the three directions this apparatus opens, each with one paragraph and no pretense of following them.

14.1 Every field is a family of oscillators

Put the free Klein–Gordon field in a periodic box of volume V (§7.3, third mechanism) and expand in the Fourier modes the box allows:

$$\phi(\vec{x},t) = \frac{1}{\sqrt{V}} \sum_{\vec{k}} q_{\vec{k}}(t)\, \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x}}, \qquad q_{-\vec{k}} = q_{\vec{k}}^{*} \tag{218}$$

(the reality condition ties the modes in $\pm\vec{k}$ pairs, each pair carrying two real degrees of freedom — the amplitude's real and imaginary parts; the steps are standard and not belabored). Insert into the Lagrangian $L = \int \mathrm{d}^3x \left[\frac{1}{2}\dot{\phi}^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{1}{2}m^2\phi^2\right]$ and let the orthogonality $\int \mathrm{d}^3x\, \mathrm{e}^{\mathrm{i}(\vec{k}-\vec{k}')\cdot\vec{x}} = V\delta_{\vec{k}\vec{k}'}$ collapse the integrals:

$$\boxed{L = \sum_{\vec{k}} \left[\tfrac{1}{2}\left|\dot{q}_{\vec{k}}\right|^2 - \tfrac{1}{2}\,\omega_{\vec{k}}^2 \left|q_{\vec{k}}\right|^2\right], \qquad \omega_{\vec{k}}^2 = |\vec{k}|^2 + m^2.} \tag{219}$$

This is the statement the whole series was built to reach: *a free field is a countable family of independent simple harmonic oscillators*, one per wavevector, each with the frequency the dispersion relation (140) assigns. The object that opened the first document [Osc. §2] — one mass, one spring — is the *atom of field theory*: every result proved there applies here, mode by mode, verbatim. The chain's normal modes becoming Fourier modes (§6.3, last dictionary row) was this fact in embryo; the Hamiltonian

$$H = \sum_{\vec{k}} \left[\tfrac{1}{2}\left|p_{\vec{k}}\right|^2 + \tfrac{1}{2}\,\omega_{\vec{k}}^2 \left|q_{\vec{k}}\right|^2\right] \tag{220}$$

is [Osc.]'s energy, summed over infinitely many springs; and the Maxwell field, in Lorenz gauge four massless copies of the same (§13.1) minus the gauge redundancy, is two such families — one per polarization (§13.2).

14.2 Toward the quantum theory

Everything canonical quantization needs is already on the table. The bracket $\{\phi(\vec{x}), \pi(\vec{y})\} = \delta^3(\vec{x}-\vec{y})$ of §10.3 becomes, in mode variables, one canonical pair per oscillator; promote brackets to commutators (§10.3's one-sentence hook, now inflated to three). The quantum harmonic oscillator has the spectrum $E_n = \omega\,(n + \frac{1}{2})$ [stated; the only quantum-mechanics input used in this outlook], so a state of the quantum *field* is an assignment of an excitation number $n_{\vec{k}}$ to every mode, with

$$E = \sum_{\vec{k}} \omega_{\vec{k}}\, n_{\vec{k}} \;+\; E_0 : \tag{221}$$

energy arrives in lumps of $\omega_{\vec{k}}$, momentum in lumps of $\vec{k}$, and each lump sits on the mass shell $\omega^2 = |\vec{k}|^2 + m^2$ — Fig. 6, at last, *literally*: the lumps are **particles**, and the field's excitations of the Maxwell family are *photons*, two polarizations per wavevector, which is the answer to why the light of §13 arrives in pieces. (The constant $E_0 = \sum \frac{1}{2}\omega_{\vec{k}}$ diverges — the theory's first ultraviolet infinity, cousin of Appendix B's warning about products of distributions, and the problem named *renormalization*. [stated; a QFT course's first month.]) From here the handoff is clean: this document's Lagrangians, currents, and brackets are the exact classical inputs of the canonical formalism of Schwartz Chs. 2–3, and Remark 10.1 already mapped the fork in the road there.

14.3 Two directions not taken

Curved spacetime. Every structure in this document stood on one fixed object: $\eta_{\mu\nu}$. Promote it to a *field*, $g_{\mu\nu}(x)$, and the template of §3.1 survives pointwise while the physics transforms: freely falling particles still extremize $\int d\tau$, but the proper time is now measured by g, and the equation of motion picks up terms encoding the metric's variation — gravity as geometry. The source of the metric's own dynamics is an object this document built: the energy-momentum tensor of §9, sitting on the right-hand side of Einstein's equations exactly where J^μ sits in (191) — and the once-harmless additive constant in the Lagrangian (§8.2) stops being harmless, because gravity weighs $\mathcal{L}$ itself: the cosmological constant. [stated; a general relativity course.]

The gauge principle. This document met gauge freedom as a property electromagnetism *has* (§5.2); modern field theory runs the logic backwards, as a property interactions *come from*. Take the global $U(1)$ of §9.3 and demand it hold *locally*, $\alpha \to \alpha(x)$: the gradient $\partial_\mu \phi$ no longer transforms covariantly, and repairing it forces the introduction of a field A_μ with exactly the transformation law (91) and exactly the minimal coupling (107) — the electromagnetic field is *what localizing a phase demands*. Replace the phase circle by larger internal symmetry groups and the same demand manufactures the weak and strong interactions: the Standard Model is three runs of this argument. [stated; the actual run is Schwartz Ch. 8 and beyond.]

14.4 Closing the series

One mass on one spring; two, coupled; N, chained; the chain, continued to a rope; the rope's equation caught transforming badly; spacetime rebuilt so that transforming well means something; mechanics re-founded on a variational principle and Noether's dictionary between symmetry and conservation; the principle handed a field, then the simplest relativistic field, then the field that carries light — and at the end, expanded in modes, it is one mass on one spring again, infinitely many times. The recycled ideas were few: quadratic Lagrangians and their normal modes, coordinates adapted to a preserved structure, symmetry converted to conservation, redundancy counted correctly. Physics ahead — quantum fields, curved spacetime, gauge theory — will add machinery, but it will replace remarkably little of what these documents built; mostly it will run the same few ideas again, on stranger stages.

A Linearity of the Transformation Between Inertial Frames

This appendix proves Lemma 2.3 and disposes of the transverse coordinates. Write the transformation as a map $f : \mathbb{R}^4 \to \mathbb{R}^4$, $x \mapsto x' = f(x)$, between the coordinates that two inertial frames assign to the same events. We assume f is a bijection (each frame labels every event exactly once) and continuous (nearby events get nearby labels).

A.1 Homogeneity forces affinity

The physical input is the *homogeneity of space and time*: no event is distinguishable from any other by the laws of physics, so the transformation law for coordinate *differences* cannot depend on where in spacetime the difference sits. Formally: for every displacement $a \in \mathbb{R}^4$ there is a displacement $g(a)$ such that

$$f(x+a) - f(x) = g(a) \qquad \text{for all } x. \tag{222}$$

(If the left side depended on x, two identically prepared experiments at different locations, separated by the same a, would transform differently — a violation of homogeneity.)

Lemma A.1. *A continuous f satisfying (222) is affine: $f(x) = \mathsf{L}x + b$ with L linear and b constant.*

Proof. Evaluate (222) at two displacements. First, $f(x + a_1 + a_2) - f(x) = g(a_1 + a_2)$ directly. Second, insert the intermediate point:

$$f(x+a_1+a_2) - f(x) = \big[f((x+a_1)+a_2) - f(x+a_1)\big] + \big[f(x+a_1) - f(x)\big] = g(a_2) + g(a_1). \tag{223}$$

Hence g satisfies the Cauchy functional equation $g(a_1 + a_2) = g(a_1) + g(a_2)$. Additivity gives $g(qa) = q\,g(a)$ for all rational q (integers by iteration, reciprocals by substitution), and continuity of g — inherited from f via $g(a) = f(a) - f(0)$ — extends this to all real scalars. So g is linear; call it L. Setting $x = 0$ in (222): $f(a) = \mathsf{L}a + f(0)$, which is the claim with $b = f(0)$. □

In the standard configuration the origins coincide, $f(0) = 0$, so $b = 0$ and the transformation is linear, proving Lemma 2.3.

Remark A.2 (An alternative route). A different argument, requiring no appeal to homogeneity as a separate principle, starts from the definition of inertial frames: free particles move on straight worldlines in both, so f maps lines to lines, and a continuous line-preserving bijection of $\mathbb{R}^4$ is affine by the fundamental theorem of affine geometry. The conclusion is the same; the theorem invoked is substantially harder than Lemma A.1, so the main text leans on homogeneity.

A.2 The transverse coordinates are spectators

Let the relative motion lie along x. Isotropy about the motion axis (no direction transverse to the boost is preferred) and homogeneity force $y' = \kappa(v)\,y$, $z' = \kappa(v)\,z$ with a single scale $\kappa(v) > 0$: the transverse map can involve no shear into t, x, or the other transverse direction, since any such term would single out a transverse direction.

Now run the reciprocity–isotropy argument of Theorem 2.4 Step 3 on κ: composing the boost with its inverse gives $\kappa(v)\kappa(-v) = 1$, while reflection symmetry of the transverse plane under $x \to -x$ gives $\kappa(-v) = \kappa(v)$. Hence $\kappa(v)^2 = 1$, and positivity (continuity from $\kappa(0) = 1$) yields

$$\boxed{y' = y, \qquad z' = z.} \tag{224}$$

Lengths transverse to the motion are absolute — which is why the entire derivation of §2 could be run in the (ct, x) plane.

B Distributions, and the License for the Delta Function

This appendix issues the license claimed in §10.2: what kind of object δ is, and why the integral notation for it is safe. Everything is stated for one dimension; three dimensions is the same theory with products of test functions.

Test functions and distributions. Let $\mathcal{D}$ be the space of *test functions*: infinitely differentiable functions $f : \mathbb{R} \to \mathbb{R}$ vanishing outside some bounded set. A *distribution* is a linear functional $T : \mathcal{D} \to \mathbb{R}$, continuous in the appropriate sense (if $f_n \to f$ with all derivatives, uniformly, supports in a common bounded set, then $T[f_n] \to T[f]$). Distributions are exactly the objects that *know how to be integrated against test functions* — and nothing more.

Functions are distributions; δ is not a function. Every locally integrable function g defines a distribution by ordinary integration, $T_g[f] = \int gf \,\mathrm{d}x$; such distributions are called *regular*, and the map $g \mapsto T_g$ loses no information. The evaluation functional

$$\delta_y[f] \;=\; f(y) \tag{225}$$

is manifestly linear and continuous — a distribution — but it is not regular: a candidate g would have to satisfy $\int gf = f(y)$ for all f, forcing $g = 0$ almost everywhere away from y, whence $\int gf = 0 \neq f(y)$ in general. The notation

$$\delta_y[f] \;=\; \int \delta(x-y)\, f(x)\, \mathrm{d}x \tag{226}$$

is therefore a *notation*: the right side is defined by the left. Its utility is that the rules of integral manipulation, applied formally to the fiction, compute correctly — because each rule is secretly a definition:

- **Derivatives.** Define $T'[f] \equiv -T[f']$ — which is what integration by parts *would* give if T were a decaying function, boundary terms and all. Every distribution is thereby infinitely differentiable, and manipulations like "integrate the delta by parts" are applications of this definition, not of calculus.
- **Multiplication by smooth functions.** Define $(gT)[f] \equiv T[gf]$ for smooth g; e.g. $g(x)\,\delta(x-y) = g(y)\,\delta(x-y)$ as distributions.
- **Change of variables, scaling.** Defined by transporting the test function; e.g. $\delta(ax) = \delta(x)/|a|$ is the statement $\int \delta(ax)f(x)\mathrm{d}x = f(0)/|a|$, i.e. substitution performed on the test-function side.

What the fiction does *not* license is multiplying two distributions: δ^2 is not defined by any of the above, and no consistent general definition exists. (Classical field theory never needs such products; quantum field theory does — products of field operators at a point — and a substantial part of its machinery, under the name *renormalization*, is the repair. One sentence of foreshadowing, no more.)

Smeared fields. The careful reading of a classical field mirrors that of δ: the physically meaningful pairing is the field *smeared* against a test function, $\phi[f] = \int \phi(x)f(x)\,\mathrm{d}x$ — an average over the resolution profile f, which is all any finite-resolution measurement returns. "The field at a point" is the idealized $f \to \delta$ limit, harmless classically for continuous ϕ and adopted without further comment in the main text; the smeared object is the one that survives, unaltered, into the quantum theory.